2.9.27

MIL-125 reciple

- 51.5 mg BDC
- 1.75 ml DMF

Purge with N_2 to mix

~~DMF~~ ~~Prop:~~

- 44.7 mg Ti cluster
- 0135 mg MeOH

Pyge MeOH with N_2

~~MeOH~~ ~~Prop:~~

R	eq	MMol	mg	ml	M	Mw
BDC	14.76	0.131	51.5		.375	166.13
Ticluster	1	0.021	44.7			2129.5
DMf	—	—	—	1.75		
MeOH	—	—	—	0.35		73.09
						32.04

a	turbid	5 days →	Clear ~~ppt~~ supernatant	white ppt
b	turbid	→	clear supernatant	white ppt
c	turbid	→	clear supernatant	white + yellow ppt
d	turbid	→	" "	white + yellow ppt

Continued on Page

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MIL-125-NH$_2$ cyp (ARH-1-2)

R	eq	mmol	mg	M	M_n	M_w
NH$_2$ cyp	1.476	0.131	77.3	–	~~375~~	249.27
Ti cluster	1	0.021	44.7	–		2128.5
DMF	–	–	–	1.75	–	–
MeOH	–	–	–	0.35	–	–

Observations

~~At four samples~~

a → yellow suspension → clear yellow supernatant

b → " " → clear yellow supernatant

c → " " → brown supernatant

d → " " → dark brown supernatant

Continued on Page

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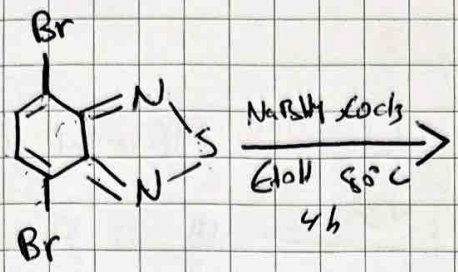
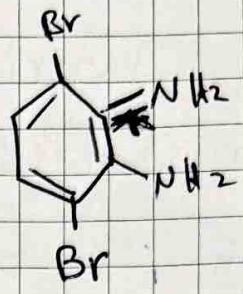


4,7-dibromo-2,1,3-benzothiadiazole

(A)

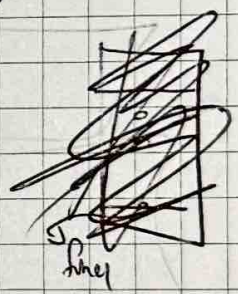


$+ S^{-2} + Co^{+2} \rightarrow CoS$ (black)

3,6-dibromo-1,2-benzenediamine

(B)

R	eq	mmol	mg	ml	mM	Mw
A	1	1.895	557	—	—	293.97
$NaBH_4$	2	3.128	143.4	—	—	32.83
$CoCl_2 \cdot 6H_2O$	0.1	0.1895	45.1	—		237.83
EtOH	—	—	—	23	—	46
B						



1- Add dithiazole in RBF
2- Add $NaBH_4$
3- Add $CoCl_2 \cdot 6H_2O$
4- Add ethanol → black
5- Stir @ 88°C 4h (this time left overnight).
6- SEP funnel
7- Silica Filter trough. dark green soln
8 - Rotavap.

Continued on Page 5

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Continued from Page

GSAS-II Tutorials

- 1) Neutron Diffraction - (resolve structure)

- open GSAS-II click → import → GSAS-II powder data file
- chose Gamet.raw
- Then click yes then chose prm file
- Date → Add new Phase

↓

General tab

- Phase type can be changed
- write on space group : a 3 d (space between).
- change unit cell a = 12.04000

After above is done

→ go to data tab (next to general).

→ go to atoms tab

- Edit atoms → Append atoms (6 times)

change X → 0.125 Z → 0.25

Continued on Page

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1st row Y

2nd row Fe

3rd row Al

4th row Al

5th row Fe

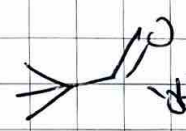
6th row O

Assign X, y, z

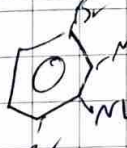
Af for words

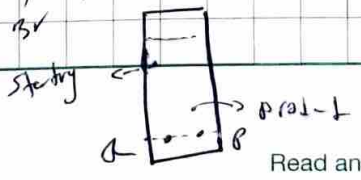
Calculate → refine

Ti_8O_8 cluster synthesis (ARH-1-5) (P.7 continu...

$Ti(isopropyl)_4$ +  → $Ti_{48}O_8(O_2C_2(CH_3)_3]_{16}$

R	eq	mmol	mg	ml	mM	Mw
Ti(isopropyl)4	1	4.22	1.2 grams	1.25		284.26
Oxalic acid	10.15	42.84	4.375 g	4.84		
Acetonitrile	—	—	—	15		
Ticks for	~~0.055~~	~~0.233~~				

TLC for  with EtOAc / Hex 1:9



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6/29/2025

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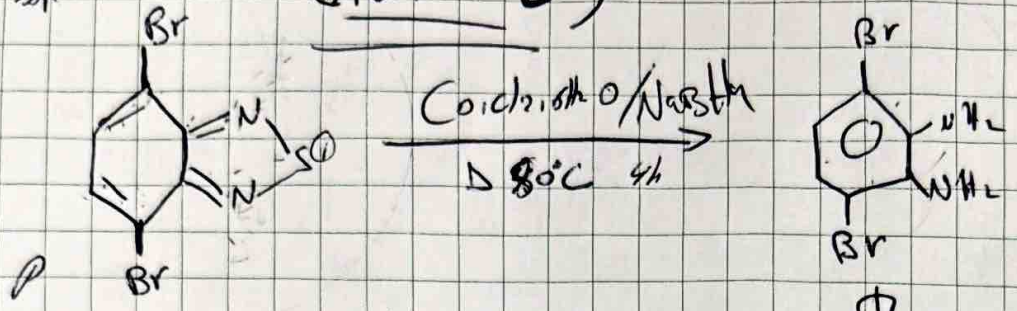
Date

(ARH-1-6)



4,7-dibromo-2,1,3 benzodithiazol

Reagent	Mass	mMoles	mM	FW	Volume	eq
SR	557	~~1.444~~	82.38	293.97		1
NaBH4	1434	3.78	164.35	37.83		2
CoCl2.6H2O	45.1	0.1895	~~823~~	237.83		0.1
EtOH	—	—	—	—	23 ml	—

① Add dithiazol in 50 ml RBF
② Add NaBH4 " " "
③ Add CoCl2.6H2O " " " (turns black when using some para)
④ Add 23 ml EtOH (turns black + bubbling)
⑤ reflux for ~~2 hr~~

reaction starts @ ~~2:08~~ 2:04

@ 2:26 rxn has dur turns dark) pink color (observed)

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continue P. 8 →

② Rf t_1 ⇒ 38 min t_2 :- 2h 24 min t_3 :. 3h 42 mins

⑥ rotavap to remove ethanol ⇒ dark pink solid t_4 :. 4h 13min

⑦ Add ≈ 10ml water t_5 : 6:50

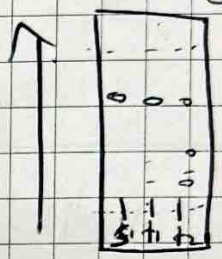
⑧ Extract with DCM ≈ 20ml (double amount of water) (aq layer orange, org yellow)

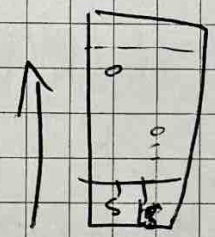
⑨ brine

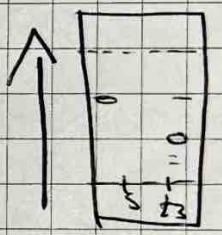
ⓑ dry with Na_2SO_4

ⓘ rotavap ⇒ yellow solid

194 mg ≈ 61%







Solvent :- EtylAc : Hexane
10%

#continue Page (13)

Rxn time :- 4h 48 min

① add 4.75 ml Pivalicacid

② add ADN from SPS

③ add Ti(OiPr)$_4$ ⇒ turns yellow

④ heat @ 120°C 6d using teflon bomb.

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continue (ARH-1-6)

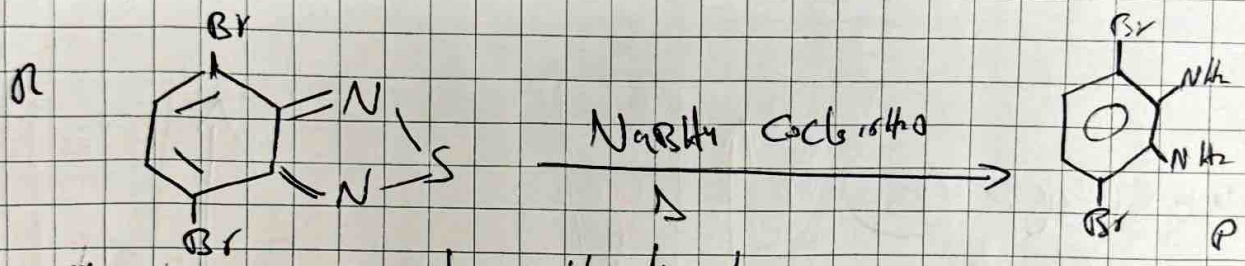
NMR for Product

- chloroform added to product

yellow solution forms

Product present but Impure

ARH-1-8



4,7-dibromo-2,1,3-benzothiadiazole

Reagent	Mass mg	mmoles	mM	eq	MW	Vol ml
R	294	1	50	1	293.91	-
NaBH4	76	2	100	2	37.83	
CoCl2.6H2O	25	0.021	105	0.107	237.83	-
EtOH	-					20
P	94	0.353	—	0.35	265.93	—

35% yield

NO Product in NMR

Continued on Page

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Add R + $NaBH_4$ + Carbonyl
↓ Add Ethanol
heat 2 h
↓
filter + celite
↓
rotavap
↓
Add 100 mL water
↓
Extract Et_2O → Dry Na_2SO_4

observations

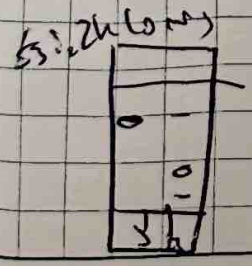
① Add R (50 mL RBF) ② Add $NaBH_4$ sticks on paper ③ Add $CoCl_2$ solution
↓
black crystals

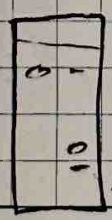
④ Add EtOH ⇒ black ⑤ heat

rxn starts @ 1:19 pm

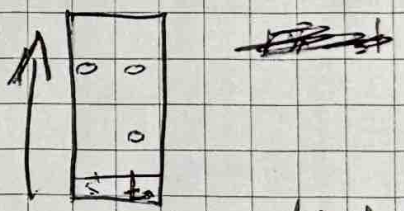
- observation: upon heating new spot emerges under product spot

- ~~t1~~ t1 :- ~~1h 30 min~~ 1h 30 min



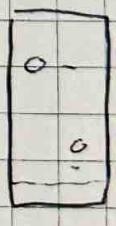


, lighter than t1



10% ethyl acetate in hexane

t1 : 1h



10% ethyl acetate in hexane

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t_4 3h | t_5 4h

⑥ filter using celite ⇒ filtrate red color

⑦ extract using EthylAcetate ⇒ emulsion ⇒ turns yellow aqueous orange

⑧ ~~[illegible]~~ dry using Na_2SO_4

⑨ filter ⑩ rotavap,

rxn yield 94mg. = 35.3% yield

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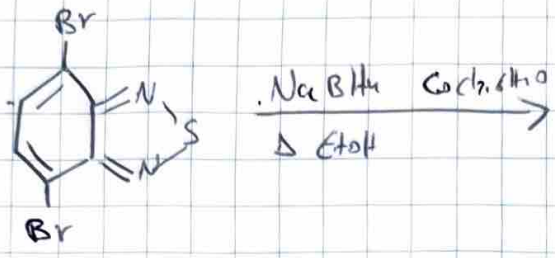
ARH-1-11



R P

Reagent	Fw	Mass	nMoles	mM	Vol	
R	293.7	294	1	50	-	
NaBH4	37.83	76	2	100	-	
CoCl2.6H2O	237.93	10	0.04	2	-	
EtoH	46	-	-	-	20	
P				-		

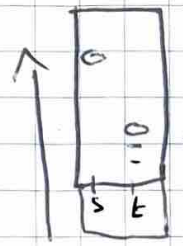
① Weigh 294mg of R in 50ml RBF

② Weigh 7 NaBH4

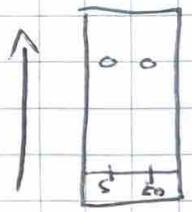
③ Weigh $CoCl_2 \cdot 6H_2O$

④ Add EtoH (some added by bottle to remove NaBH4 on neck

rxn proceeded for ~~5~~ 5h 20 mins



10% EtoAc in hexane



10% EtoAc/Acetone in hexane

rxn starts @ 9:10

⑤ rxn filtered over celite → celite turns black + blood red filtrate.

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6) the filtrate was left for 1h:20 mins in suction flask (due to class time) colored red ——→ green

7) rotavap → green pdt

8) Add water → black oil droplets form

9) extract with EtOAc ⇒ EtOAc ~~layer~~ ~~oily~~ colorless

10) brine

11) rota vap → oily green product

12) nmr ⇒ product present

13) in 1000ml RBF fill V3 with chloroform ⇒ add charcoal

14) heat until small bubbling occurs

15) add charcoal ⇒ bubbles increase then calm down

⇒ 16) swirl with occasional heating

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(17) filter with celite

continue ARH-1-6

Recrys solvent

① Methanol | yellow upon addition → soluble in cold (with sonication)

② Acetonitrile yellow upon addition → " " " (" ")

③ Ethyl Acetate → " " " " (" ")

④ water → insoluble in cold → (Melted insoluble (forms a layer on top))

Recrys (Methanol + water solvent pair)

① Add Methanol (minimal) yellow color

② heat

③ keep adding until no solid appears

④ Add water → turbid

⑤ boil → clear

} yellow solution

⑥ cool in fridge

⑦ suction.

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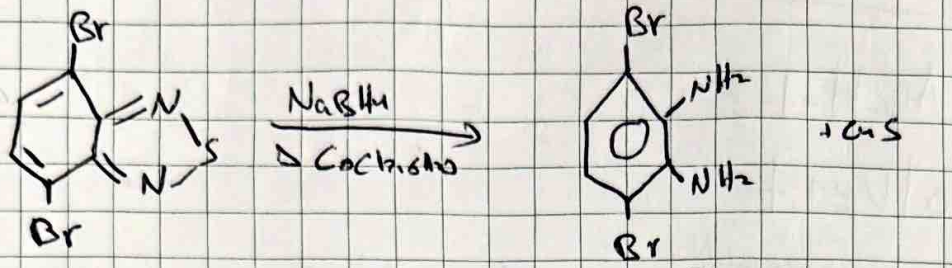
ARH-1-14



R P

Reagent	fw	Mass(mg)	mmol	mM	Vol	eq
R	293.7	294	1	66.6	–	1
$NaBH_4$	37.83	76	2	133.3	–	2
$CoCl_2.6H_2O$	237.93	15	0.06	2.67	–	0.04
EtOH	–	–	–	–	15	–

① Add R

② Add $NaBH_4$

③ Add $CoCl_2.6H_2O$

④ Add EtOH

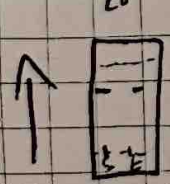
⇓

black color + bubbling

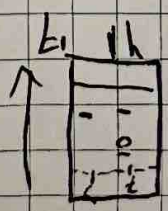
⑤ heat @ 78°C according to thermistor

(actual is 74~77)

⑥ @ t= rxn → ~~red~~ red 2h → black t=3h



10% ethyl acetate/hexane



10 % EtOAc/Hexane

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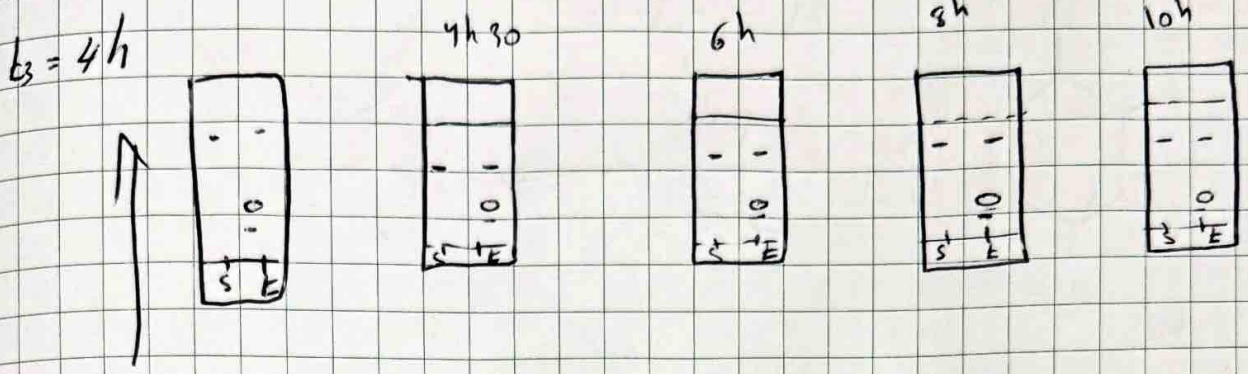

⑦ @ 13h rxn stopped ⟹ dark purple color

⑧ filter over celite ⟶ black residue dark purple filtrate

⑨ rota vap ⟶ dark purple residue

⑩ Add water ⟶ brown slurry suspension

⑨ extract with EtOAc ⟶ upper layer (org) dark purple
lower pale brown

⑩ " " brine ⟶ " " upper layer clear
lower layer clear

⑪ dry with Na_2SO_4.

⑫ rotavap ⟶ oily dark brown residue

TLC

NMR confirms
Impure Product

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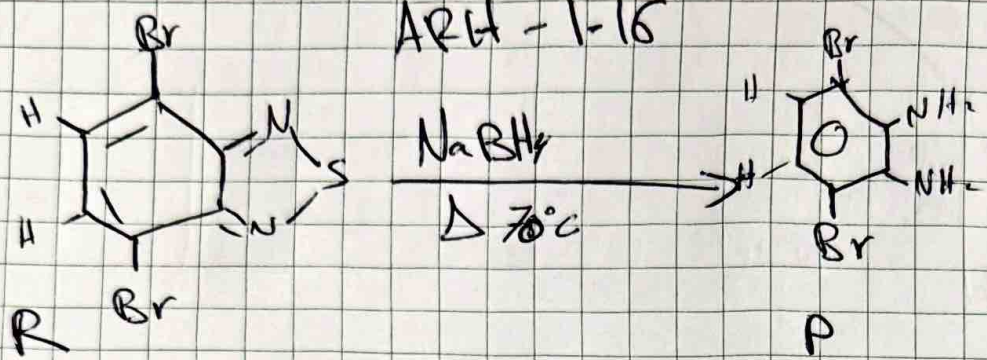
ART-1-16



R P

chemical	fw	Mass	mMol	mM	Vol	eq
R	293.7	294	1	50	-	1
NaBH4	37.83	76	2	100	-	2
CoCl2·6H2O	237.93	5	.0021	1.05	-	2.1
Ethanol	-	-	-	-	20	-

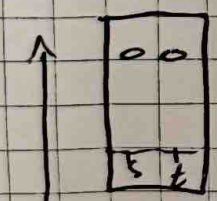
DOI
10.3389/fchem.2018.00359

① Add R ② Add NaBH4 ③ Add $CoCl_2 \cdot 6H_2O$ ↓ black crystals

④ Add EtOH → Grey bubbling → heat @ 75°C

rxn started @ 10:00 AM

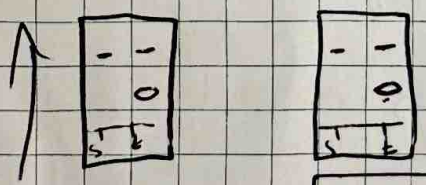
t0



10% EtOAc in hexane

* upon heating rxn turns to beige

t = 2h t = 30 min



highly impure NMR

* rxn after 20 mins turns red

- extract with water x3 EtOAc → purple aqueous

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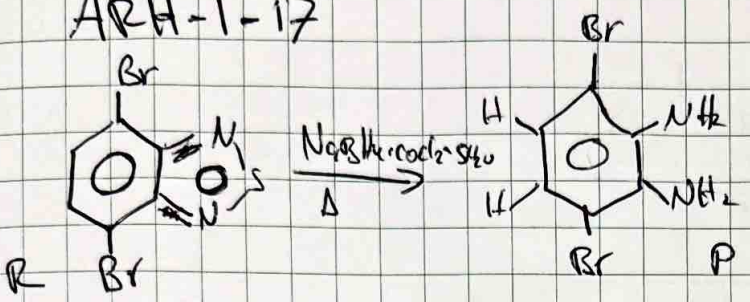
ARH-1-17



Chemical	Fw	Mass mg	mmol	mM	eq	Vol (mL)
R	293.17	100	0.34	340	1	–
$NaBH_4$	37.83	32.2	0.85	850	2.5	–
$CoCl_2 \cdot 6H_2O$	237.83	3.2	0.013	13	0.04	–
EtOH	–	–	–	–	–	1
P						

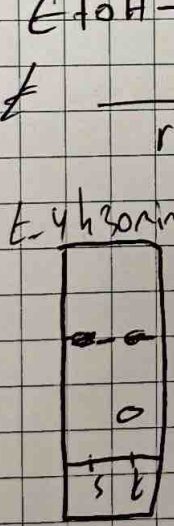
① Add R ② Add $NaBH_4$ ⇒ SM1 RBF

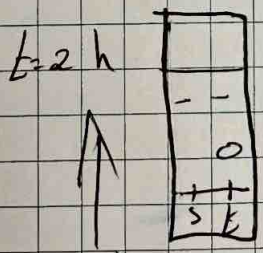
③ Add $CoCl_2 \cdot 6H_2O$

④ Add EtOH ⟶ solution turns black

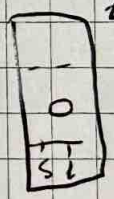
⑤ heat ⟶ turns pink
rxn starts @ 10:09

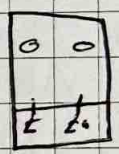
⑥





10% EtOAc/hexane





⑦ rxn after 4h 30min heated @ 60°C ⇒ thermostat

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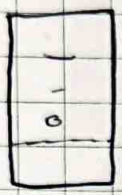
Continued from Page

tot. = 6h 30 min



rxn had no solvent

Extract with EtOAc (x4) ⇒ purple organic brom eq

- brine ⇒ clear brine / purple org.

- Na_2SO_4

- rotovap.

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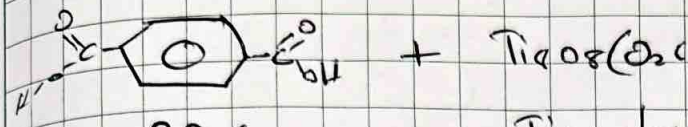
ARH-1-19

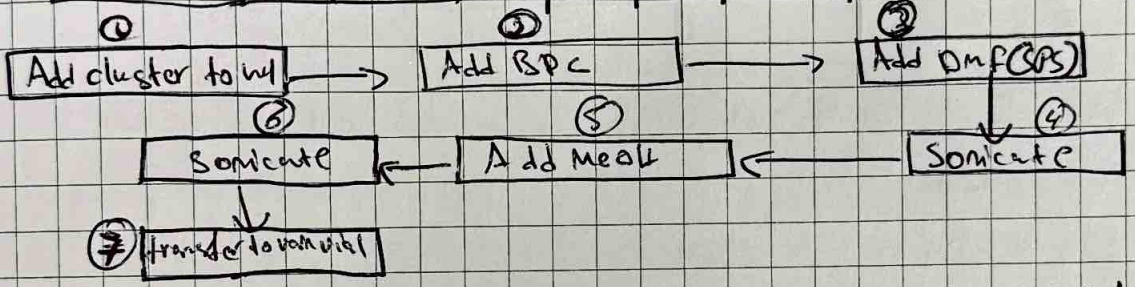


BDC

Ti-cluster

MIL-125

Chemical	Fw	Mass (mg)	mMol	mM	eq	vol
BDC	166.13	51.5	0.31	0.1375	14.75	
Ti-cluster	2128.15	44.7	0.021		1	
DMF	–	–	–	–	–	1.75
MeOH	–	–	–	–	–	0.135
MIL-125						



① Add cluster (4ml vial) ② Add linker ③ Add DMF (SOPS)

④ Add MeOH ⑤ Sonicate => turbid

⑥ heat for 3 days @ 120°C (remove @ 09/25/2025 2pm)

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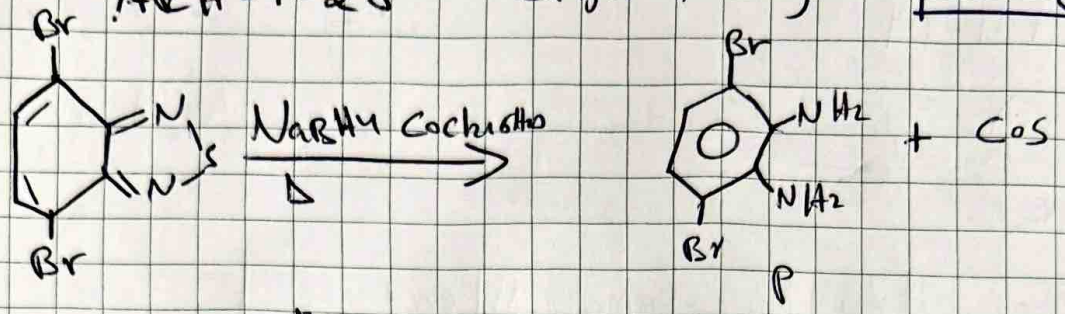
AEH-1-20 (Right of hood)

35% yield



R P

chemical	fw	mass	mmol	mM	vol	eq
R	293.7	100	0.34	226.67	–	1
$NaBH_4$	37.83	32.2	0.85	86.7	–	2.5
$CoCl_2 \cdot 6H_2O$	237.93	[redacted]	0.034	22.667	–	0.1
EtOH	–	–	–	–	1.5	–
P						

① Add R in 5ml RBF

② Add $CoCl_2 \cdot 6H_2O$

③ Add EtOH

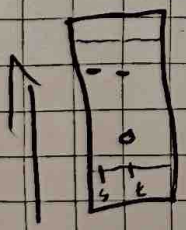
④ heat + stir until dissolving
⇓ blue suspension

⑤ Add $NaBH_4$
⇓
blue suspension turns red + bubbling

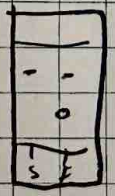
rxn starts @ 8:55 AM

TLC (Allin 6% EtOAc in hexanes)

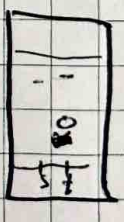
t=0



t=1h



t=2h 30 min



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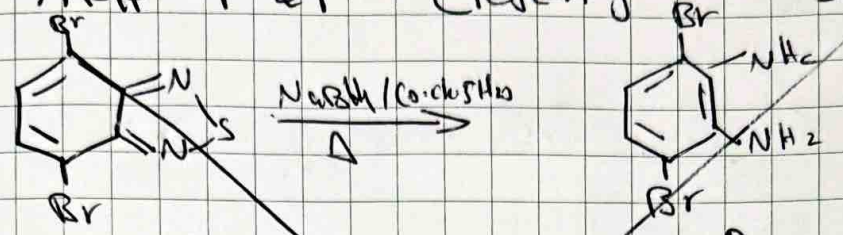
ARH-1-21 (repeating ARH-1-05)



R P

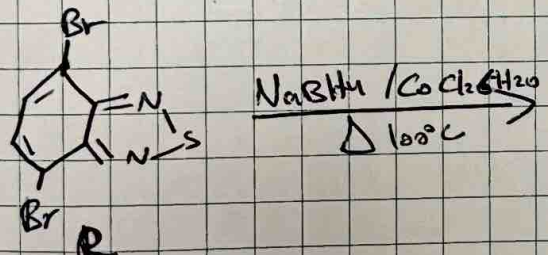
chemical	fw	Mass	mmol	mM	vol	eq
R	293.7	557	1.89	82.138	–	1
NaBH4	~~37.83~~ 37.83	143.4	3.78	164.135	–	2
CoCl2·5H2O	237.83	45.1	0.189	8.21	–	0.1
EtOH	–	–	–	–	23	–

rxn NOT DONE

① Add R ② Add $CoCl_2 \cdot 5H_2O$ ③ Add ~~EtOH~~

④ Add $NaBH_4$

ARH-1-21



R P

50% yield

chemical	fw	(avg) Mass	mmol	mM	vol	eq
R	293.7	100	0.34	22.667	–	1
NaBH4	37.8	32.2	0.85	55.67	–	2.5
CoCl2·5H2O	237.83	8	0.034	2.267	–	0.1
EtOH	–	–	–	–	15ml	–
P						

① Add R in sml RBF
② Add $CoCl_2 \cdot 5H_2O$
③ + EtOH
④ heat + stir (blue suspension)
⑤ Add $NaBH_4$ (rxn turns red to black)

Rxn Starts 11:40 AM

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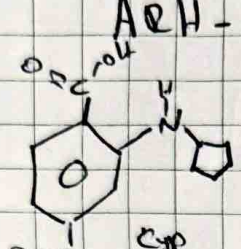
ARH-1-22



+ $Ti_8O_8(OOCC(CH_3)_3)_8$ → MIL-125-NH$_2$ cyp

Cyp Ti-C

Chemical	MW	Mass	mmol	mM	eq	Vol
Cyp	249.27	7.73	0.031	151	4.26	
Ti-C	2128.5	44.17	0.0021	10.124	1	
DMF	—	—	—	—	—	1.75
MeOH	—	—	—	—	—	0.35
MIL-125-NH$_2$ cyp						

4 vials

① Add Cyp in 2 mL vial ② Add Ti-C ③ Add 5 PS DMF

④ Add MeOH ⑤ sonicate ⑥ transfer to vials

⑦ pipette 1.75 mL to cyp sol into Ti-C to vials ⑧ add methanol ⑨ transfer to vials.

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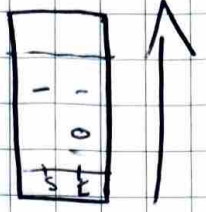
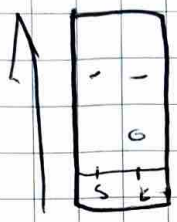
continue (ARH-1-20)

* t= 3h 30 min



Note! increasing Co(h.s·H2O is bringing rxn closer to completion.

t= 4h 30 min



t= 5h 30 min

⑥ after 5h 30 min rxn stopped

⑦ filter through celite ⇒ black residue + pink filterate

⑧ rotavap ⇒ orange ppt ⑨ Add DI yellow suspension

⑩ extract 3x EtoAc ⇒ EtoAc pale yellow / aq layer

⑪ Brine x3 for organic layer ⇒ clear aq / pale yellow EtoAc (orange)

⑫ ~~MgSO4~~ Add Na_2SO_4.

⑬ rotavap ⇒ olive green ppt ⟶ next day it was gray

NMR confirms product without Starting Material

35% yield

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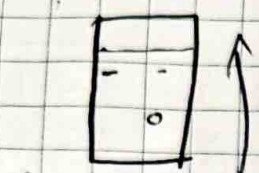
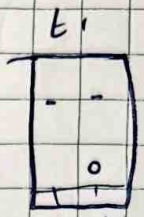


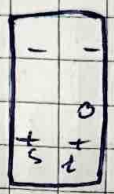
t=0

10% EtOAc/hexane

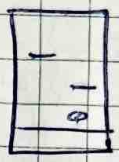


t=1h

t=2h



t=3h



t=6h

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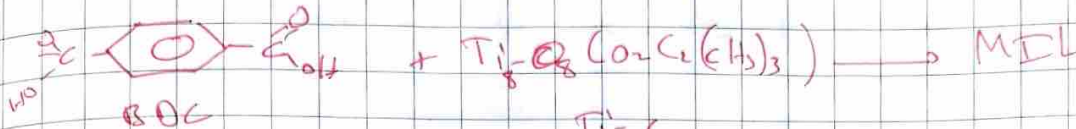
ARH-1-25



chemical	fw	Mass (g)	mmol	mM	eq	vol
BDC	166.13	51.5	0.31	0.372	1.496	—
Ti-C	2125.5	44.2	0.021		1	—
Etoh	—	—	—	—	—	1.25
MeOH	—	—	—	—	—	0.35

① Add BDC in 4ml vial

② Add Ti-C

③ Add DMf

④ Add MeOH

⑤ transfer to vain vials

⑥ heat @ 120°C 3days ~~A~~

(to be removed ~~Tuesday~~ Monday ~~A~~)

⑦ ~~sample crystallinit~~ solve exchoged x3 DMF x3 Etoh

⑧ ~~Heat~~ filter @ crystallyy confmd by PXRD

⑩ digest with 50ML DCL, 200ML DMSO

NMR gives ~~CF3~~ ~~MEF~~ 0.01% MIL-125-NHCyp

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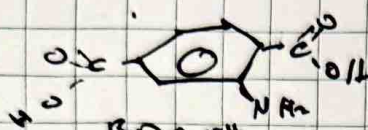
ARH-1-26



+ ZrCl4 ⟶ UiO-66-NH2

chemical	fw	(mg) Mass	mmol	M/M	eq	vol
BDC-NH2	180	266	1.48	37	1	-
ZrCl4	344	344	1.48	37	1	-
DMF	-	-	-	-	-	40
~~DEF~~	~~~~	~~~~	~~~~	~~~~	~~~~	~~~~
UiO-66-NH2						
FA	46	13.6 g	296	7400	200	11.2

① Add BDC-NH2
② Add ZrCl4
③ Add DMF (turbid yellow)
④ ~~Add DEF~~ ✱
⑤ Add F.A (solution becomes clear).
⑥ 48h oven 120°C
⑦ Pale yellow PPT ⇒ filtered
⑧ Solvent exchge DMF/✱EtOH.

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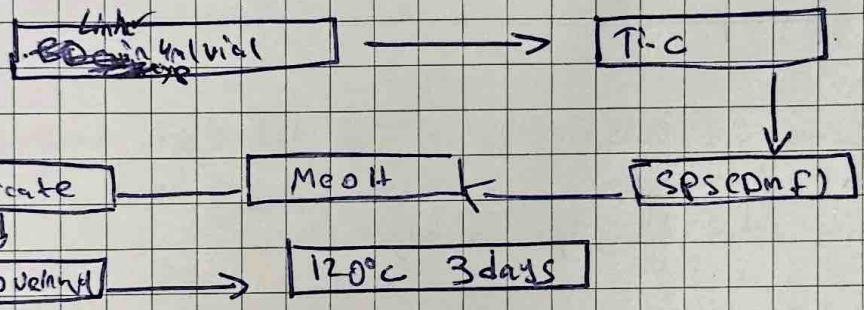
ARH-1-27

BDC NH-cyp + $Ti_8O_8(C_6H_5CO_2)_{32}$ ⟶ MIL-125 NH cyp

chemical	fw	Mass (mg)	mmol	mM	eq	Vol
BDC-NH cyp	223	223	0.131	147.1	14.7	–
$Ti_8O_8(CO_2Ph)_{32}$	2128.15	~~4417~~ 48	~~0.021~~ 0.021	10	1	–
DMF	–	–	–		–	1.75
MeOH	–	–	–	–	–	0.35
MIL-125 cyp						



① BDC NH-cyp stock ⟹ clear yellow solution (DMF) in a 20 ml vial

② Add cluster to vial in vial

③ Pipette 1.75 ml stock ④ Add MeOH

oven 3 days ⟹ some had dark brown ppt.

Note: DMF used from bottle not SPS.

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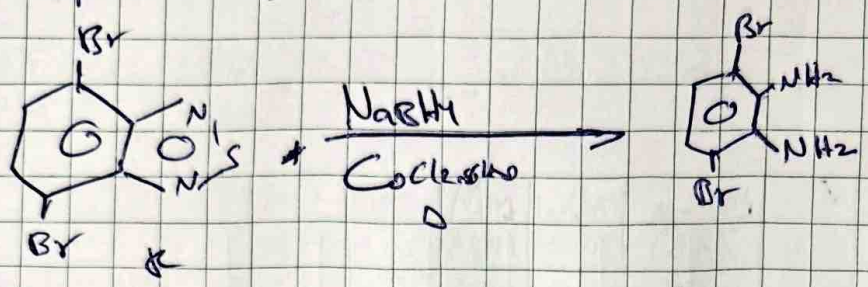
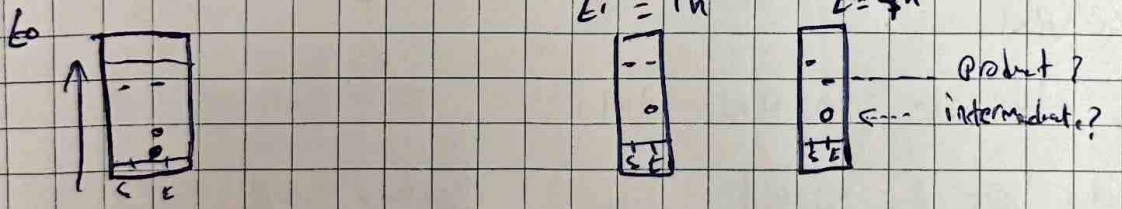
ARH-1-28



Chemical	Fw	Mass (mg)	mmol	mM	eq	Vol
R	289.7	100	0.34	226.67	1	-
$NaBH_4$	37.83	32.2	0.85	566.7	2.5	-
$CoCl_2.6H_2O$	237.93	16	0.068	45.2	0.2	-
EtOH	-	-	-	-		1.5
P						

① Add R in SM RBF
② Add $CoCl_2.6H_2O$
③ Add EtOH (blue suspension)
④ heat @ 100°C + stir
⑤ After 10 min Add $NaBH_4$ → brown color

rxn starts @ 10 AM



10% EtOAc/Hexane

Speculation circle on TLC is an intermediate that goes to product ?

⟹ rxn will continue to see if o gets paler and or base gets darker

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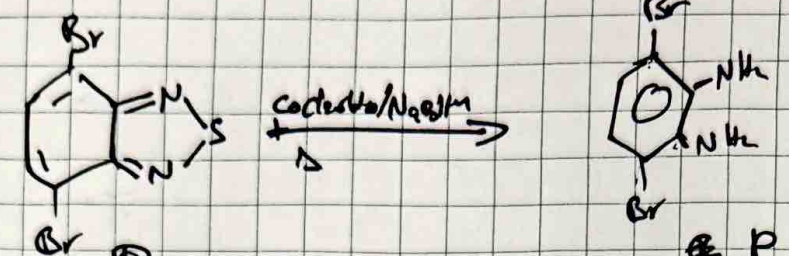
ARH-1-30



R P

6.4 g NaBH4

Chemical	fw	Mass(mg)	mmol	mM	eq	vol (ml)
R	298.2	200	2.68	226.67	1	—
$NaBH_4$	37.83	6.4g	169.17	556.67	2.5	—
$CoCl_2.6H_2O$	237.83	16	0.067	246	0.1	—
EtOH	—	—	—	—	—	3

① Add R in 5ml RBF

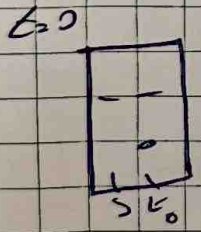
② Add $CoCl_2.6H_2O$

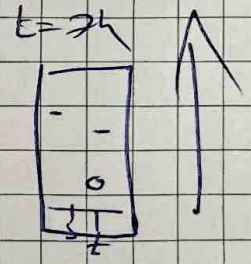
③ Add EtOH ⇒ blue suspension

④ heat @ 100°C

⑤ after 15 m.s add $NaBH_4$ ⇒ bubbly → turns red

⑥ leave for 7h





for EtOAc/hexane

⑤ filter over celite ⇒ red filtrate

⑥ rotavap ⟶ orange solid

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(7) Quench with water ⇒ yellow ppt + orange soln.

(8) extract 3 time EtOAc → 1st time yellow orange color of EtOAc
→ 2nd time pale purple - - - -
→ 3rd time - - - - - -

(9) dry with Na_2SO_4 (10) rotavap ⇒ yellow solid

80% yield

NMR pure but greasy

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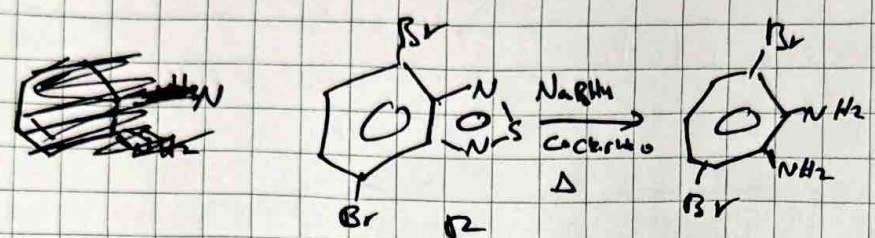
ARH-1-32



chemical	fw	Mass(mg)	mmol	mM	eq	vol
R	295.7	400	[illegible]	[illegible]	1	–
NaBH4	37.83	128.8	3.4	850	2.5	–
CoCl2.6H2O	237.93	32	0.134	34	0.1	–
EtOH	–	–	–	–	–	4 ml

① Add R in a 10 ml RBF
② Add $CoCl_2 \cdot 6H_2O$
③ Add EtOH ⇒ blue suspension
④ heat @ 100°C + stir
⑤ after 10 mins add $NaBH_4$ ⇒ bubbling + dark red color solution
⑥

rxn starts @ 5:45

rxn failed ~~Sattle~~

leak in condenser joint.

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ARH-1-33 (Actinometry Prep)

$$2\,KOH + H_2C_2O_4 \xrightarrow[80^\circ C]{H_2O} 2\,H_2O + K_2C_2O_4$$

Chemical	Fw	Mass	mmol	eq	Vol	mM
H_2O	—	—	—	—	10	3774
KOH	56.1	2.12	0.037	7.4		
$H_2C_2O_4$	126	2.27	0.018	1	—	1764
$K_2C_2O_4$	166	3	1	1	—	

1) In a 100 ml RBF add KOH
2) Add H_2O ⇒ heat is given out
3) Heat @ 80°C + stir for 30 min

$$3\,K_2C_2O_4 + FeCl_3 \xrightarrow[\Delta 88C]{} K_3Fe(C_2O_4)_3$$

Chemical	Fw	Mass	mmol	M/M	eq	Vol
$K_2C_2O_4 \cdot H_2O$	184	3.32	18.04	1804	3.6	—
$FeCl_3 \cdot 6H_2O$	270	1.62	6	600	1	—
H_2O	18	—	—	—	—	10

1) Add $FeCl_3 \cdot 6H_2O$ in a 50ml beaker
2) Add H_2O ⇒
3) sonicate
4) Add dropwise to solution of $K_2C_2O_4$
5) cool ⟶ filter ⟶ wash with cold Acetone

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0.006 M Actinometer solution in 250 ml RBF

① 0.73675 g of $K_3Fe(C_2O_4)_3$

② Add H_2O ③ Add H_2SO_4 (25ml) (0.05M) = 0.1 N

④ dilute to mark

Cuvette prep:-

① 2ml actinometer solution

② 0.67ml of phenanthroline

③ 0.33 ml buffer

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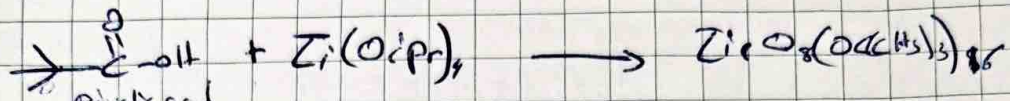
ARH-1-35



Chemical	FW	(mg) Mass	mMol	mM	Vol	eq
Ti(OiPr)4	284.16	1200	4.22	281	1.75	1
Pivalic acid	102	~~4.375g~~	42.89	2859	~~Full~~	10.2 → 4.84
ACN	-	-	-	-	15	-

4.375g

① Add ACN (SPS) 40ml Bomb ② Add pivalic acid ③ Add Ti(OiPr)4

④ heat @ 120°C 6 days by teflon bomb.

To be removed on Monday 21st october.

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Sealed tubes (Peer sps).

① use reference tube for cutting.
② mark @ middle + end.
③ keep doing it by sliding
④ cut the ends (Not the middle).
⑤ ~~Clear~~ rotate till sound ⇒ then crack by hand (like PLC).

Flame:-
① burn @ the middle ⇒ rotation.
② after split round the tips with flame.

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ARH-1-37

TAM + TPA

3MAA, Dioxane, 120°C, 3 days → COF-300

Chemical	fw	Mass (mg)	mmol	mM	eq	Vol
TAM	380.8	20	0.052	51	1	–
TPA	134	12	0.089	87.2	1.7	–
Dioxane	–	–	–	–	–	1
3MAA	–	–	–	–	–	0.12

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ARH-1-38 (CoF-300 co-pilot)

TAM + TPA ⟶ CoF-300

0.2 g

Chemical	fw	Mass (mg)	mmol	mM	eq	vol
TAM	380.8	76	0.2	47.6	1	=
TPA	134	53.7	0.4	95.4	2	-
Dioxane	-	-	-	-	-	2
Mesitylene	-	-	-	-	-	2
6M AA	-	-	-	-	-	0.2

① Add TAM in a borosilicate tube ② Add TPA
③ Add dioxane ④ sonicate ⇒ yellow solution
④ Add Mesitylene ⑤ Add 6M AA ⇒ ppt
⑥ freeze ppt ⑦ vac
⑧ put in 120°C oven after defrosting
⑨ filter

Note: fernando says attach vac ⟶ freeze ⟶ open vac (to prevent water condensing into tube).

- 3 replicates were made 1 @ same scale as table other 2 half scale (1/2 ARH-1-38).

0.2 g ARH-1-38 ⟶ collapsed xrd (peak @ 9)

0.1 g ARH-1-38(2) ⟶ collapsed but has peak @ 6.9)

ARH-1-38 ⟶ mixture

Note: ARH-1-38 was sealed by fernando (small head space).

Continued on Page

ARH-1-39

Prepare std solution of Fe^{2+} (0.1) M

$Fe(NH_4)_2(SO_4)_2$

0.1 × 0.025 = 0.00025 moles ⟹ 0.09525 g

(POINT DISSOLVE)

* to prepare 0.001 M

$V_1 \times 0.1 = 0.001 \times V_2$ | V_2 = 1 ml

0.09525 g $Fe(SO_4)_2(NH_4)_2 \cdot 6H_2O$ ⟹ 250 ml

* to prepare 0.0001 M from 0.001 M

V_1 = 10 ml | V_2 = 100 ml

* to prepare 0.00002 M from 0.001 M

V_1 20 ml | V_2 100 ml

* to prepare 0.00003 M from 0.001 M

V_1 = 30 ml | V_2 = 100 ml

* " " 0.00004 M from 0.001 M

V_1 = 40 ml | V_2 = 100 ml

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* to prepare 0.0005M from 0.001M

$V_1 = 50\ ml$ $V_2 = 100\ ml$

to prepare 0.0005M from 0.001M

$V_1 = 50\ ml$ $V_2 = 100\ ml$

WAS NOT DONE => STOCK INSOLUBLE

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② ~~Calib curve~~ 0.0001 M ⟶ 0.001 M

~~Stock 0.1 M~~ Fe^{2+} 1) Prepare 0.1 M Fe^{2+} in 0.1 N H_2SO_4

~~1 L~~ ~~250 ml~~ 50 ml flask ⇒ 0.1 M × 0.05 L = 0.005

0.005 moles Fe^{2+} × 392.1 ~~FeSO4.7H2O~~ in 50 ml 0.1 M H_2SO_4

~~0.98 g~~ 0.196 g

2) 0.105 M H_2SO_4 in 100 ml

0.105 × 0.1 = 0.005 moles

0.005 × 98 = 0.49 g

V = 0.49 / 1.83 = 0.268 ml = 268 µL in 100 ml H_2O

3) phenanthroline buffer

1) ~~1.42 g~~ NaOAc 2) ~~mg~~ 27.48 mg phenanthroline

3) ~~15 ml~~ 10 ml H_2O 4) ~~1~~ 0.05 M H_2SO_4

④ Prepare 0.0004 M Fe^{2+}

~~$C_1V_1 = C_2V_2$~~

$0.1 \times V_1 = 0.0004 \times 0.05$

$\Rightarrow V_1 = 2$ ml of 0.1 M Fe^{2+} in 50 ml 0.1 N H_2SO_4

Prepare solutions in cuvettes-

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Calib conc (mM)	Buffer Vol (ml)	Fe^{2+} 0.4mM Vol (ml)	H_2O (ml)
0	0.875	0	1.625
0.02	0.875	0.125	1.5
0.04	0.875	0.25	1.375
0.06	0.875	0.375	1.25
0.08	0.875	0.5	1.125
0.1	0.875	0.625	1
0.12	0.875	0.75	0.875
0.14	0.875	0.875	0.75
0.16	0.875	1	0.625

① Add buffer

② Add Fe^{2+} (0.4mM)

③ Add H_2O

Conc Fe^{2+} (mM)	Abs (nm)
0	~~0.001~~ 0.009
0.02	0.323
0.04	0.475
0.06	0.733
0.08	0.975
0.1	1.216
0.12	1.47
0.14	1.705
0.16	1.947

0.5 slit

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ARH-1-43 (5% MIL-125)

BDC + NH$_2$CyP + $Ti_8O_8(OC(CH_3)_3)_{16}$ ⟶ MIL125NH$_2$ 5%

Ti-C

Chemical	fw	Mass (mg)	mmol	mM	eq	vol
BDC	166.13	49	0.295		3.37	–
NH$_2$CyP	249.27	3.9	0.0156		0.054	–
Ti-C	2128.5	44.7	0.021		1	–
DMF	–	–	–	–	–	1.75
MeOH	–	–	–	–	–	0.35

① Add Ti-C

② Add BDC.

③ Add NH$_2$CyP

④ Add DMF (SPS)

⑤ Add anhydrous MeOH

⑥ heat for 4 days

⑦ Purge with N_2.

⑧

to be removed Wednesday morning

Low Crystallinity

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1984 double speak.

ARH-1-44

~~① 19 mg TAM (0.05 mmol) ② 6.7 TPA (0.05 mmol) ③ 1.5 dioxane~~

① 19 mg TAM (0.05 mmol) + 6.7 mg TPA (0.05 mmol) 0.5ml dioxane/0.5ml mesitylene 100 μM AA

② ~~9.5~~ 9.5 mg TAM (0.025 mmol) + 6.7 mg TPA (0.05 mmol) 0.5ml dioxane/Mesitylene 100 μM AA

③ 9.5 mg TAM (0.025 mmol) + 13.4 mg TPA (0.1 mmol) 0.5ml dioxane/Mesitylene 100 μM AA

Note: upon sealing ~~tubes~~ tubes under vac, very low pop sound was heard.

to be removed => ~~Wednesday~~ ~~Tuesday~~ wednesday by noon every

record on excel

condition no.2 gave peak @ 7 (others were amorphous or collapsed).

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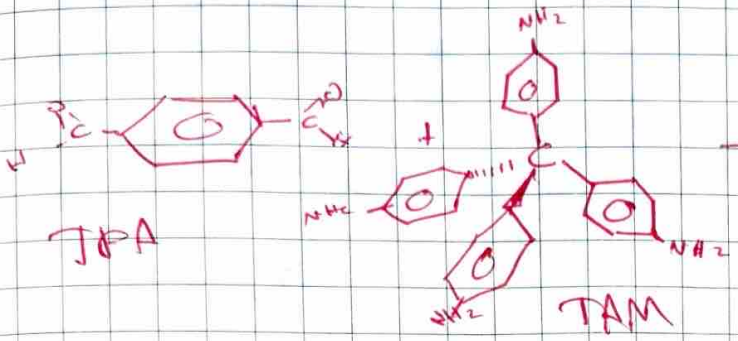
ARH-1-45-1 COF-300 ~~Vial Vial~~ ~~Vial vial~~ vial's DOI: doi.org/10.1021/jacs.8b13091

Yue Bin Zhang



TPA + TAM ⟶ COF-300

TPA

TAM

Chemical	Fw(g)	mass(mg)	mmol	mM	(ml) Vol	eq
TPA	134	12	0.0894	37.08	—	1.7
TAM	348.18	20	0.052	21.67	—	1
Dioxane	—	—	—	—	1.6	—
Cyclohexane	—	—	—	—	0.4	—
3M AA	—	—	—	—	0.4	—

~~① in a 4ml vial vial~~

~~add T~~

① in a 4ml vial Add ~~TPA~~ TAM

② Add TPA

③ Add Dioxane

④ sonicate

⑤ Add Cyclohexane

⑥ sonicate

⑦ Add 3M AA sonicate

⑧ 65°C 3d (Monday)

ARH-1-45-2

TPA + TAM ⟶ COF-300

Chemical	Fw(g)	mass(mg)	mmol	mM	vol (ml)	eq
TPA	134	14	0.104	43.34	—	2
TAM	348.18	20	0.052	21.67	—	1
Dioxane	—	—	—	—	~~1.6~~ 1.6	—
Cyclohexane	—	—	—	—	0.4	—
3M AA	—	—	—	—	0.4	—

① 4ml vial Add TAM

② Add TPA

③ Add Dioxane

④ Sonicate

⑤ Add cyclohexane

⑥ Sonicate

⑦ Add 3M AA → Sonicate

⑧ 65°C. 3d. (Monday)

(both have crystallization)

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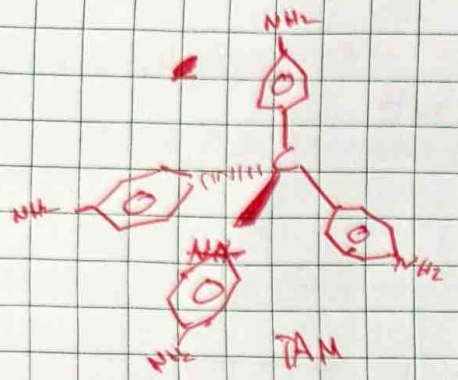
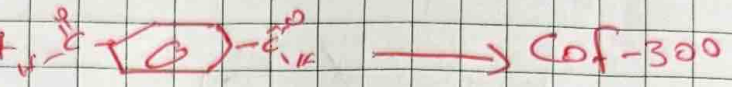
ARH-1-46-1 COF-300 copilot



TAM



Chemical	fw (g)	mass (g)	mmol	mM	vol ml	eq.
TPA	134	~~1g~~ 13.4	0.1	90.9		
TAM	380.8	19	0.05	45.45		
Dioxane	–	–	–	–	0.5	–
Mesitylene	–	–	–	–	0.5	–
6M AA	–	–	–	–	100	–

ARH-1-46-2

Chem	fw	mass (g)	mmol	mM	vol ml	eq
TPA	134	13.4	0.1	90.9	–	2
TAM	380.8	19	0.05	45.45	–	1
Dioxane	–	–	–	–	0.5	–
3M AA	–	–	–	–	0.1	–
Mesitylene	–	–	–	–	0.5	–

① in a sealed borosilicate tube add TAM

② add TPA

③ Add Dioxane → yellow

④ Add Mesitilene

⑤ Add ~~AA~~ AA

⑥ Freeze

⑦ Vac

⑧ Seal (remove on Friday)

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1.09 g/cm³

ARH-1-47-1 Cof-300 copilot

TPA + TAM ⟶ Cof-300

#	Chemical	FW (g)	mass (mg)	mmol	mM	vol	eq
	TPA	134	13.4	0.1	909	-	2
	TAM	38.08	19	0.05	45.45	-	1
	Dioxane	-	-	-	-	0.5	-
	Mesitylene	-	-	-	-	0.5	-
	6MAA	-	-	-	-	0.1	-
	Aniline	93.13	46.6	-	454.5	0.046	10

ARH-1-47-2 Cof-300 copilot

Chemical	FW (g)	mass (mg)	mmol	mM	vol	eq
TPA	134	13.4	0.1	909	-	2
TAM	38.08	19	0.05	45.45	-	1
Dioxane	-	-	-	-	0.5	-
Mesitylene	-	-	-	-	0.5	-
6MAA	-	-	-	-	0.1	-
Aniline	93.13	46.6	1	454.5	0.046	10

① in a borosilicate tube add TAM

② Add TPA

③ Add Dioxane ④ Add Mesitylene

⑤ Add Aniline

⑥ Add AA

⑦ Freeze

⑧ Vac

⑨ seal

(removed on Fridy).

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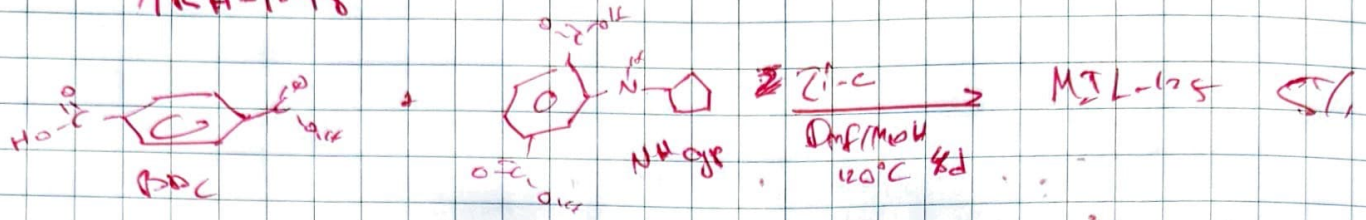
ARH-1-48



chemical	Fw (g)	mass (mg)	mmol	mM	vol	eq
BDC	166.13	490	2.95		-	1.33
NH2gp	249.27	390	1.56		-	0.74
Zr-C	2128.5	447	0.21		-	1
DMF	-	-	-	-	9.5	-
MeOH	-	-	-	-	38	-

① in a 50 ml jar add Zr-c
② Add BDC
③ Add NH2gp
④ Add DMF → SPS → yellow color
⑤ Add MeOH
⑥ Purge with N2
⑦ 120°C (reaction on stirring).

↓ big bubbles form!

Sodium sulfate 50ML DCL
200 ML DMSO.

works!

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AQH-1-49 MIL-125 10%

BDC + NH-cyp $\xrightarrow{\text{Ti-C}}$ MIL-125

Chemical	fw g	mass mg	mmol	mM	vol	eq
Ti-C	217.85	447	0.21	10	—	1
BDC	166.13	~~460~~ 463	2.79	13.19	—	1.3.19
NH-cyp	249.22	~~773~~ 775	~~3.1~~ 3.11	15.76		~~1.55~~ 1.559
DMF	—	—	—	—	17.5	—
MeOH	—	—	—	—	3.5	—

773 mg

460 / 166.13 = 2.7689 mmol BDC

3.1 - 2.7689 = 0.331

0.331 × 249.22

① Add Ti-C in a 50ml jar

② Add BDC

③ Add NH-cyp

④ Add DMF → sonicate → yellow solution (turbid)

⑤ ~~Purge~~ Add MeOH ~~→ 50 ml~~

⑥ Purge N_2 → big bubbles from

works!

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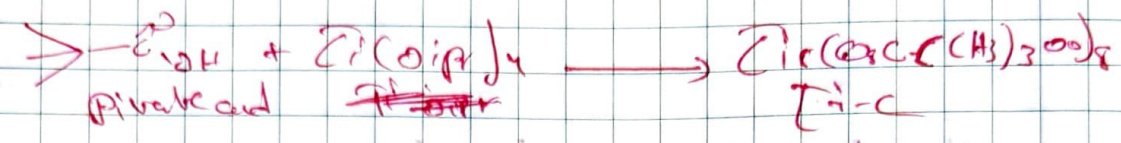
ARH-L-50 (Ti-cluster)



chemical	Fw(g)	mass(mg)	mmol	mM	vol(ml)	eq
pivalic acid	102	~~4325~~ 4325	42.89	2859	4.84	10:2
Ti(oipr)4	284.26	1200	4.22	284	1.25	
Acetonitrile	-	-	-	-	15	-

① in a 45 ml teflon bomb add ACN
② Add pivalic acid (4.84 ml)
③ Add ~~ACN~~ Ti(oipr)4
④ heat @ 120°C for 6 days
(removed on Monday)

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ARH-1-51 (reproducing ARH-1-45.1 Yue Bing)

TPA + TAM ⟶ COF-300

Chemical	fw(g)	mass(mg)	mmol	mM	vol (ml)	eq
TPA	134	12	0.089	37.08	-	1.7
TAM	380.8	20	0.052	21.67	-	1
Dioxane	-	-	-	-	1.6	-
Cyclohexane	-	-	-	-	0.4	-
3M AA	-	-	-	-	0.4	-

① in a 4ml vial add TAM
② Add TPA
③ Add Dioxane
④ Sonicate
⑤ Add cyclohexane
⑥ Sonicate
⑦ Add 3M AA
⑧ Sonicate
⑨ 65°C 3 days

~~(to be removed Tuesday 4/11)~~ A

ARH-1-51-1 dark yellow => may added x2 TPA
ARH-1-51-2 Pale yellow upon adding dioxane
ARH-1-51-3 Pale yellow " " "

ARH-1-51-1 ⟶ filter → PXRD ✓ ⟶ sol exch by MeOH anhydrous

ARH-1-51-2 → filter → PXRD ✓ → sol exch cyclohexane

ARH-1-51-3 → filter → PXRD ✓ → sol exch Acetic anhydride + Dioxane

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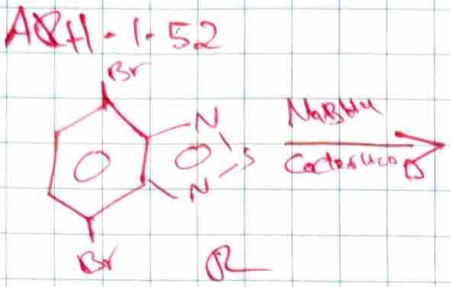
ARH-1-52



R

Chemical	Mass (mg)	Mass (g)	mmol	mM	Vol	eq
R	~~800~~ 295.7	800	2.72	340	-	1
NaBH4	~~37.96~~ 25.76	~~1.485~~	6.6	850	-	2.5
CoCl2·6H2O	232.83	64	2236	34	-	-
EtOH	-	-	-	-	8	

① Add R in 50 ml RBF
② Add CoCl2·6H2O in
④ heat @ 100°C + stir

③ Add EtOH -> blue suspension
⑤ Add NaBH4 after 6 hrs
↓
bubbling + dark red color

79% yield

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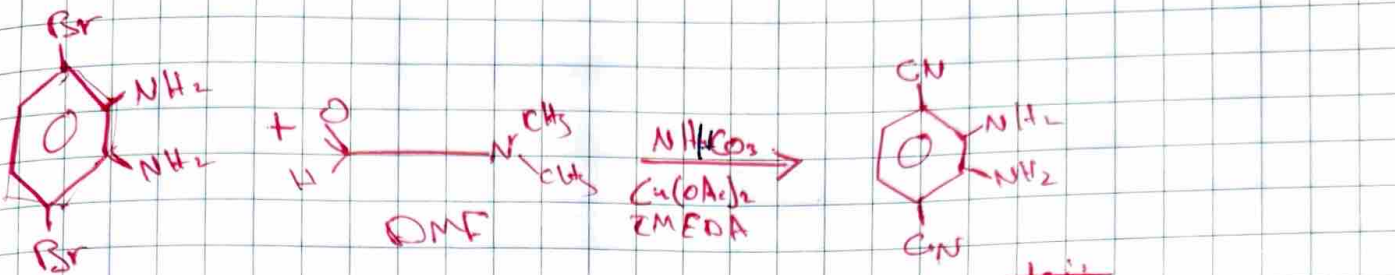

R

doi: https://doi.org/10.1021/ol20173b

chemical	Fw	mass(mg)	mmol	mM	vol	eq
R	266	53.2	0.2	100	—	1
NH_4HCO_3	~~79~~	63.2	0.8	400	—	4
$Cu(OAc)_2$	181.6	~~174~~	0.96	480		4.8
DMF	—	—	—	—	2ml	—
TMEDA	116	~~185.6~~	~~0.16~~	80	0.024	0.8

① in ~~4ml~~ 3ml conical vial Add R

② Add NH_4HCO_3

③ ADD SPS DMF → black

④ ~~TMEDA~~ Add TMEDA

⑤ heat @ 150°C →

⑥ evaporate with ~~DMF~~ EtoAc (2x50 ml)

⑦ ~~stir~~ Add brine → extract with EtoAc

→ after 1gh rxn turns brown.

⑧ NMR shows i) DMF present

ii) Singlet peak of benzene in R shifted to left

⇒ deacylated (cyanide possible).

iii) Amine peak not visible (maybe overlaps DMF).

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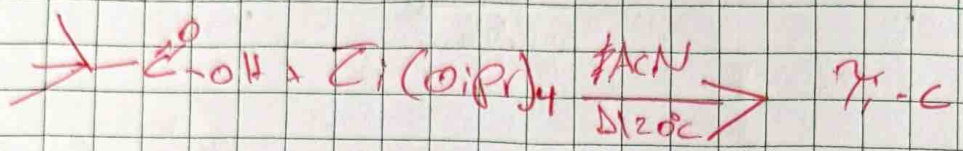
ABH-1-54



chemical	FW (g)	mass (g)	mmol	M	vol (mL)	eq
Ti(OiPr)4	284.26	1200	4.22	281	1.25	1
Picolic acid	102	4.375	42.69	2859	4.184	10.2
ACN	—	—	—	—	15	—

① Add picolic in teflon bag ② Add ACN

③ Add Ti(OiPr)4 ⇒ turn yellow ④ heat @ 120°C 1 dys

remove on 11/18/2025

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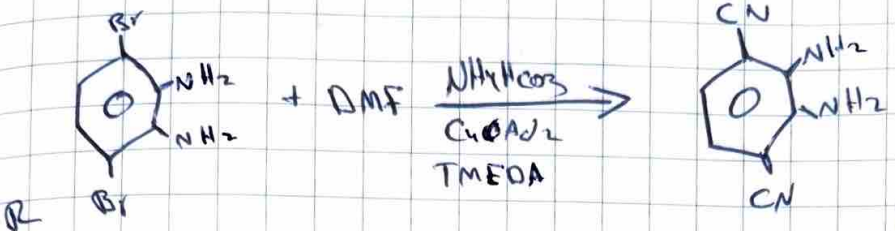
ARH-1-55



doi: https://doi.org/10.1021/o12d73b

Chemical	{w(g)	Mass(mg)	mmol	mM	Vol(ml)	eq
R	266	53.2	0.2	100	-	1
NH4HCO3	79	63.2	0.8	400	-	4
DMF	~~79~~ 8.773	9444	12.9	12900	1	64.5
Cu(OAc)	~~49.7~~	~~872~~ 430	0.24	240	-	1.2
~~TMEDA~~ (8.06)	116	9.28	0.08	80	0.012	0.4

ARH-1-52

① in a 3ml conical vial add Cu(OAc) ② Add NH_4HCO_3

③ Add ~~TMEDA~~ DMF → turquoise green sol ④ ~~Add R~~ Add ~~DMF~~ TMEDA → royal blue solution

⑤ ~~Add R~~ Stir the rxn @ 50°C ⑥ ~~Stir the rxn @ 50°C~~ Add R

↓

solution turns black (combn of blue green).

to be removed tomorrow @ 2:30 pm.

TLC

after 22h

10% EtOAc/hexane

20% EtOAc/hexane

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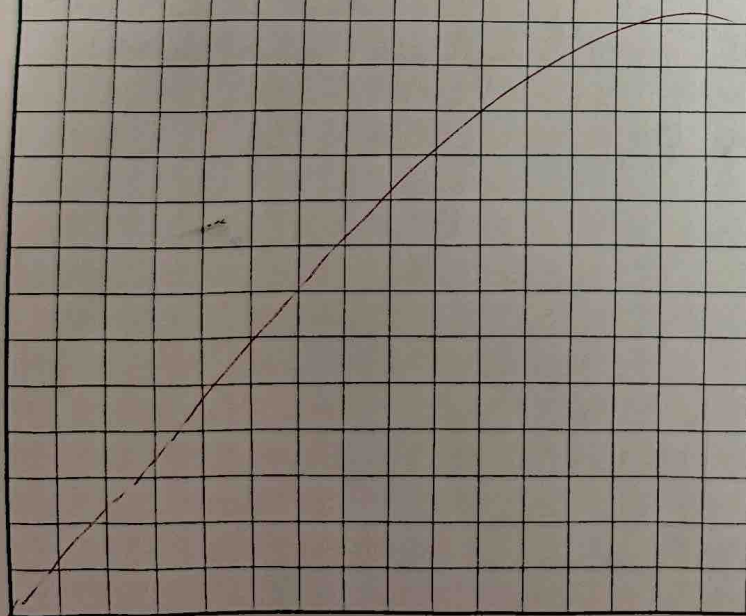
- filter over celite (wet with EtOAc)
- extract filtrate 3x water followed by 3x brine to get rid of DMF
- dry with Na_2SO_4 - rotavap

NMR No ~~product~~ run. ~~product~~ impure



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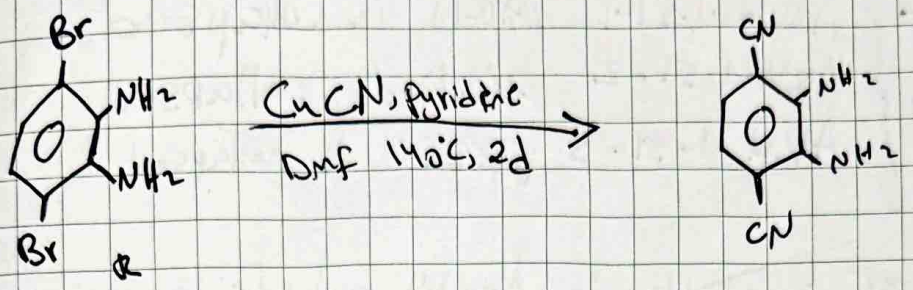
ART-1-57



R

chemical	Fw	Mass(mg)	mmol	mM	eq	mmol
R	266	62.5	0.235	~~470~~ 0.235	1	-
~~CuCN~~ CuCN	89.56	~~2.35~~ 52.9	0.588	1175.2	2.5	-
~~Pyridine~~ Pyridine	79.1				2.5	~~0.047~~ 0.59
DMF	-	-	-	-	-	~~0.5~~ 1 ml DMF

582.6

① Add CuCN in a RBF

② Add DMF

③ Add pyridine

④ Add R

⑤ heat ~~④~~ 140°C 2d

* rxn is air free.

rxn workout

EDTA + Na_2CO_3 → Filter celite → charcoal

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~~ARH-1-51-1~~

PXRD After activation. $\Rightarrow$

- ARH-1-51-1 PXRD is collapsed.
- ARH-1-51-2 PXRD is collapsed
- ARH-1-51-3 PXRD is collapsed

conclusion COF is stable in Acetic anhydride

Cavs left in Acetic Anhydride for (weak) $\Rightarrow$ collapsed form

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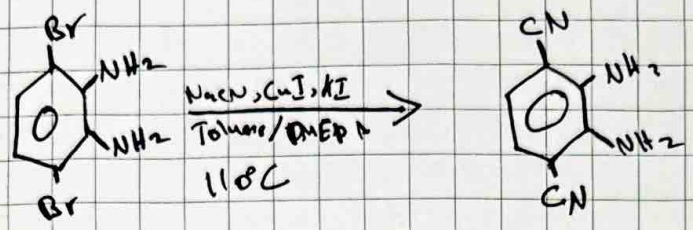
ARH-1-59



R

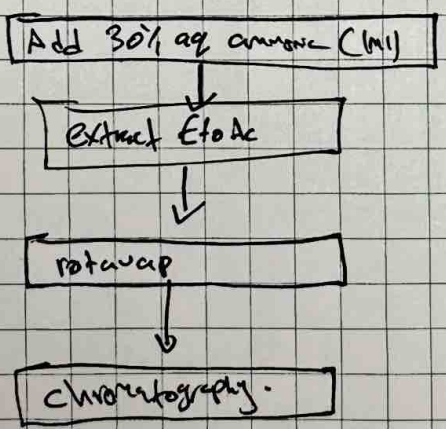
chemical	fw g	mass mg	mmol	mM	L	eq
R	266	57.8	0.218	181		1
NaCN	49	12.8	0.261	217.7	—	1.2
CuI	190.5	4.125	0.0217	18.1	—	0.1
KI	166	7.125	0.0429	35.8	—	0.2
DMEDA		-	0.218	181	0.0234	1

DMEDA:- Dimethyle ethylene diamine.

① Add NaCN in a
② Add CuI
③ Add R
④ Add KI
⑤ evac + purge N_2 x3
⑥ Add toluene + DMEDA
⑦ heat @ 110°C

* rxn is air free.

~~After~~ rxn work up



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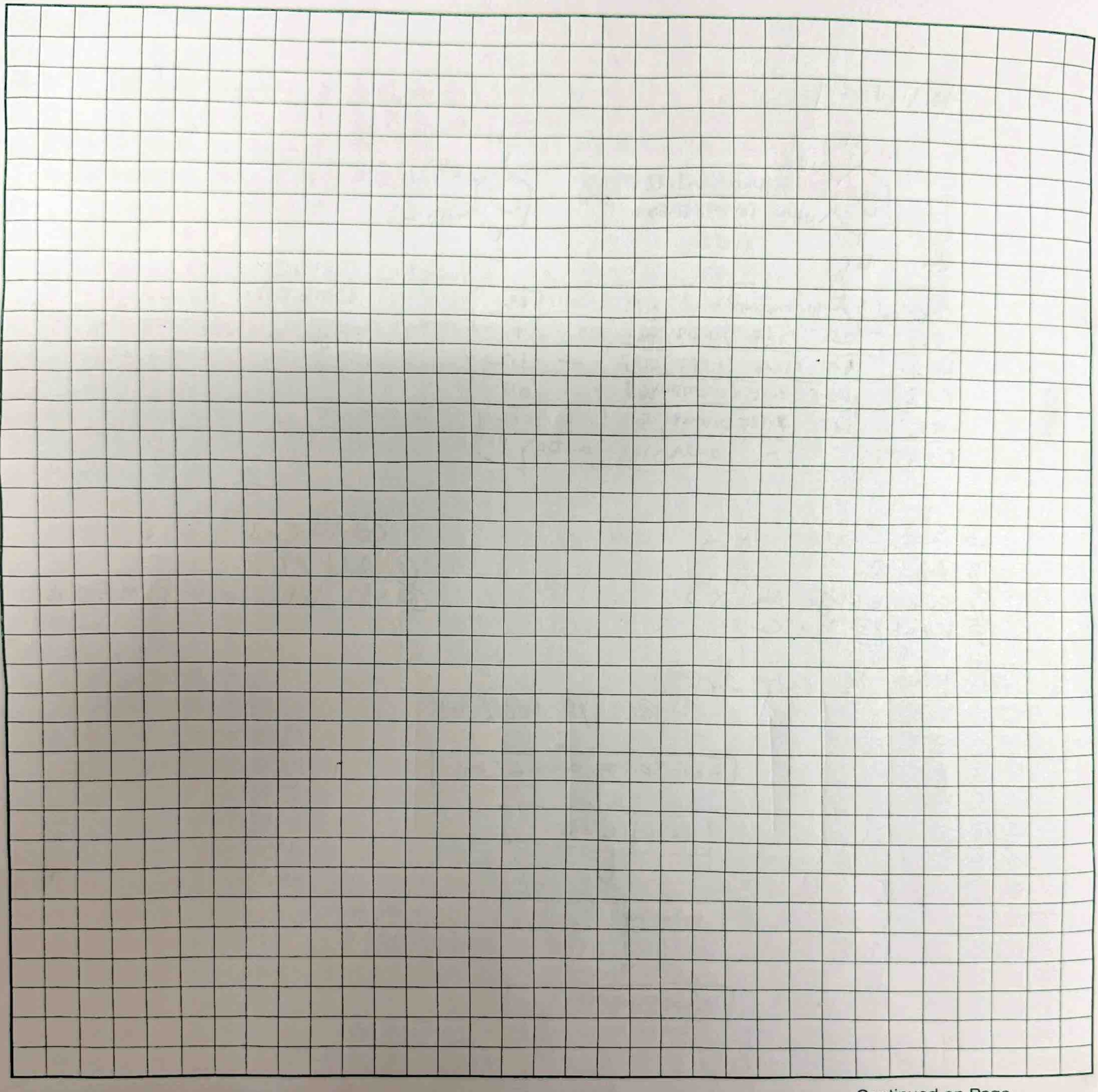

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Project

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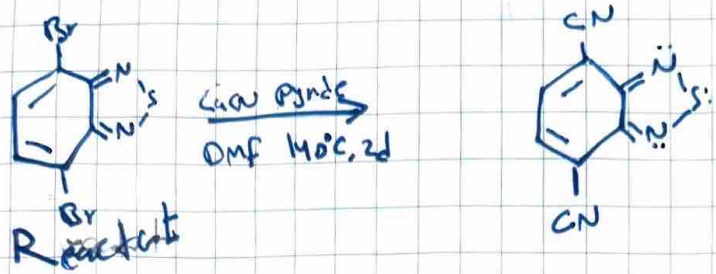
ARH-1-61



Reactant

Chemical	wt(g)	Mass(mg)	mmol	mM	Vol	eq
R	193.7	69	0.235	0.235	–	1
CuCN	89.56	~~52.7~~ 59	0.588	588	–	2.5
Pyridine	79.1	46.5	0.588	588	0.047	2.5
DMF	–	–	–	–	1	–

① Add CuCN in a conical flask

② Add R

③ Add DMF

④ Add Pyridine

⑤ heat @ 140°C 2d

* rxn is air free

MSDS CuCN

- Solid m.p 474°C
- Don't store next to Acids
- reacts with Mg, oxidizers, nitrates, bases
- Explode with Acids → HCN → toxic.

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after 2d rxn turns from yellow ⟶ red brown

~~⑥ Add 1 ml of NH4OH to 9 ml of water~~

~~⑦ Add~~

⑥ Add 3ml of $NH_3 \cdot H_2O$ to 3ml toluene

⑦ ~~Add~~ pour rxn into ~~solution~~ mixture & stir ↑

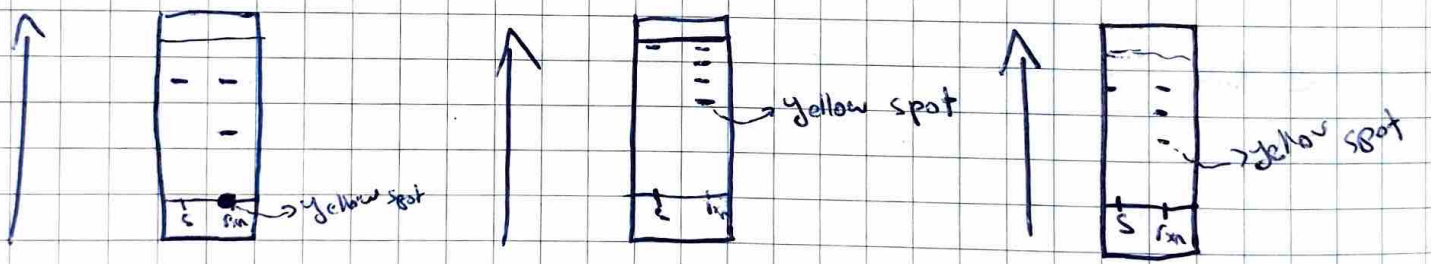
⑧ Stir for 20 min ⑨ filter

⑩ extract with toluene x3

⑪ brine ⟷ ⑫ Na_2SO_4

NMR confirms product, but impure } ⇒ ARH-01 crude
NMR confirms presence of starting material }

TLC:



10% EtOAc/hexane 40% EtOAc in hexane 20% EtOAc/hex

* TLC confirms starting material

- Colum with 10% EtOAc/hexane ⟶ middle fractions ~~Rare~~ (yellow green color) contain product according to NMR ≤ 1%

Continued on Page

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ARH-1-63 MIL-125 (15% NH$_2$-yp)

BDC + BDC-NH$_2$-yp + Ti-C ⟶ MIL-125-NH$_2$-yp 15%

chemical	Fw(g)	mass(mg)	mmol	mM	eq	vol
~~BDC~~ Ti-C	217.85	447	0.21	10	1	–
BDC	166.13	437.8	2.635		12.5	–
NH$_2$-yp	249.22	116.1	0.459		2.18	–
DMF					175	
MeOH					35	

① In a 100 ml Jar add Ti-C

② Add BDC

③ Add DMF(SBS) ~~→~~ → yellow

④ Add NH$_2$-yp

⑤ Add anhydrous MeOH ⟹ purge N_2

⑥ heat 3d 120°C

PXRD confirms crystallinity

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14.175

eq Mw Water

ARH-1-64

BDC + BDCNHoyp + Ti-c ⟶ MIL-125 NHoyp (20%)

chemical	Fw (g)	mass (mg)	mmol	mM	eq	Vol
Ti-C	228.05	44.7	0.21	10	1	–
BDC	166.13	411.4	2.48		11.80 eq	
NHoyp	249.27	154.5	0.62		2.952 eq	–
DMF	–	–	–	–	~~133~~	17.5
MeOH	–	–	–	–	~~85~~	3.5

① Add Ti-C in 2.100 ml flask

② Add BDC

③ Add NHoyp

④ Add DMF ⟹ yellow sol ⟶ sonicate 5 min

⑤ Add MeOH (anhydrous) → ~~Add~~ ~~Add~~ ~~N2 pyr~~

⑥ Heat (120°C 3d.

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PiV >—CO_2– $Ti_8O_8(PiV)_{12-16}$

14.76 total eq of linker

3.0996 total moles of –

ARH-1-65

Ti-C BDC + BDCNHcyp ⟶ MIL-125 (25%)

chemical	FW (g)	mass (mg)	mmol	mM	eq	vol
Ti-C	2129.5	447	0.21	10	1	–
BDC	166.13	348.2	11.07		11.07	
NHcyp	249.27	193.2	3.69		3.69	
DMF	–	–	–	–	–	17.5
MeOH	–	–	–	–	–	3.5

① Add Ti-C

② Add BDC

③ Add NHcyp

④ Add DMF → yellow suspension (SOS) → sonicate

⑤ Add MeOH only dissolves ⇔ sonicate

⑥ ~~purge with N_2 → big bubbles~~

⑦ heat @ 120°C (3d).

crystallinity confirmed by PXRD.

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ARH-1-66

114.76 total linker eq
3.10996 total moles of linker

BDC + BDC-NHcyp + Ti-C ⟶ MIL-125 (50%)

Chemical	Fw(g)	Massing	mmol	mM	eq	Vol
Ti-C	212.85	947	0.24	9.96	1	-
BDC	166.13	257.5	1.55	73.4	7.38	
BDC-NHcyp	249.27	386.4	1.55	73.8	7.38	
DMF	—		—		~~15~~	175
MeOH	—		—			3.5

1. Add Ti-C into 50ml jar
2. Add BDC
3. Add NHcyp
4. Add DMF (sds) → yellow suspension
5. Sonicate
6. Add Anhydrous MeOH
7. Sonicate
8. ~~Purge with N_2 → tiny bubbles~~
9. Heat @ 120°C (3d)

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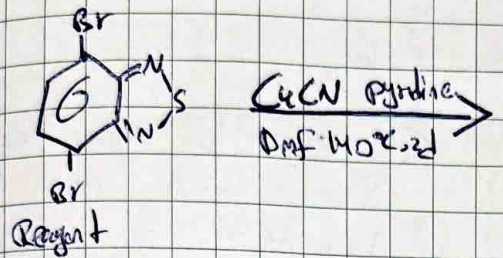
g/mL

APH-1-87



Reagent

Product.

Chemical	Fw g	mg (mg)	mmol	mM	eq	vol (mL)
Reagent	393.7	139	0.47	0.47	1	–
CuCN	89.56	169	1.88	940	4	
Pyridine	79.1	149	1.88	940	4	0.151
DMF	–	–	–	–	–	2

① in a 5ml RBF add R

② Add CuCN

③ ~~purge~~ Vac → purge N_2 (x3)

④ Add purged DMF

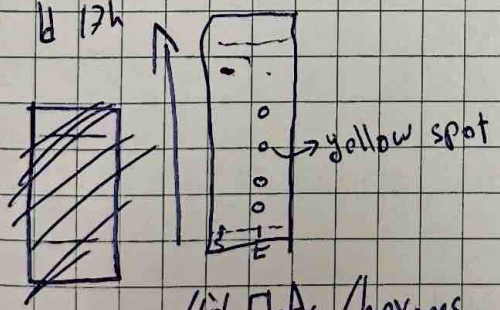
⑤ Add purged pyridine

yellow color

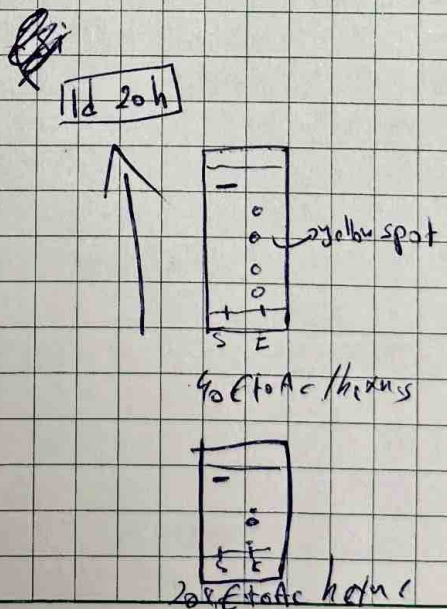
⑥ heat @ 140°C (rxn becomes darker over time).

TLC:-

after 1d 17h



4d, EtOAc/hexanes



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ARH-1-68

$Ti(OiPr)_4$ + ACN 6d ~~Ti-C~~ Ti-C

chemical	Fw(g)	mass(mg)	mmol	mM	eq	vol
Pivalic acid	102	4375	42.89	2859	10.2	4.84
$Ti(OiPr)_4$	284.26	1200	4.22	281	1	1.25
ACN	–	–	–	–	–	15

① in a teflon bomb add pivalic → feet smell
② Add ACN
③ Add $Ti(OiPr)_4$ → Pale yellow color
④ heat @ 120C → 6d.

~~Crystal~~

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ARH-1-69

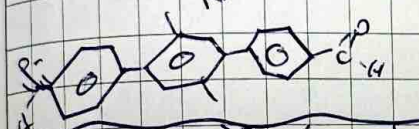
Zr cluster + DMF → Zr formate

120°C 2d

chemical | mass(mg) | mass(g) | mmol

ARH-1-69 (expanded COF-300)

Ter Ald



+ TAM → cof (300) expanded

chemical	fw(g)	mass(mg)	mmol	mM	eq	vol(ml)
Ter Ald	314	28	0.089	37.08	1.7	—
TAM	360.6	20	0.052	21.67	1	—
Dioxane	-	-	-	-	-	1.6
Cyclohexane	-	-	-	-	-	0.4
3MAA	-	-	-	-	-	2.4

DOI: doi.org/10.1021/~~jacs.8b13~~ Jacs. 8b13 [illegible]

Procedure similar to ARH-1-45-1

① in a 4ml vial Add TAM

② Add Ter Ald

③ Add dioxane

④ Sonicate

⑤ Add cyclohexane

⑥ Sonicate

⑦ Add 3MAA

⑧ Sonicate

⑨ 65°C / 3d.

⑩ light ppt + 2 phases (bot color pink / upper yellow)

- PXRD confirms crystallinity of product

As synthesized

- to do solvent exchange Dioxane / THF. → upon solvent exchange → Cof collapses.

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ARH-1-70

chemical	Fw g/l	mg/mg	mmol	mM	eq	(kg)
Ter Ald	314	314.4	0.1		2	—
TAM	360.8	23	0.05		1	—
Diox	—	—	—	—	—	0.25
Pyridine	—	—	—	—	—	0.15
Aniline	—	—	—	—	—	0.06
6MAA	—	—	—	—	—	0.2

① Add TAM
② Add Ter Ald
③ Add Diox → sonic → yellow suspension
④ Add pyridine → sonic → " "
⑤ Add Aniline → sonic → " "
⑥ Add 6MAA → sonic → orange suspension
⑦ heat @ 85°C → 3d

* DOI:- doi.org/10.1073/pnas.2320237121 (yvr bico)
ppt with purple supernatant after 3d.

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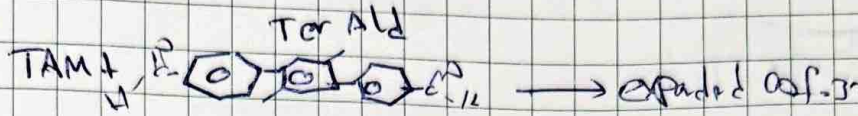
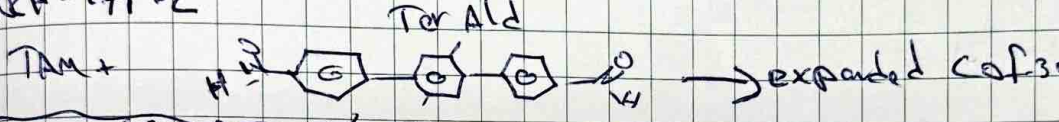
ARH-1-71-1



Chemical	Fw (g)	mass (mg)	mmol	mM	eq	vol
TAM	380.44	20	0.052	52	1	–
TerAld	314.38	29	0.089	89	1.7	–
Diox	–	–	–	–	–	0.145 ml
Mesitylene	–	–	–	–	–	0.55 ml
6MAA	–	–	–	–	–	0.12 ml

① Add TAM in Borosilicate tube
② Add TerAld
③ Add Diox 0.145 ml
④ Add Mesitylene 0.55 ml
⑤ Add 6MAA (0.12) ml
⑥ heat @ 120°C 3d

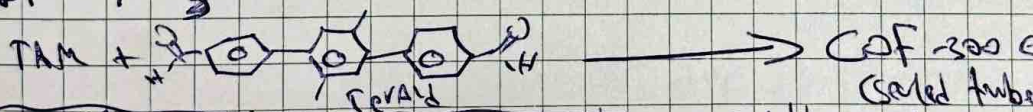
ARH-1-71-2



Chemical	Fw (g)	mass (mg)	mmol	mM	eq	vol
TAM	380.44	20	0.052	52	1	–
TerAld	314.38	29	0.089	89	1.7	–
Diox	–	–	–	–	–	0.75
Mesitylene	–	–	–	–	–	0.75
6MAA	–	–	–	–	–	0.1

① Add TAM in Borosilicate tube
② Add TerAld
③ Add Diox (0.125 ml)
④ Add Mesitylene (0.75 ml)
⑤ Add 6MAA (0.1 ml)
⑥ heat @ 120°C 3d

ARH-1-71-3



Chemical	Fw (g)	mass (mg)	mmol	mM	eq	vol
TAM	380.44	20	0.052	52	1	–
TerAld	314.38	32.7	0.104	104	2	–
Diox	–	–	–	–	–	0.145
Mesitylene	–	–	–	–	–	0.55
6MAA	–	–	–	–	–	0.1

① Add TAM in Borosilicate
② Add TerAld
③ Add Diox → sonic (0.145 ml)
④ Add Mesitylene → sonic (0.55 ml)
⑤ Add 6MAA → sonic
⑥ heat @ 120°C 3d

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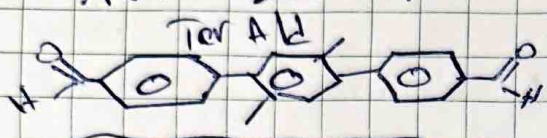
ARH-1-71-4



chemical	Fw(g)	Mass(mg)	mmol	mM	eq	vol
TAM	390.18	20	0.052	52	1	-
Ter Ald	314.36	32.7	0.104	104	2	-
Diox	-	-	-	-	-	0.45
Mesitylene	-	-	-	-	-	0.55
6MAA	-	-	-	-	-	0.2

① Add TAM in Borosilicate tube
② Add Ter Ald
③ Add Diox → sonic
④ Add Mesitylene → sonic
⑤ Add 6MAA → sonic
⑥ heat @ 120°C 3d.

ARH-1-71-5

chemical	Fw(g)	Mass(mg)	mmol	mM	eq	vol
TAM	390.18	20	0.052	52	1	-
Ter Ald	314.36	32.7	0.104	104	2	-
Diox	-	-	-	-	-	0.45
Mesitylene	-	-	-	-	-	0.55
6MAA	-	-	-	-	-	0.2

① Add TAM in Borosilicate tube
② Add Ter Ald
③ Add Diox → sonic
④ Add Mesitylene → sonic
⑤ Add 6M AA → sonic
⑥ heat @ 120°C 3d

PXRD

ARH-1-71-1 → Poor crys

ARH-1-71-2 → Poor Acceptable → solvent exch Dox. → COF decomposes

ARH-1-71-5 PXRD Acceptable → solvent exch Dox → COF decomposes

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ABH-1-73

NH$_2$-BDC + Ti-C + BDC ⟶ MIL-125 (75%)

Chemical	Fw (g)	mass (g)	mmol	mM	eq	vol
Ti-C	2120.5	447	0.121	9.26	3.69	—
BDC	166.13	128.7	[redacted]	369	14.07	—
NH$_2$-BDC	249.27	579.5	2.3247	11.07		
DMF	—	—	—	—	—	17.5
MeOH	—	—	—	—	—	3.5

0.22-g

① Add Ti-C in a 50 mL jar
② Add BDC ~~NH$_2$-BDC~~
③ Add NH$_2$-BDC
④ Add DMF ⟶ yellow suspension
⑤ Add MeOH ⟶ deep yellow suspension
⑥ ~~Add~~ N$_2$ purge + sonicate ⟶ clear yellow suspension
⑦ Heat @ ~~75°~~ 120°C 3d.

Confirmed crystallinity by PXRD.

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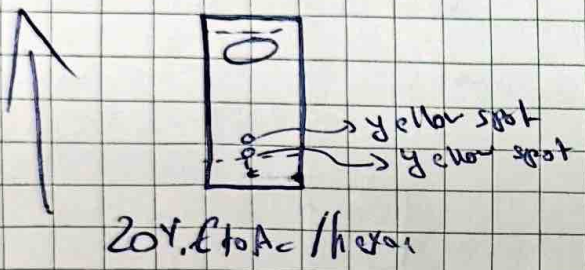
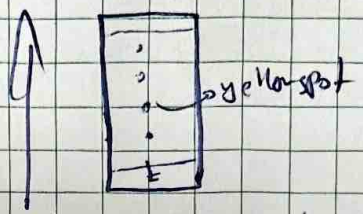
(7) dump the rxn in mixture of 3ml NH4OH 3ml toluene (2 phases; organic is red, ammonium hydroxide green).
(8) stir for 20 mins
(9) filter over celite plug by toluene solvent → orange filtrate
(10) extract 3x water. → organic became orange → yellow
aq orange → yellow.

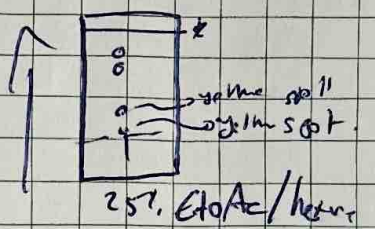
TLC:-



20% EtOAc/Hexane

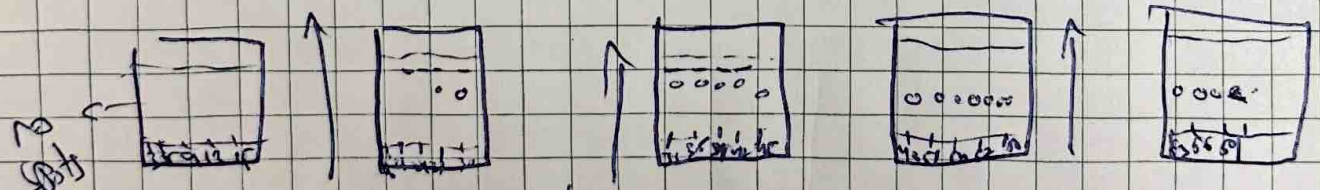


40% EtOAc/hexane



25% EtOAc/Hexane

- Load column with 10% EtOAc/Hexane
- run with 25% EtOAc/hexane
- collection



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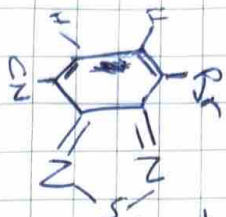
- NMR confirms ~~spot~~ top spot ~~of~~ yellow spot is product.

- NMR confirms  elutes with product

⇒ try allowing rxn 3d to proceed.

~~try~~

- NMR of ~~lower~~ upper spot recollected.

~~- residue washed with hexanes~~ yield ≤ 1%

-

Notes: Some residue remains on top of column (insoluble)

⇒ try to recrys from mixture of hexane and ethyl acetate

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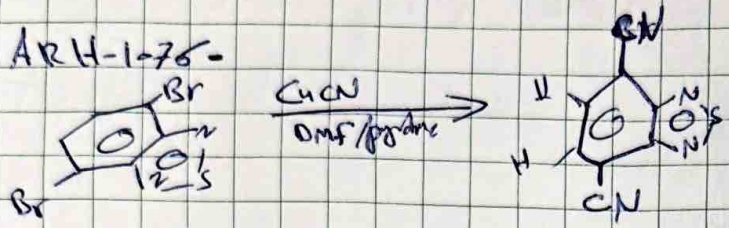
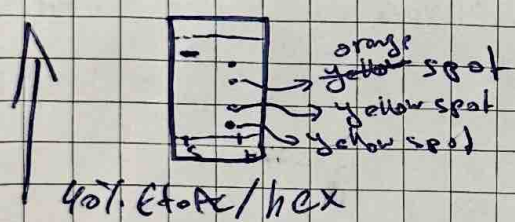
ARH-1-76-



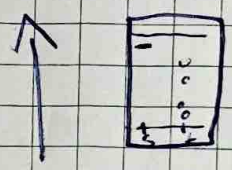
Chemical	fw (g)	massing	mol	mM	volume	eq
reactant	293.7	~~1570~~	~~0.17~~	~~3400~~		1
CuCN	89.56	~~1215~~ 304	~~0.265~~ 0.068	~~1560~~ 340	—	~~4~~ 1
DMF	—	—	—	—	—	1
product 1986						
pyridine	—	—	—		0.025	

① ~~Part~~ Add Reactant (70mg) small RBF
② Add CuCN (24mg)
③ Degass flask + Pump, N2 (x5)
④ degass DMF (20 min)
⑤ Add DMF to flask → pale yellow
⑥ leave for ~ 3d.
(remove on thursday)
heat @ 140°C

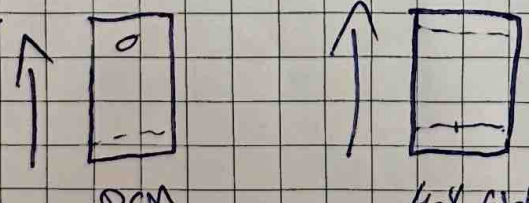
⑦ TLC



TLC: 3d



⑧ dump in 3ml NH4OH 3ml toluene suspension → mix for 20 mins → green solution with red brown solid.
⑨ Extract x3 water → aq turns orange organic tiny yellow
⑩ Extract x3 brine
⑪ add Na2SO4 → filter on celite → rotovap
⑫ TLC



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ARH-1-77

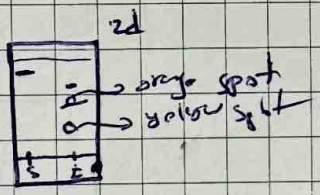
reactant

Chemical	Fw (g)	mass (mg)	mmol	mM	vol (mL)	eq
reactant	203.7	100	[illegible]	[illegible]	-	1
CuCN	89.56	121.8	1.36	1360	-	4
DMF	-	-	-	-	1	-
Prod	168	3	-	-	-	-

① Add Reactant to 5ml RBF (100) mg
② Add CuCN (121.8) mg
③ Degass flask N_2 x5
④ degass DMF (20 min)
⑤ Add DMF to flask
⑥ leave for ~~3~~ 2 days
heat @ 140°C

Note:- rxn done under shlenk line not ~~tube~~ balloon.

⑦ TLC:-



⑧ NMR from the rxn is is taken after 2d:-

↓↓

$ - DMF (solvent of rxn)
- Singlet @ 8.12 => product
- aromatic junk.
- ~~intermediate present~~

⑨ - dump rxn in 3ml toluene 3ml NH4OH mixture.
⑩ - extract x3 water → organic layer = yellow color
⑪ - rotavap → orgo pot
⑫ recrys with hexanes
⑬ filter (adsorbed on filter paper → use nylon membrane)
⑭ collect residue on filter paper with EtOAc → rotavap

NMR confirms product but impure

4.76% yield

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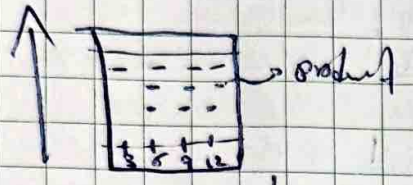
13 pack
~~load~~ colom with 25% EtoAc/hexane
14 - run column with 40% EtoAc

15 TLC :-



40 EtoAc hexn.

coloution)
- on colom it is streaking
(but less than 25% EtoAc)

note that crude is soluble in 40% EtoAc/hexane but not 25% EtoAc.

16 - collect fractions → nmr.
17 - recrys with hexane + EtoAc solvent pair.

yield ≤ 1%

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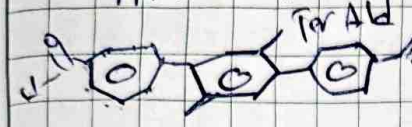
ARH-1-79



+ TAM → Expanded COF-300

check page 69

Chemical	FW (g)	mass (g)	mmol	mM	eq	vol (mL)
TAM	314	20	0.052	21.67	1.7	–
TerAld	380?	26	0.068?	32.08		–
Dioxane	–	–	–	–	–	1.6
Mesitylene	–	–	–	–	–	0.4
~~Cyclohexane~~ 3M AA	–	–	–	–	–	0.4

① in a vial Add TAM
② Add Ter Ald
③ Add Diox → sonicate
④ Add Mesitylene → sonicate
⑤ Add 3M AA → sonicate
⑥ heat @ ~~120°C~~ 65°C → 3d

low cryst by PXRD.

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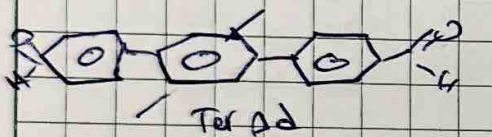
ARH-1-80-1

→ Expanded COF-300 (Sealed tube)

Chemical	Fw(g)	mass(mg)	mmol	mM	eq	vol (mL)
TAM	380.18	20	20	0.052	1	–
TerAlH	314.38	32.7	32.7	0.104	2	–
DiOx	–	–	–	–	–	0.135
Mesitylene	–	–	–	–	–	0.25
6M AA	–	–	–	–	–	0.12

① in a borosilicate tube Add TAM
② Add TerAlH
③ Add ~~DiOx~~ Mesitylene → sonic
④ Add dioxane → sonic
⑤ Add 6M AA
⑥ Heat @ 120°C 3d

ARH-1-80-2



→ Expanded COF-300 (sealed tube)

Chemical	Fw(g)	mass(mg)	mmol	mM	eq	vol (mL)
TAM	380.18	20	20	0.052	1	–
TerAld	314.38	32.7	32.7	0.104	2	–
DiOx	–	–	–	–	–	0.125
Mesitylene	–	–	–	–	–	0.25
12M AA	–	–	–	–	–	0.1

① in a borosilicate tube Add TAM
② Add TerAld
③ Add ~~DiOx~~ Mesitylene → sonic
④ Add dioxane → sonic
⑤ Add 12M AA
⑥ Heat @ 120°C 3d

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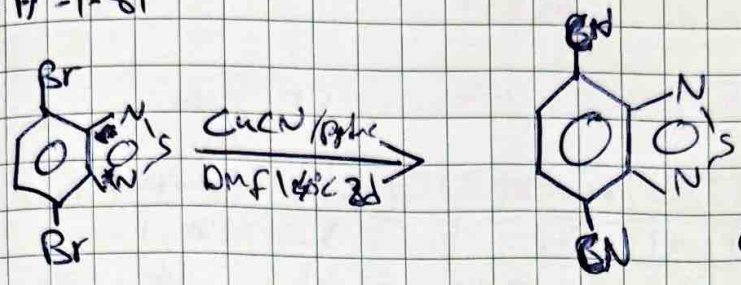
ARH-1-81



Reactn

chemical	FW g/mol	mass mg	mmol	mM	eq	vol mL
Reactn	293.7	69	0.235	235	1	-
CuCN	89.56	84	0.94	940	4	-
Pyridine	79.1	46.5	0.588	588	2.5	0.05
DMF	-	-	-	-	-	1
product						

① Add CuCN in a vial 3mL

② Add R

③ Add DMF (SPS) + freeze pump thaw

④ Add pyr (freeze pump thaw)

(*) rxn is air free.

- after 3d rxn turns black
- ~~add water~~ (by mistake) - dump into 3mL NH4OH : 5mL toluene mixture
- stir for 20 mins → turns red-brown
- add water (by mistake forgot to filter first).
- filter over celite
- wash with toluene. Celite residue to make sure compound is dissolved
- extract with water x3 → aq dark brown, organic yellow
- extract with Brine x3
- Na2SO4 is added
- Rotavap → NMR → product + starting
- ~~wash with hexane~~
- wash with 20% EtOAc/hex → filter
- ~~do~~ NMR.

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ARH-1-82-1 expanded COF-300

+TAM → expanded COF-300

TerAld

Chemical	mass (g)	mass (mg)	mmol	mM	eq	vol (ml)
TerAld	3M	28	0.089	89	1.7	-
TAM	3568	20	0.052	52	1	-
Dioxane	-	-	-	-	-	1.6
Mesitylene	-	-	-	-	-	0.4
~~Aniline~~						
3M AA	-	-	-	-	-	0.4

·4ml vial

1. In a ~~vial~~ Add TerAld
2. Add TAM
3. Add Dioxane → sonic
4. Add Mesitylene → sonic
5. Transfer to vacuum vial
6. Add 3M AA
7. Vent @ 65°C 3days

~~ARH-1-82-2 expanded COF-300~~
~~Aniline~~

crystallinity low. by PXRD

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ARH-1-83-

TPAA

+ TAM → cof-300

Chemical	fw (g)	mass (mg)	mol	mM	eq	Vol
TPAA	134	12	0.089	37.06	1.7	—
TAM	3808	20	0.052	21.67	1	—
Dioxane	—	—	—	—	—	1.6
cyclohexane	—	—	—	—	—	0.4
Aniline		59				0.06
3MAA						0.4

① Add TPA in a 4ml vial
② Add TAM
③ Add Dioxane → yellow sol
④ Add cyclohexane → " "
⑤ Add Aniline
⑥ transfer to vial in vial
⑦ Add 3MAA
⑧ heat @ 65°C → 3d

3d no cryst → leave it for 1 week.

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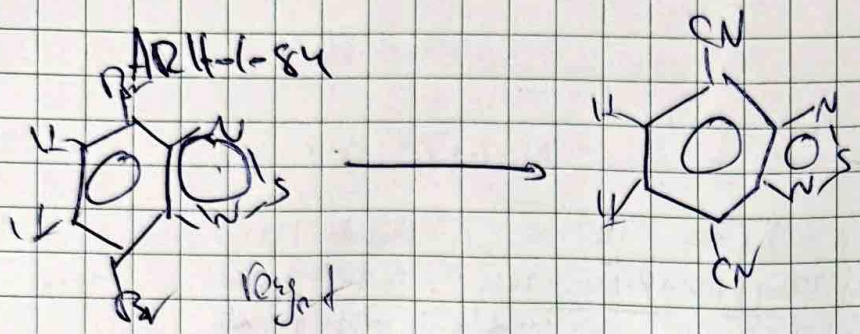
ARH-1-84



Reagent

chemical	Fw (g)	mass (mg)	mmol	mM	vol	eq
Reagent	248.7	59	0.235	235	-	1
CuCN	89.56	84	0.94	940	-	4
~~Pyridine~~	~~Xml~~	~~485~~	~~0.588~~	~~588~~	~~0.5~~	~~0.5~~
~~DMA~~ DMF	—	—	—	—	1	-
Pyridine	79.1	465	2.588	588	2.012	

~~5ml RBF~~ ~~Microwave tube~~

① Add R to ~~5ml crimp~~ flask

② Add CuCN

③ purge-vac x3 N_2

④ Add DMF Mixture mixture (freeze pump thaw x3) → yellow sol

⑤ heat @ ~~140°C~~ → 2d

~~200°C~~

~~160°C~~

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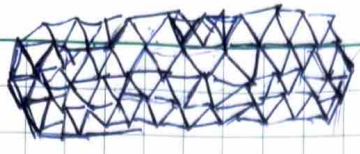
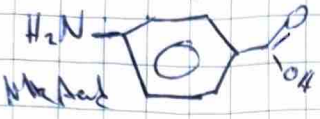


ADCH-1-85



+ $Ti(OiPr)_4$ → $Ti_6O_6(C_7H_8NO_2)_6(C_3H_2O)_6$ (100°C, 3d)

Chemicals	F.w.(g)	mass(mg)	mmol	M	eq	vol(mL)
NH2 Acid	~~137.14~~ 139	~~202.1~~ 140.1	1.4	233	4	—
$Ti(OiPr)_4$	284	—	~~0.35~~	~~583~~	1	0.103
2-propanol	—	—	—	—	—	6

① in a ~~[illegible]~~ teflon bomb add NH2 Acid
② Add 2-propanol
③ Add $Ti(OiPr)_4$
④ Vortex
⑤ heat@100°C 3d

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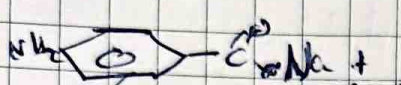
ARH-1-86



chemical	f.w.(g)	mass(mg)	mmol	mM	eq	vol
~~NH2 Acid Nitrile~~	~~160~~	~~5.475~~	~~34.2~~	3.60		
Ti(OiPr)4	~~284.26~~	~~10.26~~	~~0.36~~	2d	1	~~0.105~~ 0.093
ACN	—	—	—	—	—	~~5.1~~ 1

① In a teflon bomb add Acid
② Add ACN
③ ~~Add NH2~~ Ti(OiPr)
heat @ 120°C 3d

chemical	f.w.(g)	mass(mg)	mmol	mM	eq	vol
Acid NH2	160	~~5.475~~ 14.22	3.4	0.20		—
Ti(OiPr)4	284.26	~~10.26~~	~~0.36~~ 0.092		1	0.0926
ACN	—	—	—	—	—	1

① In a ~~teflon bomb~~ borosilicate tube Add Acid
② Add ACN
③ Add Ti(OiPr)
heat → 100°C 15h

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ARH-1-87-1 https://doi.org/10.1021/ja409033p

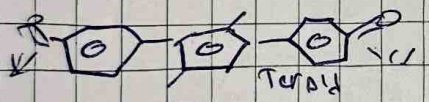
TerAld

+ TAM → expanded COF-300 Solvothermal

Chemical	fw (g)	mass (mg)	mmol	mM	eq	vol (ml)
TerAld	314	30	0.0952	79.3	1.8	
TAM	3808	20	0.052	433	1	
Diox	–	–	–	–	–	1
3MAA	–	–	–	–	–	0.2

① in a Borosilicate tube Add TAM
② Add TerAld
③ Add Diox
④ sonic
⑤ Add 3M AA
⑥ sonic
⑦ heat @ 120°C 3d

ARH-1-87-2



+ TAM → Expanded COF-300

Chemical	fw (g)	mass (mg)	mmol	mM	eq	vol (ml)
TerAld	314	30	0.0952	1.8	1.8	–
TAM	3808	20	0.052	1	1	–
Diox	–	–	–		–	1
3MAA	–	–	–		–	2
Aniline	93.13	29.056			6	0.0328

① in a Borosilicate tube Add TAM
② Add TerAld
③ Add Diox
④ Sonic
⑤ Add 3 Aniline
⑥ Add 3M AA
⑦ heat @ 120°C 3d

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ARH-1-88-1

Ter Ald + TAM ⟶ expanded CoF-300

chemical	Fw(g)	mass(mg)	mmol	mM	eq	Vol(ml)
Ter Ald	314	28	0.089	89	1.7	–
TAM	380	20	0.052	52	1	–
Diox	–	–	–	–	–	1.6
Cyclohex	–	–	–	–	–	0.4
3MAA	–	–	–	–	–	0.4
Aniline	93.1	29.056	0.312	312	6	0.026

vcm vial

① in a 4ml vial Add TAM
② Add Ter Ald
③ Add dox → sonic → yellow
④ Add cyclohex → sonic
⑤ Add Aniline → orange
⑥ Add 3MAA

Heat @ 65°C → 3d 4d ☆

ARH-1-88-2

~~Vcm vial~~ ~~4ml vial~~ vcm vial ☆

chemical	Fw(g)	mass(mg)	mmol	mM	eq	Vol(ml)
Ter Ald	314	28	0.089	89	1.7	–
TAM	380	20	0.052	52	1	–
Diox	–	–	–	–	–	~~1.6~~
Cyclohex	–	–	–	–	–	0.4
3MAA	–	–	–	–	–	0.2
Aniline	93.1	29.056	0.312	312	6	0.026

① in a 4ml vial Add TAM
② Add Ter Ald
③ Add dox → sonic
④ Add cyclohex → sonic
⑤ Add aniline → orange
⑥ ~~Add~~ Add 3MAA
⑦ 65°C → 3d 4d ☆

ARH-1-88-1 ~~Aniline~~ ⟶ ~~x~~ low crys PXRD.

ARH-1-88-2 ⟶ low crys but ~~n~~ smaller peaks

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Project

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ARH-489

ATAM → Cat-300

chemical	fw (g)	mass (g)	mmol	mmol	eq	vol
TPA	134	12	0.089	89	1.7	—
TAM	380.6	20	0.052	52	1	—
Dox	—	—	—	—		1.6
~~AA~~	~~—~~	~~—~~	~~—~~	~~—~~	~~—~~	~~0.4~~
~~cat~~	~~—~~	~~—~~	~~—~~	~~—~~	~~—~~	~~—~~
Amine	93.5	59	—	632	6	0.06
3MAA	—		—	—		0.4

low eqs by PXRD.

① In 4mL HCl Add TPA
② Add TAM
③ Add Dox → sol'n
④ Add catalyst sol'n
⑤ ~~Add 3M AA → fresh brown~~
⑥ Add 3M AA ~~→ sol'n~~
⑦ Add Amine
⑧ Heat @ 65°C → 3d 4d

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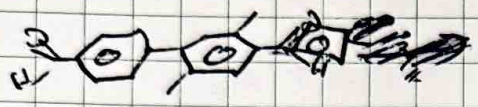
ARH-L-90-1



→ Expected Cof-300

chemical	Fw(g)	mass(mg)	mmol	mM	eq	vol(mL)
Ter Ald	314	28	0.089	89	1.7	—
TAM	380	20	0.052	52	1	—
Diox	—	—	—	—	—	1.6
Cyclohex	—	—	—	—	—	0.4
6MAA	—	—	—	—	—	0.2
Aniline	93.1	29.056	0.312	312	6	0.028

① in a vial Add TAM
② Add Ter Ald
③ Add diox → sonic → yellow
④ Add cyclohex → sonic
⑤ ~~Add A~~ transfer to vial
⑥ Add Aniline
⑦ Add 6MAA
⑧ heat @ 85°C 3d

ARH-L-90-2

chemical	Fw(g)	mass(mg)	mmol	mM	vol(mL)
Ter Ald	314	28	0.089	89	—
TAM	380	20	0.052	52	—
Dox	—	—	—	—	1.6
cyclohex	—	—	—	—	0.4
6MAA	—	—	—	—	0.4
Aniline	93.1	—	0.312	6	0.028

① in a vial Add TAM
② Add Ter Ald
③ Add diox → sonic → yellow
④ Add cyclohex → sonic
⑤ ~~Add~~ transfer to vial
⑥ Add Aniline
⑦ Add 6MAA
⑧ heat @ 85°C.

ARH-1-90-1 : white ppt yellow supernatant. → Acceptable crys 2nd crop higher than 1st
ARH-1-90-2 : pale yellow ppt pink supernatant → low crys

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ARH-1-91-1

Chemical	Fw(g)	mass(g)	mmol	mM	eq	vol
TerAld	314	28	0.089	89	1.7	–
TAM	350	20	0.052	52	1	–
Diox	–	–	–	–	–	1.6
Cyclohexane	–	–	–	–	–	0.4
6M AA	–	–	–	–	–	0.4
Aniline	93.1	29.056	0.312	312	6	0.029

① Add TerAld in 4ml vial
② Add TAM
③ Add diox → sonic → yellow
④ Add cyclohexane → sonic
⑤ transfer to vac vial
⑥ Add Aniline
⑦ Add 6M AA
⑧ heat @ 65°C 3d

ARH-1-91-2

~~TerAld~~

chemical	Fw(g)	mass(g)	mmol	mM	eq	vol
TerAld	314	28	0.089	89	1.7	–
TAM	350	20	0.052	52	1	–
Diox	–	–	–	–	–	1.6
Cyclohexane	–	–	–	–	–	0.2
Mesitylene	–	–	–	–	–	0.2
6M AA	–	–	–	–	–	0.2
Aniline	93.1	29.056	0.312	312	3.12	~~0.029~~ 0.056

① Add terAld in 4ml vial
② Add TAM
③ Add diox → sonic → yellow
④ Add cyclohexane → sonic
⑤ ~~transfer to vac~~ Add Mesitylene → sonic
⑥ transfer to vac vial
⑦ Add Aniline
⑧ Add 6M AA
⑨ heat @ 65°C 3d.

ARH-1-91-3

chemical	Fw(g)	mass(g)	mmol	mM	eq	vol
TerAld	314	28	0.089	89	1.7	–
TAM	350	20	0.057	52	1	–
Diox	–	–	–	–	–	1.6
Cyclohexane	–	–	–	–	–	0.2
Mesitylene	–	–	–	–	–	0.2
6M AA	–	–	–	–	–	0.2
Aniline	93.1	29.056	0.312	312	6	0.029

① ~~in~~ 4ml vial Add terAld
② Add TAM
③ Add diox → sonic yellow
④ Add cyclohexane → sonic
⑤ Add Mesitylene → sonic
⑥ transfer to vac vial
⑦ Add Aniline
⑧ Add 6M AA
⑨ heat @ 65°C

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ARH-1-91-1 : yellow ppt after supernatant → low cry

ARH-1-91-2 : yellow ppt dark super → Acceptable crys ~~blood~~ peaks.

ARH-1-91-3 ~~Brown~~ ppt = = → = = good peaks

conclusion, ~~0.2 5mM 6MAA better than 0.4 5mM 6MAA~~

0.2ml 6MAA 0.4ml 6MAA

- ARH-1-86-2 is best recipe for expanded CoF-30
- Mesitylene doesn't help crystalization.
-

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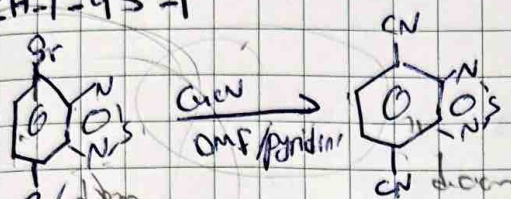
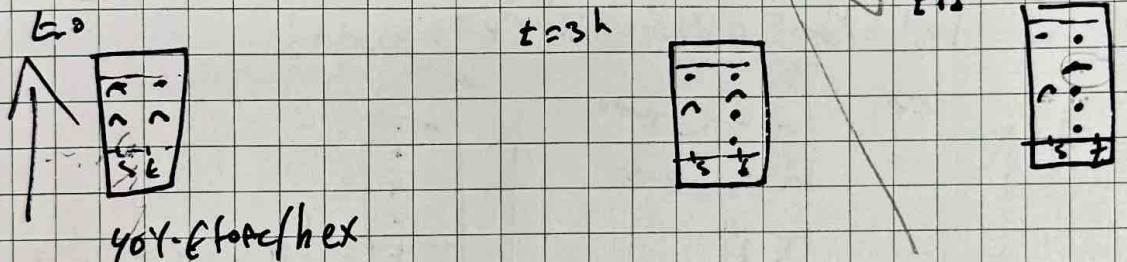
ARH-1-93-1



chemical	weight (mg)	mass (mg)	mmol	mM	eq	vol (mL)
Reagent	203.2	59	0.235	235	1	—
CuCN	89.56	84	0.94	940	4	—
Pyridine	79.1	965	0.588	[illegible]	2.5	0.047
DMF	—	—	—	—	—	1

① Add R in 5mL RBF
② Purge DMF N_2 x3
③ Purge Pyridine N_2 x3
④ Purge RBF N_2 x3
⑤ Add DMF/pyridine mixture-yellow
⑥ heat @ 160°C using thermometer on heating

* rxn after 1 day balloon popped



balloon was replaced and rxn allowed to continue.

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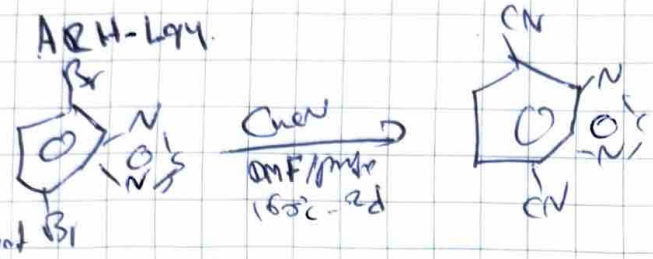
ARH-L94

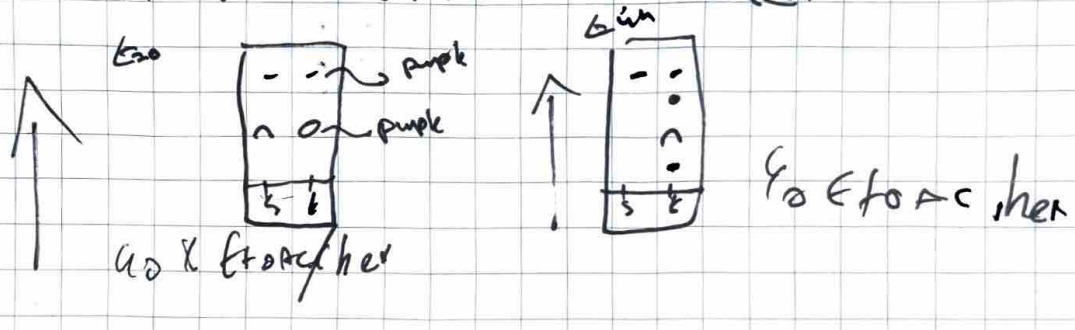


Reagent B1

chemical	fw(g)	massing	mmol	mM	eq	vol (ml)
Reagent	293.7	69	0.235	235	1	—
CuCN	89.56	84		940	4	→
DMF	—	—	—	—	—	1
pyridine	79.1	74.6	0.94	940	4	0.076

① Add Reagent in a 5mL RBF
② Add CuCN.
③ purge x N2 x3
④ Add DMF/pyridine x N2 x3
⑤ heat @ 165°C 2d

⑥ after 1d 1ml of solvent mixture is added.



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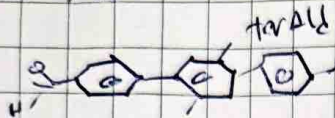
ARH-1-95



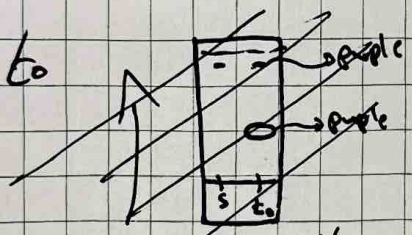
→ Expanded CoF-300 (X2 scale)

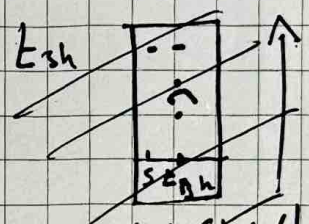
Chemical	fw(g)	mass(g)	mmol	M/M	eq	vol
TerAld	314	58	0.176	89	1.7	
TAM	330	40	0.104	52	1	
DMF	–	–	–	–	–	3:2
cyclohex	–	–	–	–		0.8
6MAA	–	–	–	–	–	0.4
Aniline	93.1	58.812	0.624	624	6	0.056

① in a 4ml vial Add TAM
② Add terAld
③ Add DMF → sonic
④ transfer to vial
⑤ Add cyclohex
⑥ Add Aniline
⑦ Add 6MAA ⑧ heat @ 65°C 3 d

check ARH-185-2

~~Mass = 75.2~~ Mass = 25.2





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ARH-1-96

Pivalic acid $Ti(OiPr)_4$ $\xrightarrow[120°C\ 6d]{ACN}$ $Ti_8O_8(OOCtBu)_6$

chemical	fw(g)	mass(g)	mmol	mM	eq	vol
$Ti(OiPr)_4$	284.26	1200	4.22	281	1	1.25
Pivalic acid	102	4.375	42.89	2859	10.2	4.84
ACN	–	–	–	–	–	15

① In a teflon bomb add ACN
② Add Pivalic (Melt down) → feel smell
③ Add $Ti(OiPr)_4$ → pale yellow
④ heat @ 120°C → 3d

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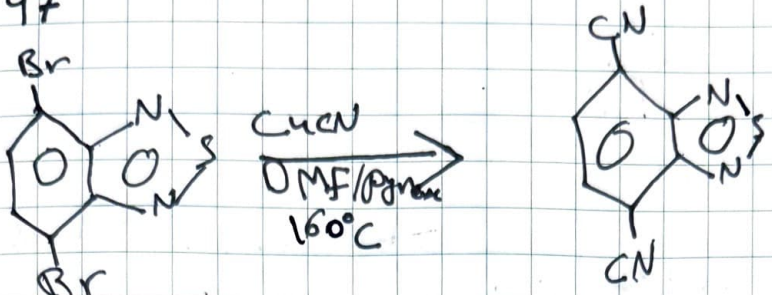
ARH-1-97



Reagent

chemical	Fw (g)	mass (mg)	mmol	M/M	eq	vol (ml)
Reagent	293.7	69	0.235	235	1	—
CuCN	89.56	84	0.94	440	4	—
DMF	—	—	—	—	—	1
Pyridine	79.4	18.2	1.88	1880	8	0.152

① Add R in a sml RBF
② Add CuCN
③ Add ~~DMF~~ DMF/Pyridine (15% Pyridine in DMF)
④ heat @ 160°C → yellow
⑤ rxn turn red brown → black ~~after~~ 1 day.

⑥ mr taken after 1 day.

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21/1/2016

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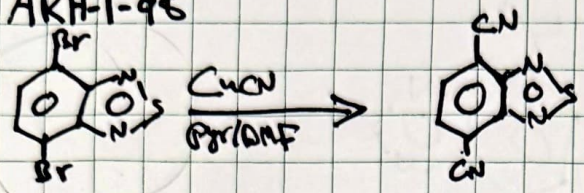
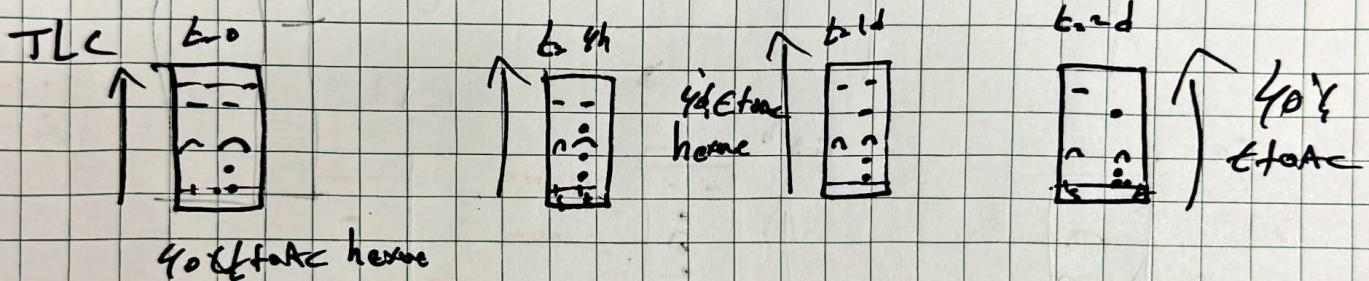
ARH-1-96

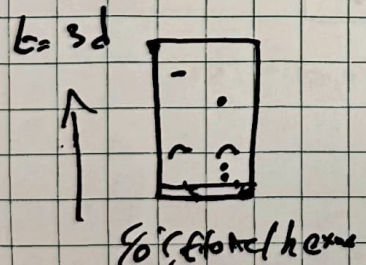


Reagent

chemicals	fw(g)	mass(g)	mmol	mM	eq	vol(ml)
Reagent	298.7	69	0.235	235	1	-
CuCN	89.56	84	0.940	940	4	-
Pyridine	79.1	46.5	0.582	582	2.5	0.047
DMF	-	-	-	-	-	1

① Add Reagent in a 5ml RBF
② Add CuCN.
③ purge N_2 and Vac for 30 mins
④ prepare 4.8% DMF Pyridine solution
⑤ purge N_2 for 20 mins
⑥ Add ~~solvent~~ 1ml solution DMF/Pyr → yellow color
⑦ heat @ 160°C





⑧ quench by

Note: Spot on UV

⑨ rxn added fractions of 5ml toluene / 3ml Nitrogen color → ~~brown~~ black-green
⑩ stir for 20 minutes
⑪ filter over celite.

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Project

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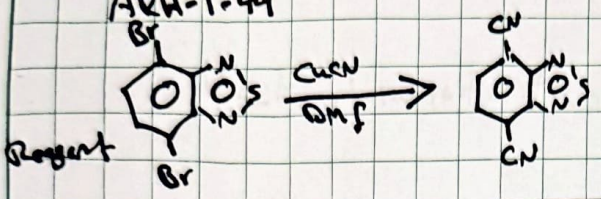
ARH-1-99



Chemical	fw(g)	Mass(mg)	mmol	mM	eq	vol(mL)
Reagent	298.7	69	0.235	235	1	—
CuCN	89.56	89	0.94	940	4	—
DMF	—	—	—	—	—	1

① Add Reagent & SM RBF
② Add CuCN
③ purge N_2 / Vac 30 min
④ Purge DMF 20 min
⑤ Add DMF to RBF
⑥ heat @ 160°C.

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Activating COF-300 :-

Cake ADH-1-38 was placed in a soxlet extraction with MeOH

① heated @ 80°C ⟶ 5hr

③ after 2d heat increased to 120°C 1d+ for 1d

PXRD confirms crystallinity of collapsed form

④ sample is being activated by Hood Vaccuum. for 2d

⑤ sample remains collapsed (PXRD measurement) with ↓ crystallinity.

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ARH-101

~~NaBH4~~ $NaBH_4 + (NH_4)_2SO_4 +$ amine → $N\text{-}BH_3$ Product

37.83 g

chemical	fw(g)	mass(mg)	mmol	mM	eq	vol(mL)
$NaBH_4$	~~37.84~~	190	5	~~380.0~~	2	–
$(NH_4)_2SO_4$	132.14	330	2.5	~~660~~	1	–
amine	99	247.5	2.5	~~500~~	1	0.179
THF ~~1.0~~	–	–	–	–	–	~~2.5~~
product						

① in a 10 ml RBF add $NaBH_4$
② Add $(NH_4)_2SO_4$
③ ~~Add THF~~ amine to THF (SPS)
④ Add Amine/THF to mixture
⑤ heat @ 80°C

NMR taken after 1h ⇒

quintet ~~at~~ at ~~-50~~ -50ppm ← stdy.
quintet @ -20ppm.
↳ product

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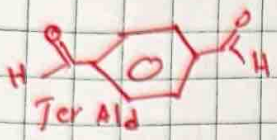
ARH-6-102

COF-300



+ TAM → COF-300

chemical	fw[g]	mass[mg]	mmol	mM	eq	vol[ml]
Ter Ald	140	12	0.084	32.08	1.7	
TAM	380	20	0.052	21.7	1	
Diox	→	~~12~~	–	–	–	1.6
cyclohex	→	~~0.2~~	–	–	–	0.4
3M AA	–	–	–	–	–	0.4

① in a 4 dram vial Add Ter Ald
② Add TAM
③ Add Dox → sonicate
④ Add cyc → sonicate
⑤ ~~Add Acetic Acid~~
⑥ Add 3M AA
⑦ heat @ 85°C 3d.

- Assemble MeOH soxlet @ 140°C
- when frit compartment is halfly filled dump As-syn COF-300
- soxlet for 4d

- pXRD
- small peak @ 7 ~~deg~~, tall peak @ 9
- Activate (100°C, shlenk vac) → collapsed

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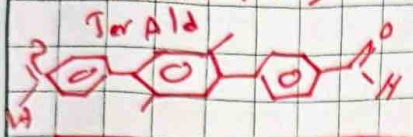
ARH-1-103



+ TAM → COF-300 (Expanded).

Chemical	Fw(g)	mass(mg)	mmol	mM	eq	Vol(ml)
Ter Ald	314	28	0.089	37.09	1.7	–
TAM	380	20	0.052	21.67	1	–
Dioxane	–	–	–	–	–	1.6
Cyclohex	–	–	–	–	–	0.4
Aniline	93.1	28.612			6	0.025
6M AA	–	–		312	–	0.12

① in a 4 mL vial add Ter Ald
② Add TAM
③ Add Dioxane → sonic → yellow
④ Add Cyclohex → sonic
⑤ Add Aniline.
⑥ Add 6M AA
⑦ transfer to vial
⑧ heat @ 65°C 3d

⑨ ~~*~~ Crystallinity confirmed by PXRD

⑩ solvent exchange with THF x3.

⑪ Activate ~~PXRD~~, (Schlenk line vac 100°C)

PXRD ~~g~~ → low age peaks almost absent.

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(8) Extract xs toluene → organic yellow aq → brown

(9) ethanol xs water → organic yellow aq light brown

(10) extract xs hexane → organic yellow aq cloudy

(11) filter over celite (to remove organic contains junk). (junk remains on celite)

(12) rotavap → brown ppt (yield 5(11)%) NMR doesn't show product.

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Signed 11/8/2016 Date

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Actinometry experiment

① soak 5 * 10ml flask overnight with 1M HCl

② synthesize ferrioxalate.

~~K_3FeC_6~~ $2KOH + H_2C_2O_4 \longrightarrow 2H_2O + K_2C_2O_4$

Chemical	Fw(g)	massing	mmol	mM	eq	Vol(ml)
KOH	56	2120	37.9	3790	2.1	–
$H_2C_2O_4$	126	[illegible]	18.07	1804	1	–
H_2O	–	–	–	–	–	10

2220 mg

copied from

① Add KOH to 50ml RBF

② Add $H_2C_2O_4$

③ Add H_2O → heat + bubbles

④ heat @ 80°C for 30min + stir

$K_2C_2O_4 + FeCl_3 \xrightarrow{H_2O} K FeC_2O_4$

chemical	fw(g)	massing	mmol	mM	eq	Vol(ml)
$K_2C_2O_4$	166.2	3000	18.04	902	3	–
~~$Fe(NO_3)_3 \cdot 9H_2O$~~	~~404.0~~	~~1625~~	6.01	300.6	1	–
H_2O	–	–	–	–	–	10

~~④ filter crystals + wash with the water~~

⑤ Add $FeCl_3$ solution slowly. (dropwise).
colorless → green.

⑥ Keep stirring ~~to~~ for 10 min

⑦ remove RBF and place in dry (crystallization will start overnight)

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Read and Understood By

Signed 27/9/2026 Date

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Sy: ARH-1a-106 synthesize Amine-Borane

$NH_4iPr + NaBH_4 + (NH_4)_2SO_4 \xrightarrow{THF} NH_4iPrBH_3$

Chemical	fw(g)	mass(mg)	mmol	mM	eq	vol(mL)
NaBH4	37.8	1000	26.43	979	1	—
$(NH_4)_2SO_4$	132.06	3500	13.22	489	0.5	—
NH4iPr	59.11		26.43	979	1	2.25
THF (SPS)	—	—	—	-	—	27
Product	72	1.902	26.43		1	

Atomfrac

① ~~in a loom RBF~~

② grind $(NH_4)_2SO_4$

③ Add to 100 mL RBF

④ grind $NaBH_4$

⑤ Add to RBF

⑥ Add NH4iPr

⑦ Add THF (SPS)

⑧ heat @ 80°C

Rxn stat @ 10:45 AM

⑧ NMR taken @ 4:45 :- No $NaBH_4$ peak @ -80 ppm
reductant @ -22 ppm present
hump after -10 ppm ⇒ borosilicate nmr tube
small peak @ 30 ppm.

⑨ Rxn stopped :- left to cool / heater off.

⑩ filter over celite ⑪ wash with SPS + THF.

⑫ rotavap. → turbid liquid remains

1.711 g isolated $\frac{1.7}{1.9} \times 100$

89.5%

⑬ nmr confirms Pure product w/ 1% THF

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MSDS
- Sulfuric Acid
- corrosive + irritation
- if contact with skin → Wash with water
- if inhaled call poison center
- NOT flammable
- Don't use water to extinguish nearby fires.

Incompatible
- ~~Don't mix~~ with strong oxid agents, combustible materials, bases
- reducing agents, finely powdered metals, peroxides

MSDS - hydrogen peroxide:-
- Oxidant, corrosive, irritant.
- call poison center if inhaled
- wash skin if in contact with skin.
- can be extinguished with water
- incompatible with + fine powdered metals, copper, reducing agents strong bases, combustible material, organic.

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ARH-1-108

+ TAM → Expanded COF-300

Chemical	fw(g)	mass(mg)	mmol	m/M	eq	vol(ml)
TerAld	314	28	0.089	40.45	1.7	-
TAM	380	20	0.052	23.79	1	-
Dioxane	-	-	-	-	-	1.6
Cyclohexane	-	-	-	-	-	0.4
Aniline	93.9	28.612	0.304	142.8	6	~~0.304~~ 0.028
6M AA	-	-	-	-	-	0.2

① Add TerAld in 2 vials
② Add TAM
③ Add Dioxane → sonicate → yellow
④ Add cyclohexane → sonicate → yellow
⑤ Add Aniline → orange
⑥ Add 6M AA →
⑦ heat @ 65°C 3d.

⑧ PXRD confirms Crystallinity as Syn. (30% yield).

⑨ Solvent exchange THF → Activate @ 100°C Shlenk line ⇒ [illegible] orange peaks disappear

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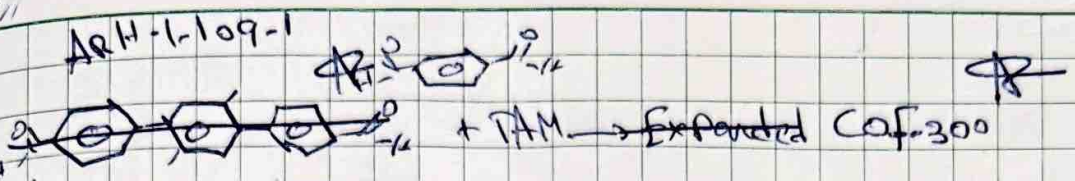
ARH-1-109-1

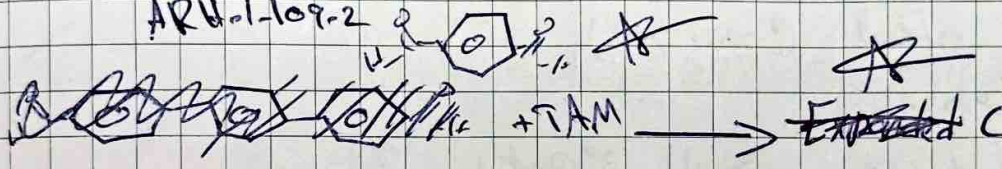


+ TAM → ~~Expected~~ CoF-300

chemical	Fw(g)	mass(mg)	mmol	mM	eq	vol(ml)
Ter Ald	~~324.30~~	~~24~~ 12	0.0891	90.45	1.7	–
TAM	380	20	0.0526	23.74	1	–
Diox	–	–	–	–	–	1.6
Cyclohex	–	–	–	–	–	0.4
Aniline	93.9	28.612	0.571	142.8	6	0.028
12MAA	–	–	–	–	–	0.1

① in a ~~[illegible]~~ 4ml vial Add TAM
② Add Ter Ald
③ Add diox → sonic → yellow
④ Add cyclohex → sonic
⑤ Add ~~Aniline~~ fresh toluene vial
⑥ Add Aniline
⑦ Add 12MAA
⑧ heat @ 65°C 3d.
⑨ wash w:

ARH-1-109-2



+ TAM → ~~Expected~~ CoF-300

chemical	Fw(g)	mass(mg)	mmol	mM	eq	vol(ml)
Ter Ald	~~324.30~~	~~24~~ 12	0.0891	90.45	1.7	–
TAM	380	20	0.0526	23.74	~~1~~	–
Diox	–	–	–	–	–	1.6
Cyclohex	–	–	–	–	–	0.4
Aniline	93.9	28.612	~~0.1069~~	245.6	12	0.028
6MAA	–	–	–	–	–	~~0.2~~

① in a ~~vial~~ vial Add TAM
② Add Ter Ald
③ Add ~~cyclohex~~ Diox → sonic
④ Add Cyclohex → sonic
⑤ Add Aniline
⑥ Add ~~12MAA~~, 6MAA

~~ARH-1-109-3~~ ~~ARH-1-109-3~~ ⑦ both solvent exchg. & with diox without filtration

chemical ⑧ DMO then after centrifuge ("wet")

ARH-1-109-1 → narrow peaks.

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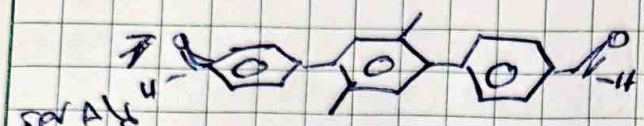
~~ARH-1-110~~ ARH-1-110



→ Expanded COF-300

chemical	Fw(g)	mass(mg)	mmol	mM	eq	vol (ml)
TerAld	341	23	0.089	40.45	1.7	—
TAM	380	20	0.052	23.79	1	—
Diox ~~Expanded COF~~	-	-	-	-	-	1.6
Cyclohexane	-	-	-	-	-	0.4
Aniline	939	20.512	0.332	~~0.6~~	6	0.028
6MAA	-	-	-		-	0.2

① Add TAM in ~~Dioxane~~ Diox
② Add TerAld
③ Add Diox → sonic (yellow)
④ Add Cyclohex
⑤ Add Aniline
⑥ Add 6MAA
⑦ Heat @ 65°C 3d.

⑧ Sample was centrifuged (beige ppt)
⑨ solvent exch acetone
⑩ PXRD → low cryst → 1st Peak @ 7.1

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AQH-1-111-1

Chemical	Fw(g)	Mass(mg)	mmol	mM	eq	vol(ml)
TenAH	140	12	0.089	40.5	1.7	–
TAM	360	20	0.053	24.0	1	–
Diox	–	–	–	–	–	1.6
Cyclohex	–	–	–	–	–	0.4
Aniline	93.1	28.012	0.318	1445	6	0.1028
3MAA	–	–				0.1

① Add TAM in ~~volumetric~~ conical
② Add TenAH
③ Add diox → sonic → yellow
④ Add cyp → sonic.
⑤ ~~Pipette~~ transfer to conical
⑥ Add 3MAA.

AQH-1-111-2

Chemical	Fw(g)	Mass(mg)	mmol	mM	eq	vol(ml)
TenAH	140	12	0.089	40.5	1.7	–
TAM	360	20	0.053	24.0	1	–
Diox	–	–	–	–	–	1.6
CyHex	–	–	–	–	–	0.4
3MAA	–	–	–	–	–	0.4
Aniline	93.1	–	–	–	6	0.1028

① Add TAM in conical
② Add TenAH
③ Add diox → yellow → sonic
④ Add cyclohex → sonic
⑤ transfer to conical
⑥ Add Aniline
⑦ Add AA
⑧ heat@55°C - 3d.

* AQH-1-111-2 w- ~~for~~
⑨ centrifuge product
(No filteration)

⑩ Solvent evaporate
⑪ take product.

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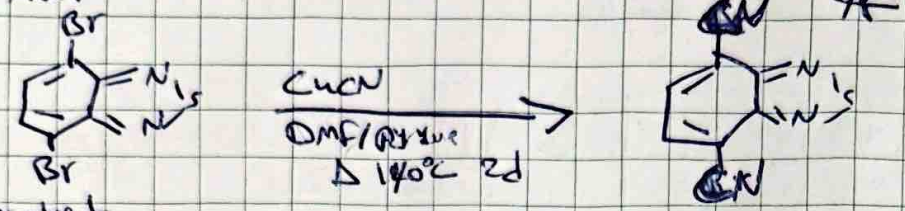
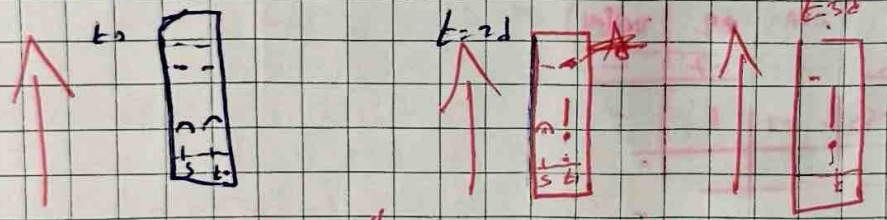
ARH-1-112



Reactant

chemical	fw g/mol	(mg/mg) mass	mmol	mM	eq	vol(mL)
Reactant	293.7	69	0.235	235	1	—
CuCN	89.56	84	0.94	940	4	
Pyridine	79.1	46.5	0.588	588	2	0.047
DMF	—	—	—	—	—	1

① in 4 5mL RBF add Reactant
② Add ~~a~~ CuCN + stirbar
③ Vac/N2 x3 (Schlenk)
④ ~~Add~~ freeze pump-thaw x3 Pyridine/DMF
⑤ Add DMF/Pyridine to RBF
⑥ heat @ 140°C



after 2d rxn turns dark brown.

40% EtOAc/hex

40% EtOAc/hex

⑦ rxn ~~stopped~~ quenched by ~~adding~~ dumping rxn in (NH4OH)/Toluene 1:1 mixture => stir for 20 mins ⟶ turns green
⑧ filter over celite
⑨ extract x3 water, ~~yellow~~ yellow organic brown aqass
⑩ " x3 brine ⟶ " " / colorless aq

Continued on Page 114

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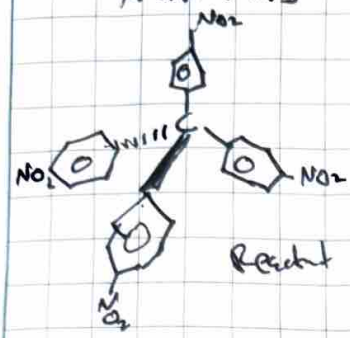
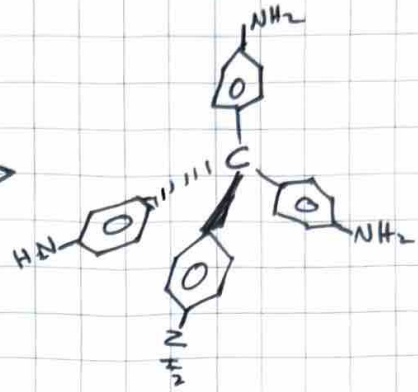
ARH-1-113



Reactant

$H_2NNH_2 \cdot H_2O$, Pd/C 5%
EtOH Δ 85°C 8h



Chemical	fw(g)	mass(mg)	Mmol	mM	eq	vol(ml)
Reactant	500	50	0.1	67	1	—
$H_2NNH_2 \cdot H_2O$	50	200	4	2.7	40	~~200~~ 0.206 *
Pd/C (5%)	–	20	–	–	–	–
EtOH	45	–	–	–	–	1.5

① in a 5ml RBF Add R
② Add Pd/C (5%)
③ Add EtOH → black.
④ Add $(NH_2-NH_2) \cdot H_2O$ ~~*~~
⑤ heat @ 85°C ~~(8h)~~ overnight
⑥ rxn was filtered

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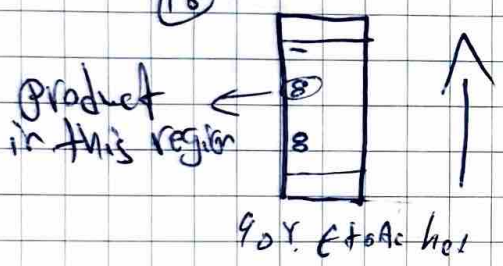
Continued from Page 11&2 AP

(12) Add Nacl sol to organic layer
(13) rotavap → orange solids
(14) crude NMR ⇒ product present (slight impurities)
(15) recrystalize from EtOAc/hexane solvent pair
(15) take NMR of residue ⇒ not product
(16) rotavap filtrate →
(17) run column (packed with 25% EtOAc/hex ran with 40% EtOAc/hex
(17) collect fractions
(18)



(19) NMR ran for every fraction ⇒ product present in region shown in (18)

(20) collect product fractions → re run column (packed with 5% EtOAc/hex ran with 20% EtOAc/hex)
(21) 1st fraction was collected → rotavap (orange ppt) NMR →
(22) 2nd " " " → " (orange ppt) NMR →

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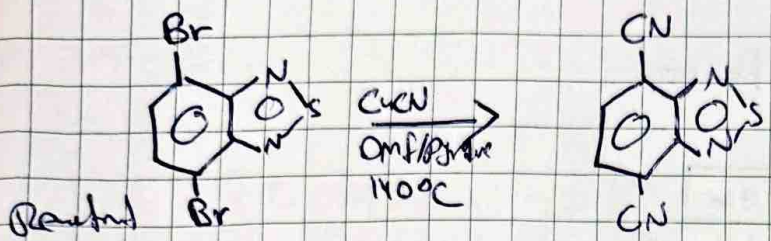
ARH-1-115



Reactant

Product

chemical	fw(g)	mass(g)	mmol	mM	eq	vol(ml)
Reactant	293.74	0.138	0.470	235	1	—
CuCN	89.56	168	1.88	940	4	—
DMF	—	2	—	—	—	2
pyridine	79.1	93	1.176	5.88	2	0.094
product		—			—	—

① in a 5ml RBF add Reactant
② Add CuCN
③ Add DMF/pyridine mixture (from a 3% ↓ heat org. base sol)
④ heat @ 140°C
⑤ rxn had a leak
- solvent dried - Not air free

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ARH-1-116-1

TerAld

+ TAM ⟶ COF-300

chemical	fw g/mol	mass mg	mmol	mM	vol/drop	eq
TerAld	140	12	0.0857		—	1.7
TAM	380	20	0.053		—	1
Dioxane	—	—	—		1.6	—
Cyclohex	—	—	—		0.4	—
3MAA	—	—	—		20	

① Add TAM to 20 mL vial
② Add TerAld
③ Add Diox → sonic → yellow
④ Add Cyclohex → sonic
⑤ transfer to vacuum vial
⑥ Add 3MAA → ppt
⑦ heat @ 85°C 3d

ARH-1-116-2

+ TAM ⟶ COF-300

chemical	fw g/mol	mass mg	mmol	mM	vol/drop	eq
TerAld	140	12	0.0857	8.45		1.7
TAM	380	20	0.053	28.21		1
Dioxane	—	—	—	—	1.6	—
Cyclohex	—	—	—	—	0.4	—
3MAA	—	—	—	—	0.1	—
Aniline	93.1	28 drops	0.307	neg	0.020	6

① Add TAM to 20 mL vial
② Add TerAld
③ Add dioxane → sonic → yellow
④ Add cyclohex → sonic
⑤ transfer to vacuum vial
⑥ Add Aniline
⑦ Add 3MAA
⑧ heat @ 85°C → 3d.

① yellow ppt (small amount) crystalline

⑩ DOX ×5 (Naftifine)

5 mg

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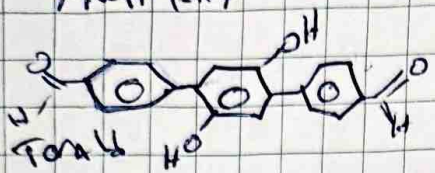
ARH-1-117



+ TAM → Expanded COF-300(OH)$_2$

chemical	Fw(g)	Mass(mg)	mmol	mM	eq	vol(ml)
TerAld	318	28	0.0089	40.16	1.7	—
TAM	380	20	0.0052	23.6	1	—
Diox	—	—	—	—	—	1.6
cyclohex	—	—	—	—	—	0.4
6MAA	—	—	—	—	—	0.2
Aniline	93.9	—	—	—	—	0.08

① Add TAM in 20 ml vial
② Add Ter Ald (DCF-JT-153)
③ Add Diox → sonic
④ Add cyclohex → sonic
⑤ transfer to vial vial
⑥ Add Aniline
⑦ Add 6M AA
⑧ heat @ 65°C → 3d

⑨ after 3d rxn solution turned dark brown and yellow ppt formed
⑩ ⑫ rxn solution was not fluorescent.
⑪ centrifuge x3 Acetone

(4mg)

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5/30/2024

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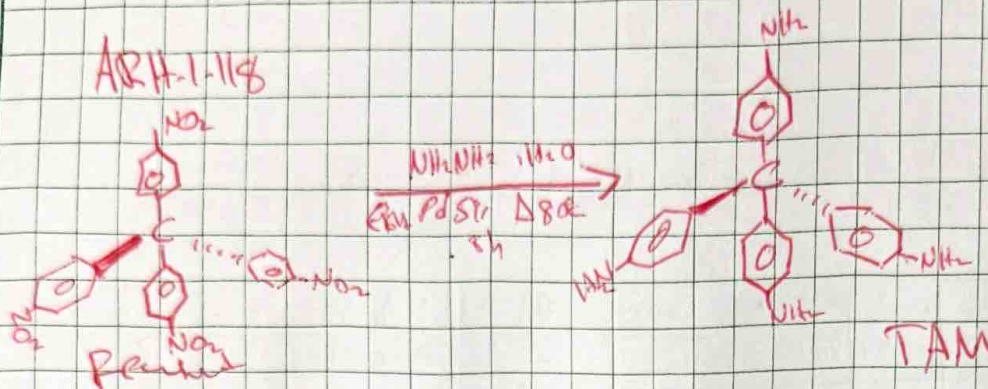
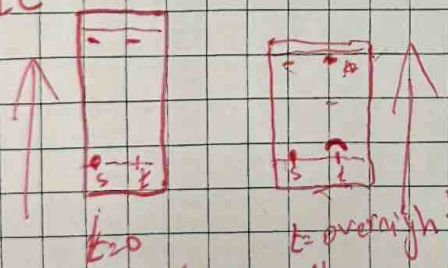
ARH-1-118



Chemical	Fw(g)	Mass(mg)	mmol	mM	eq	Vol(mL)
Reactant	500	50	0.1	67	1	–
H_2NNH_2	50	200	4	2680	40	0.200
EtOH	–	–	–	–	–	1.5
Pd/C (10%)	2 –	5%	–	–	20% Pd	
TAM	240	24				

① In a 5mL RBF Add Reactant
② Add hydrazine
③ Add EtOH
④ Add Pd/C (10%) runs black.

TLC



50% EtOAc/Hex

⑥ Rxn was HOT filtered over celite
⑦ Hot ethanol was used to rinse flask and filter funnel
⑧ ~~5~~ Hot THF was added to wash any residue on filter.

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* (a) filtrate was turbid beige color
(10) sample was rotavapped
(11) NMR shows pure product (beige brown). 64%

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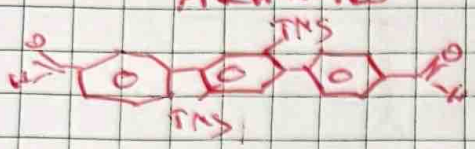
ARH-1-100



+ TAM → expanded COF-300 $(TMS)_{x2}$

chemical	FW(g)	mass(mg)	mmol	mM	eq	vol(mL)
TerAld	427	34	0.089	40.45	1.7	–
TAM	800	20	0.052	23.8	1	–
Diox	–	–	–	–	–	1.6
cyclohex	–	–	–	–	–	0.4
Aniline	93.1	28.012	0.312	142.7	6	0.025
6MAA	–	[illegible]	–	–	–	0.2

A

① in a 20mL vial Add TAM
② Add TerAld
③ Add Diox → sonic (transfer to vial)
④ Add cyclohexane → sonic
⑤ transfer to vial
⑥ Add Aniline
⑦ Add 6MAA
⑧ heat @ 65°C 3d

⑨ After 3d vial had a milky suspension (solid was in suspension not ppt).

⑩ hot centrifuge with Diox x3 → ppt from @ bottom of centrifuge tube → resuspended → centri x3.

⑪ XRD confirms crystallinity

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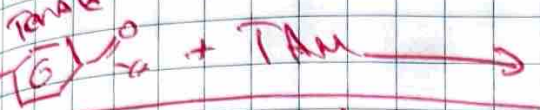
ARH-1-124
Terald



chemical	Fw(g)	mass(mg)	mmol	mM	eq	vol(ml)
Terald	140	120	0.86	34.3	1.17	—
TAM		200	0.95	29.3	1	—
Dtex		46.68		—	—	1.6
Cyclohex		40.4		—	—	4.44
3MAA		40.4		—	—	4.44

① in a 50ml Jar
Add TAM
② Add Terald
③ ~~Add Cyclohex~~ Add Dtex
→ sonic (5 min)
④ Add Cyclohex → sonic (5 min)
⑤ Add 3MAA → ppt forms.
⑥ Heat @ 65°C
3d.

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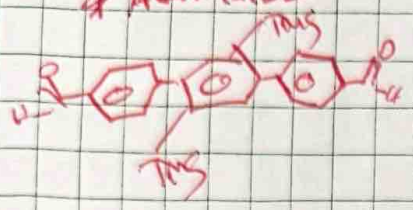
AcH-I-122-1



→ AM → expanded COF-300 (TMS)$_2$

Chemical Formula	Fw	Missing	mmol	mM	eq	vol(ml)

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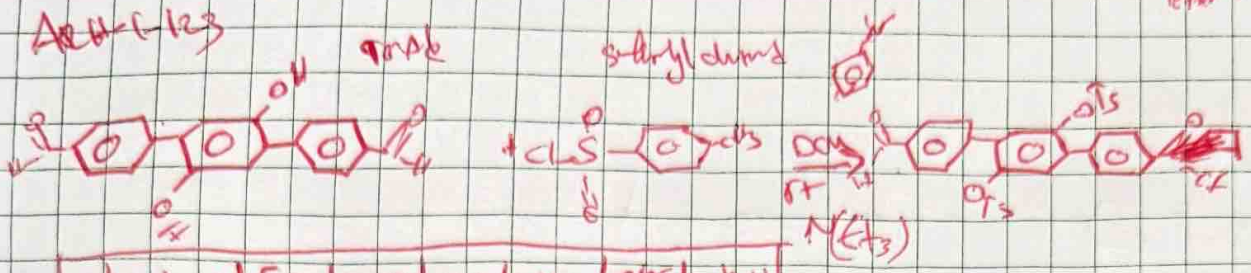
DOI: 10.1021/acs.joc.3c03130

ARH-I-123

TerAld

p-Tolyl chloride



chemical	Fw	mass/g	mmol	mM	eqv	vol(ml)
TerAld	314	30	0.0943	94.3	1	–
Sulfonyl chloride	190.65	54	0.2834	283.4	3	–
CHCl3	–	–	–	–	–	1
DMAP	122	1.15	0.0094	9.43	0.1	
Et3N	101.19	29	0.2834	283.4	3	0.040

i) 1 gram of p-TsCl dissolved in 2.5 ml CHCl3 (nasty smell)

ii) Add 13.5 * of hexane → turbid

iii) filter

iv) evaporate filterate. white solid forms on flask

② in a 5ml RBF Add TerAld, DMAP, Tosylchloride

③ Add CHCl3 → yellow suspension

④ Add stirbar + Et3N

TLC

Left: EtOAc/Hex

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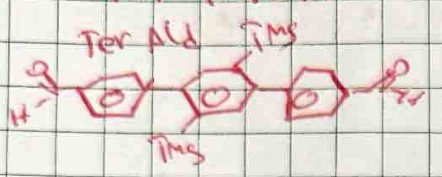
ARH-1-124-1



+ TAM → COF-330(TMS)x2

Chemical	fw(g)	mass(g)	mmol	mM	eq	vol(ml)
TerAld	427	38	0.089	40.45	1.7	–
TAM	380	20	0.052	23.8	1	–
Diox	–	–	–	–	–	1.6
DMA	–	–	–	–	–	0.4
6MAA	–	–	–	–	–	0.2
Aniline	93.9	28.6	0.312	142.7	6	0.028

① Add TAM in a 4ml vial
② Add TerAld
③ Add Diox → sonic
④ Add DMA → sonic
⑤ degas for 10 min
⑥ Add Aniline → shake
⑦ Add 6MAA → shake
⑧ Heat @ 65°C 3d

TerAld(TMS)2 + TAM → COF-330(TMS)2

ARH-1-124-2

Chemical	fw(g)	mass(g)	mmol	mM	eq	vol(ml)
TerAld	427	38	0.089	40.45	1.7	–
TAM	380	20	0.052	23.8	1	–
Diox	–	–	–	–	–	1.6
~~Mesitylene~~ DMA		–	–	–	–	0.4
6MAA	–	–	–	–	–	0.2
Aniline	93.9	28.6	0.312	142.7	6	0.028

① Add TAM in a 4ml
② Add TerAld
③ Add Dox → sonic
④ Add Mesitylene → sonic
⑤ degas for 10 min
⑥ Add Aniline → shake
⑦ Add 6MAA shake
⑧ Heat @ 65°C 3d

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ARH-4-125-1

TerAH(TMS)$_2$ + TAM ⟶ COF-330(TMS)$_2$

Chemical	fw(g)	Mass(g)	mmol	mM	eq	vol(mL)
TerAH(TMS)$_2$	427	38	0.089	4045	1.7	—
TAM	380	20	0.0527	238	1	—
Diox	—	—	—	—	—	1.6
Mesitylene	—	—	—	—	—	0.8
DMA	—	—	—	—	—	0.8
6MAA	—	—	—	—	—	0.2
Aniline	93.13	28.842	0.3132	1422	6	0.029

① Add TAM in 4mL vial
② Add TerAH(TMS)$_2$
③ Add Diox → sonic
④ Add DMA → sonic
⑤ Add Mesitylene → sonic
⑥ transfer to vacuum
⑦ Add Aniline
⑧ Add 6MAA
⑨ heat @ 65°C 3d.

~~TerAH(TMS)$_2$~~ ARH-125-2

TerAH +TAM ⟶ COF-330(TMS)$_2$
(TMS)$_2$

Chemical	fw(g)	Mass(g)	mmol	mM	eq	vol(mL)
TerAH(TMS)$_2$	427	38	0.089	4045	1.7	—
TAM	380	20	0.0527	238	1	—
Diox	—	—	—	—	—	1.6
chlorobenzene	—	—	—	—	—	0.4
6MAA	—	—	—	—	—	0.2
Aniline	93.13	28.842	—	—	6	0.029

① Add TAM in 4mL vial
② Add TerAH(TMS)$_2$
③ Add Diox → sonic
④ Add ~~DMA~~ → sonic
⑤ Add chlorobenzene ~~Aniline~~ transfer to vacuum
⑥ Add Aniline
⑦ Add 6MAA
⑧ ~~Add~~ heat @ 65°C 3d.

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ARH-1-126-1

TerAld(OH)$_2$ + TAM → COF-330(OH)$_2$

chemical	fw(g)	mass(mg)	mmol	mM	eq	vol(ml)
TerAld(OH)$_2$	318	30	0.0881	40.45	1.7	
TAM	380	20	0.052	23.8	1	
Diox	←	1.6	—	—	—	1.6
DMF	—	0	—	—		0.4
6MAA	—	—	—	—	—	0.2
Aniline	93	28.012	0.1312	1427	6	0.1028

① Add TerAld in a vial
② Add TAM
③ Add DMF → some → red brown → yellow
④ Add Diox → sonic → red brown
⑤ transfer to vial
⑥ Add Aniline
⑦ Add 6MAA.
⑧ bubble N_2

@ yellow mustard ppt formed @ bottom

ARH-1-126-2

TMS, TMS + TAM → COF-330(TMS)$_2$

Chemical	fw(g)	mass(mg)	mmol	mM	eq	vol(ml)
TerAld	427	35	0.0819	40.45	1.7	—
TAM	380	20	0.052	23.8	1	—
Diox	—	—	—	—	—	1.6
DMF	—	—	—	—	—	0.4
6MAA	—	—	—	—	—	0.2
Aniline	93	28.012	0.312	1427	6	0.028

① Add TerAld
② Add TAM
③ Add DMF → sonic → turbid
④ Add Diox → sonic → turbid
⑤ transfer to vial
⑥ Add Aniline
⑦ Add 6MAA → pink color with slight suspension
⑧ bubble N_2 → dull beige.

@ white ppt in bottom supernatant turned from pink

- good crystallinity ⇒ can be imaged

16 mg obtained ⇒ 30% yield

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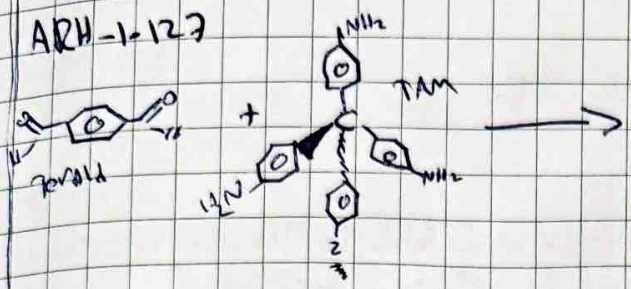
2:30

ARH-1-127



Chemical	fw(g)	mass(g)	mmol	mM	eq	vol(ml)
TerAld	140	12	0.085	85.7	1.7	–
TAM	380	20	0.05	5M	1	–
Diox	–	–	–	–	–	1.6
Cyclohex	–	–	–	–	–	0.4
3MAA	–	–	–	–	–	0.4

~~in vial~~ in a ~~40 mL jar~~ in vial

Add TAM

② Add TerAld

③ Add Diox sonic → yellow

④ Add cychex sonic → yellow

⑤ transfer to vial

⑥ Add 3MAA → yellow ppt

⑦ heat @ 65°C (3d).

⑧ yellow ppt pink supernatant

⑨ transfered to a centrifuge tube x3 Dioxane.

⑩ didn't crystalize.

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4/23/2025

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Date

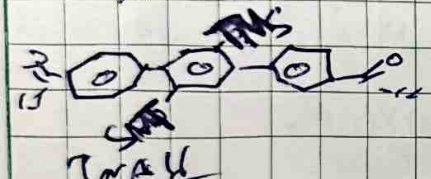
ARH-1-128-1

+ TAM → COF-330(TAM)2

chemical	FW(g)	Mass(mg)	mmol	mM	eq	Vol(ml)
TerAld	330	38	0.089		1.7	–
TAM	380	20	0.052		1	–
Diox	–	–				1.2
DMF	–	–				0.8
6MAA	–	–				0.2
Aniline	93.9	28.6	0.317	–	6	0.028

① Add TerAld to 4ml vial
② Add TAM
③ Add DMF
④ sonic → turbid
⑤ Add diox → sonic
⑥ transfer to vacuum
⑦ Add Aniline
⑧ Add 6MAA → yellow ppt
⑨ heat @ 65°C 3d
⑩ beige supernatant + ppt on bottom
⑪ Acceptable crystallinity

ARH-1-128-2



+TAM → COF-330(TAM)2

chemical	FW(g)	mass(mg)	mmol	mM	eq	Vol(ml)
TerAld	330	38	0.089		1.7	–
TAM	380	20	0.052		1	–
Diox	–	–	–	–	–	1.6
DMF	–	–	–	–	–	0.4
12MAA	–	–	–	–	–	0.1
Aniline	93.9	28.6	0.317	–	6	0.028

① Add TerAld to 4ml vial
② Add TAM
③ Add DMF → turbid
④ sonic →
⑤ Add diox → sonic → turbid (consistent ppt crash out)
⑥ transfer to vacuum
⑦ Add Aniline
⑧ Add 12MAA → slight ppt forms!
⑨ heat @ 65°C 3d.

Note: when putting in oven ⇒ ppt dissolves
⑩ dark beige supn + white ppt on bottom.
⑪ Low crys

185, yield

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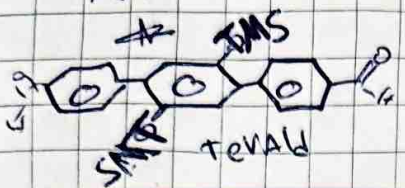
ARH-1-129-1



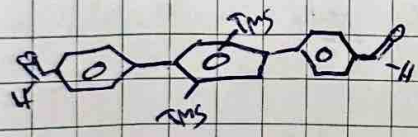
+TAM ⟶ cof-330(TMS)2

chemical	Fw(g)	mass(g)	mmol	mM	eq	vol(ml)
TerAld	422	38	0.1089		1.7	-
TAM	380	20	0.052		1	-
Diox	-	-	-		-	1.6
DMA	-	-	-		-	0.4
6MAA	-	-	-		-	0.2
Aniline	93.9	28.6	0.317		6	0.028

① Add terAld into vial
② Add TAM
③ Add DMA → sonic
④ Add Dioxane → sonic
⑤ transfer to vial
⑥ Add Aniline
⑦ Add 6MAA
⑧ heat @ 65°C 3d

106 mg cof ⑨ brown supernatant + white ppt => looks rough texture
good crystallinity

ARH-1-129-2



+TAM ⟶ cof-330(TMS)2

chemical	Fw(g)	mass(g)	mmol	mM	eq	vol(ml)
TerAld	422	38	0.1089		1.7	-
TAM	380	20	0.052		1	-
Diox	-	-	-		-	1.2
DMA	-	-	-		-	0.8
6MAA	-	-	-		-	0.2
Aniline	93.9	28.6	0.317		6	0.028

① Add TerAld into vial
② Add TAM
③ Add DMA → sonic
④ Add Dioxane → sonic
⑤ transfer to vial
⑥ Add Aniline
⑦ Add 6MAA
⑧ heat @ 65°C 3d

106 mg cof ⑨ brown supernatant + white ppt => looks rough texture.

Note - when placed in gun ppt dissolves

- Acceptable crystallinity

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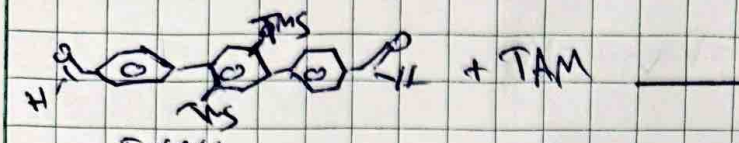
ABH-1-130-1



TerAld + TAM ⟶ COF-330 (TMS)$_2$

TerAld

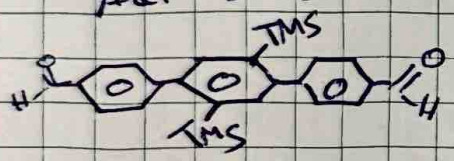
Chemical	Fw(g)	mass(mg)	mmol	mM	eq	vol(mL)
TerAld	427	38	0.089		1.2	—
TAM	380	20	0.052		1	—
Diox	—	—	—		—	1.6
DMA	—	—	—		—	0.4
6MAA	—	—	—		—	0.4
Aniline	93.9	28.6	0.317	-	~~excess~~	0.028

~~68.380 cof~~

① Add TerAld to 4mL vial
② Add TAM
③ Add DMA → sonic → suspension
④ Add Diox → sonic → suspension
⑤ transfer to vial
⑥ Add Aniline
⑦ Add 6MAA
⑧ heat @ 65°C 3d.
⑨ pink supernatant + grey solid

⑩ high crys

ABH-130-2



+ TAM ⟶ COF-330 (TMS)$_2$

Chemical	Fw(g)	mass(mg)	mmol	mM	eq	vol(mL)
TerAd	427	38	0.089		1.7	—
TAM	380	20	0.052		1	—
Diox	—	—	—		—	1.2
DMA	—	—	—		—	~~0.8~~
6MAA	—	—	—		—	0.4
Aniline	93.9	28.6	0.312		6	0.028

① Add TerAld to 4mL vial
② Add TAM
③ Add DMA → sonic
④ Add Diox → sonic
⑤ transfer to vial
⑥ Add Aniline
⑦ Add 6MAA
⑧ heat @ 65°C 3d.

⑨ high crys

Extra Peaks ≠

Note: when ~~ppt~~ put in oven ⇒ ppt dissolved
① Pink-brown supernatant + solid looks amorphous

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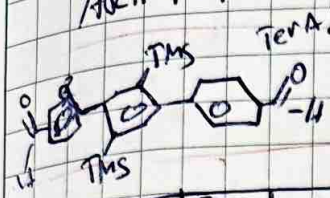
ARH-131-1

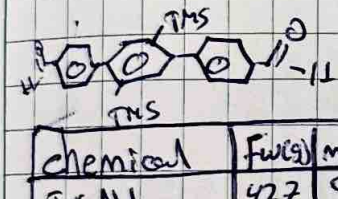


+ TAM → COF-330 (TMS)x2

chemical	fw(g)	mass(g)	mmol	mM	eq	vol (mL)
TerAld	427	38	0.089		1.7	—
TAM	230	20	0.052		1	—
Diox	—	—	—		—	1.6
DMA	—	—	—		—	0.4
12MAA	—	—	—		—	0.2
Aniline	93.13	28.6	0.317		6	0.028

① in a 4mL vial Add TerAld
② Add TAM
③ Add DMA → sonic
④ Add Diox
⑤ transfer to vial/vial
⑥ Add Aniline
⑦ Add 12MAA
⑧ heat @ 65°C 3d
⑨ crystallized (got better peaks in different condition). phase is the reported one.

ARH-1-131-2



+TAM → COF-330 (TMS)x2

chemical	Fw(g)	mass(g)	mmol	mM	eq	vol(mL)
Ter Ald	427	38	0.089		1.7	—
TAM	380	20	0.052		1	—
Diox	—	—	—		—	1.6
DMA	—	—	—		—	0.4
12MAA	—	—	—		—	0.4
Aniline	93.13	28.6	0.317		6	0.028

① in a 4ml vial Add TerAld
② Add TAM
③ Add DMA → sonic
④ Add Diox → sonic
⑤ transfer to vials
⑥ Add Aniline
⑦ Add 12MAA
⑧ heat @ 65°C 3d
⑨ crystallized (got better peaks in ARH-131-1)

phase is the reported one

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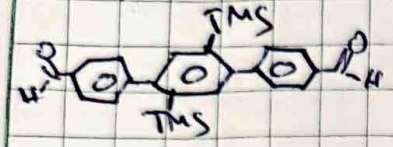
ARH-I-132-1



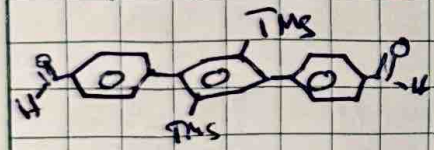
+TAM ⟶ COF-330 TMS (x2)

Chemical	fw(g)	Mass(mg)	mmol	mM	eq	Vol(mL)
TerAld	427	38	0.089		1.7	—
TAM	380	20	0.052		1	—
DMA	—	—	—		—	1.2
Diox	—	—	—		—	0.4
6MAA	—	—	—		—	0.2
Aniline	93.9	28.6	0.317		6	0.028

① Add TerAld in vial
② Add TAM
③ Add DMA → sonic
④ Add Diox → sonic
⑤ transfer to vial
⑥ Add Aniline
⑦ Add 6MAA
⑧ heat @ 65°C 3d
@ crystallized in reported phase

⑩ (ARH-I-130-1 gave better crys).

ARH-I-132-2



+TAM → COF-330 (TMS)x2

Chemical	fw(g)	Mass(mg)	mmol	mM	eq	Vol(mL)
TerAld	427	38	0.089		1.7	—
TAM	380	20	0.052		1	—
DMA	—	—	—		—	1.6
Diox	—	—	—		—	0.4
6MAA	—	—	—		—	0.2
Aniline	93.9	28.6	0.317		6	0.028

① Add TerAld in vial
② Add TAM
③ Add DMA → sonic
④ Add Diox → sonic
⑤ transfer to vial
⑥ Add Aniline
⑦ Add 6MAA
⑧ heat @ 65°C 3d.

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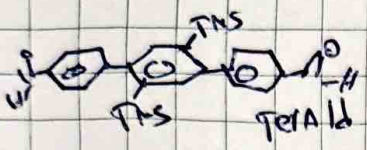
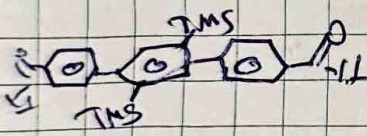


+ TAM → Cof-330(TMS)$_{1.2}$

Chemical	fw(g)	massing	mmol	mM	eq	vol(mL)
TerAld	427	38	0.089		1.7	—
TAM	380	20	0.052		1	—
~~AA~~ ~~DEF~~	—	—	—		—	~~1.2~~
Diox	—	—	—		—	~~0.6~~ 1.2
6MAA	—	—	—		—	0.12
Aniline	93.13	93.9	0.537		6	0.028

① Add Ter Ald in 4mL vial
② Add TAM
③ Add ~~DEF~~ → sonic
④ Add Diox → sonic
⑤ ~~Add~~ transfer to vial
⑥ Add ~~6MAA~~ ~~Aniline~~
⑦ Add 6MAA
⑧ heat @ 65°C 3d.

ARH-1-133-2



+ TAM → COF-330(TMS)$_{1.2}$

Chemical	fw(g)	massing	mmol	mM	eq	vol(mL)
Ter Ald	427	38	0.089		1.7	—
TAM	380	20	0.052		1	
DEF	—	—	—		—	~~1.2~~ 1.8
Diox	—	—	—		—	~~0.6~~ 1.8
6MAA	—	—	—		—	0.12
Aniline	93.9	93.9	0.537		6	0.028

① Add Ter Ald in 4mL vial
② Add TAM
③ Add DEF → sonic
④ Add Diox → sonic
⑤ transfer to vial with vial
⑥ Add Aniline
⑦ Add 6MAA
⑧ heat @ 65°C 3d.

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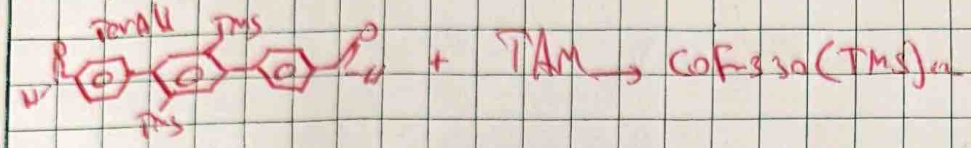
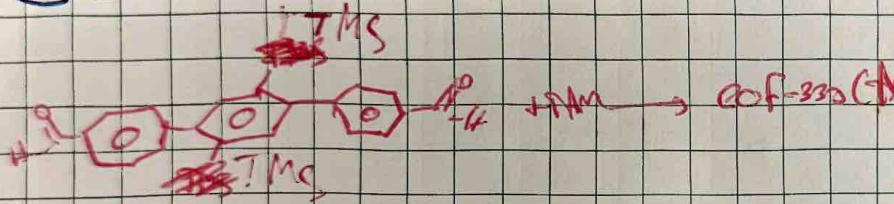
ARH-I-134-1



chemical	Fw(g)	mass(g)	mmol	mM	eq	vol(mL)
TerAld	424	38	0.089		1.7	—
TAM	350	20	0.052		1	—
DMA	—	—	—	—	—	1.6
Diox	—	—	—	—	—	0.4
Aniline	93.1	0.024	0.312	312	6	0.028
6MAA	—	0.4	—	—	—	0.1

① In a 4mL vial Add TerAld
② Add TAM
③ Add DMA → sonic → clear
④ Add Diox → sonic → clear
⑤ transfer to main vial (sonicate)
⑥ Add Aniline
⑦ Add 6MAA → dense ppt then
⑧ heat @ 65°C 3d

⑨ bad crystallinity.



chemical	Fw(g)	mass(g)	mmol	mM	eq	vol(mL)
TerAld	34	28	0.089	89	1.7	—
TAM	350	20	0.052	52	1	—
Diox	—	—	—	—	—	1.6
DMA	—	—	—	—	—	0.4
Aniline	93.1	2.896	0.312	—	6	0.028
6MAA	—	—	—	—	—	0.4

① in a 4mL vial Add TerAld
② Add TAM
③ Add DMA → sonic → clear
④ Add Diox → sonic → clear
⑤ transfer to main vial (sonicate)
⑥ Add Aniline
⑦ Add 6MAA → slight ppt then

⑧ solvent dry.
⑨ bad crystallinity.

⑩ Note :- upon transfer to main vials ⇒ vials are hot from oven ⇒ TMS) dissolves.

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ARH-1-135-1

Dioxane

chemical	fw g	massing	mmol	mM	eq	vol (mL)
TerAld	314	314	1.0 mmol		2	
TAM	380	19	0.1 mmol		1	
Diox	—	—	—	—	—	0.5
Mesitylene	—	—	—	—	—	0.5
6MAA	—	—	—	—	—	0.1

① in a borosilicate tube add TAM
② Add TerAld
③ Add Diox → sonic → cloudy Turbid Yellow
④ Add Mesitylene → sonic
⑤ Add 6MAA
⑥ Freeze → pull vac ⑦ Seal ⑧ Heat @ 120°C 3d

chemical	fw g	massing	mmol	mM	eq	vol (mL)
TerAld	314	31.4	0.1 mmol		2	
TAM	380	19	0.05 mmol		1	
Diox	—	—	—	—	—	0.5
DMA	—	—	—	—	—	0.5
6MAA	—	—	—	—	—	0.1

① in a borosilicate tube add TAM
② Add TerAld
③ Add Diox → sonic → Turbid Yellow
④ Add Mesitylene → sonic
⑤ Add 6MAA
⑥ Freeze → pull vac ⑦ Seal ⑧ Heat @ 120°C 3d.

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ARLH-1-136-1

+ TAM ⟶ COF-300

Chemical	Fw(g)	Mass(mg)	mol	mM	eq	vol(mL)
TerAld	140	12	.089	37	1.7	–
TAM	380	20	.052	21.6	1	–
Diox	–	–	–	–	–	1.6
cyclohex	–	–	–	–	–	0.4
3MAA	–	–	–	–	–	0.4

① in a vial Add TerAld
② Add TAM
③ Add Diox → sonic → yellow sol
④ Add ~~DMA~~ cyclohex → sonic → yellow sol
⑤ ~~Add 3MAA~~ transfer to main vial
⑥ Add 3MAA → yellow ppt
⑦ heat @ 65°C 3d

ARLH-136-2 ⑧ not crystalline

Chemical	Fw(g)	mass(mg)	mol	mM	eq	vol(mL)
TerAld	140	12	0.089	36.7	1.7	–
TAM	380	20	0.052	21.6	1	–
Diox	–	–	–	–	–	1.6
DMA	–	–	–	–	–	0.4
~~Aniline~~	93.1	24.6	0.312	–	–	0.028
6MAA	–	–	–	129.6	6	0.4

① in a vial Add TerAld
② Add TAM
③ Add Diox → sonic → yellow sol
④ Add cyclohex → sonic → yellow sol
⑤ transfer to main vial
⑥ Add Aniline
⑦ Add 6MAA
⑧ heat @ 65°C 3d.

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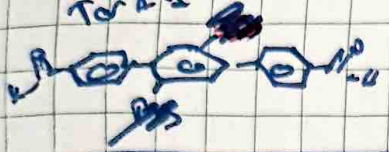
ARH-1-137-1

Ter All



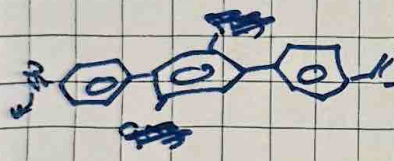
+TAM → ~~COF-330(Me)~~ COF-330 (Me)$_{x2}$ ☆

chemical	FW(g)	mass(mg)	mmol	mM	eq	vol(ml)
Ter All	~~424~~ 380	~~38~~ 38	0.089		1.7	–
TAM	380	20	0.052		1	–
~~Dio~~ Dio	–	–	–		–	1.2
~~Dio~~ DMA	–	–	–		–	0.8
Aniline	93.1	28.6	0.312		6	0.078
6MAA	–				–	0.4

① in a 4ml vial Add Ter All
② Add TAM
③ Add DMA → sonic → yellow
④ Add Dioxane → sonic yellow
⑤ ~~Add~~ transfer to vanier
⑥ Add Aniline
⑦ Add 6MAA → No ppt
⑧ heat @ 65°C 3d

☆ - did not give ppt → leave for a ppt is found

ARH-1-137-2



+TAM → COF-330(Me)$_{x2}$ ☆

chemical	FW(g)	mass(mg)	mmol	mM	eq	vol(ml)
Ter All	422	~~38~~ 28	0.089		1.7	–
TAM	380	20	0.052	–	1	–
~~Dio~~ Dio	–	–	–	–	–	1.6
~~Dio~~ DMA	–	–	–	–		0.4
Aniline	93.1	28.6	0.312		6	0.078
6MAA	–	–	–	–	–	0.4

① in a 4ml vial Add Ter All ☆
② Add TAM
③ Add DMA → sonic → yellow
④ Add Dio x → sonic
⑤ transfer to vanier
⑥ Add Aniline
⑦ Add 6MAA → No ppt hrs
⑧ heat @ 65°C 3d.

☆ - did not give ppt → leave for a ppt is found

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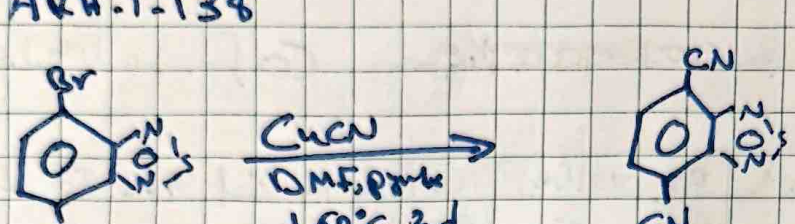
ARH.1.138



Reactions

chemical	FwGm	mass/vol	mol	mM	eq	vol(mL)
Reactant	293.7	138	0.476	235	1	–
CuCN	89.56	168	1.475	940	4	–
DMF	–	–	–	–	–	2
Pyridine	79.1	93	1.176	588	2	0.094

① In a 5ml RBF Add Reactant

② Add CuCN

③ ~~Add DMF~~ prepare DMF/pyridine mixture

④ freeze pump thaw x3

⑤ Add 2ml RBF in → yellow solution.

⑥ rxn after 1d ⇒ no sensor → turn on heat plate cyan

TLC

t=0

U.V

40% EtOAc/hex

t=1d

U.V

very light

40% EtOAc/hex

re-stain

Time

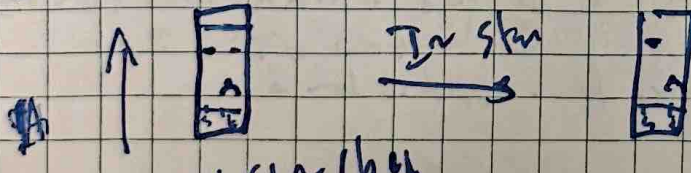
slight

medium

heavy dark

lower rxn for 3h

t=3d



25% EtOAc/hex

(stain not visible under UV ⇒ stain for purity).

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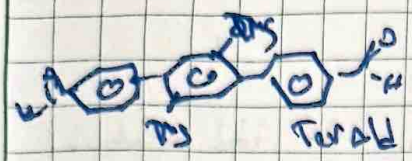
ARH-1-139 -1



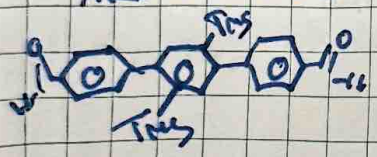
+ TAM ⟶ COF-330(TMS)x2

chemical	fw(g)	mass(mg)	mmol	mM	eq	vol(ml)
TerAld	422	38	0.09		1.2	—
TAM	380	20	0.052	-	1	-
DiOX	-	-	-	-		1.6
DMA	-	-	-	-		0.4
6MAA	-	-	-	-		0.4

① in a 4ml vial add TerAld
② Add TAM
③ Add DMA → sonic
④ Add Dioxane → sonic
⑤ transfer to vial with
⑥ Add 6MAA
⑦ heat @ 65°C 3d
⑧ ~~g~~ solvt dry.

⑧ solvt dry. ⑩ bad crys.

ARH-1-139 -2



+ TAM ⟶ COF-330(TMS)x2

chemical	fw(g)	mass(mg)	mmol	mM	eq	vol(ml)
TerAld	422	45	0.105	-	2	—
TAM	380	20	0.052	-	1	-
Dioxane	—	—	—	—	—	1.6
DMA	—	—	—	—	—	0.4
6MAA	—	—	—	—	—	0.4

① In a 4ml vial add TerAld
② Add TAM
③ Add DMA → sonic (cloudy)
④ Add Dioxane → sonic (cloudy)
⑤ transfer to vial vial
⑥ Add 6MAA
⑦ heat @ 65°C 3d.
⑧ g, 1 solvt dry.

⑧ moderate crystallinity.
⑩

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5/5/2016

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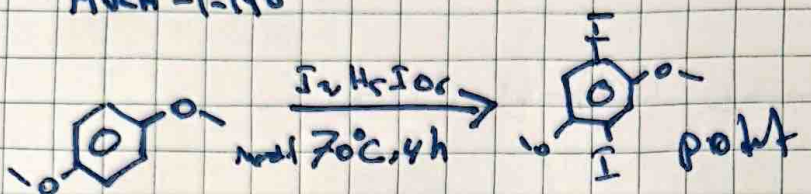
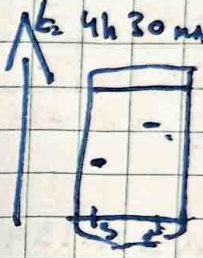
ARH-1-140



chemical	fw (g)	massing	mmol	mM	eq	vol (mL)
Reactant	138	600	4348	86960	1	—
I_2	253.8	1544	6.08	12217.4	1.4	—
H_5IO_6	227.9	594	2.606	521.76	0.6	—
product	344.8	1704	4.348	8696	1	—
MeOH	—	—	—	—	—	5

① in a 20ml RBF Add Reactant
② Add I_2
③ Add H_5IO_6 → powder dry to ensure over time
④ Add MeOH → bad homogenous solution
⑤ heat for 70°C

TLC.



~~Product~~ - quench the rxn with $Na_2S_2O_3$ → turns product white solid forms).
- Add 1M KOH
- filter → pale ~~pink crys~~ orange crystals
- wash with H_2O and MeOH.
- take NMR. → looks good (slight impurity)

extract with EtOAc and Water

- take NMR.

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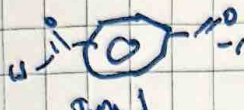
ARH-1-141



+ TAM → COF-300

Terald

chemical	[mg]	mass [mg]	mmol	mM	eq	vol [mL]
Terald	140	12			1.2	
TAM	380	20			1	
Diox	–	–	–	–	–	1.6
cyclohex	–	–	–	–	–	0.4
3MAA	–	–	–	–	–	0.1

① Add TAM to 4mL vial
② Add Terald
③ Add 6 Diox → sonic → yellow
④ Add cyclohex → sonic → yellow
⑤ transfer to vial vial
⑥ ~~Add~~ Add 3MAA → PPT forms
⑦ heat @ 65°C 3d

+ TAM → COF 300

chemical	mass [mg]	mmol	mM	eq	Vol [mL]	[mg]
Terald	12			1.2	~~[illegible]~~	140
TAM	20			1	~~[illegible]~~	380
Diox	–	–	–	–	1.6	–
DMA	–	–	–	–	0.4	–
~~6~~ 6MAA	–	–	–	–	0.4	–

① Add TAM to 4ml vial
② Add TerAld
③ Add Diox → sonic → yellow
④ Add DMA → sonic → yellow
⑤ transfer to vial vial
⑥ Add 6MAA
⑦ heat @ 65°C 3d

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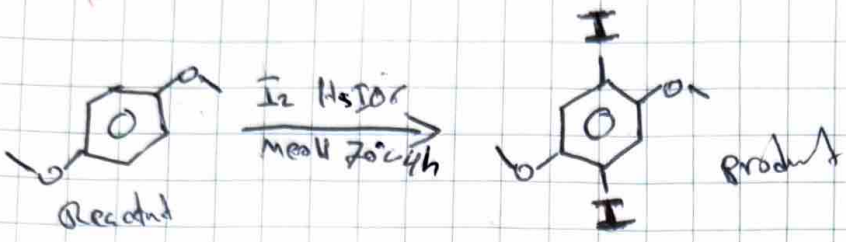
ARH-1-142



chemical	Fw(g)	mass(mg)	mmol	mM	eq	Vol(ml)
Reactant	138	600	4.348	869.6	1	—
I_2	253.8	1544	6.08	1217.4	1.4	—
$HsIO_6$	227.9	594	2.606	521.76	0.6	—
Product	391.9	1704	4348	869.6	1	—
MeOH	—	—	—	—	—	5

① in a 20 ml RBF Add Reactnt
② Add I_2
③ Add $HsIO_6$
④ Add MeOH → brown solution
⑤ heat for 70°C 4h

TLC

- quench rxn with 2 ml of sat. $NaHSO_3$ → white ppt forms + lighter yellow
- Add 2ml 1M KOH
- filter → pale pink powder **93.3%**
- high vac
- take NMR $CDCl_3$ ~~good~~ pure

$HsIO_6$ MSDS:

- white solid - MP 122°C - oxidat, irritat, corrosive
- avoid contact with organic materials strong oxidizing agents, strong bases, Reducing agents, combustible material

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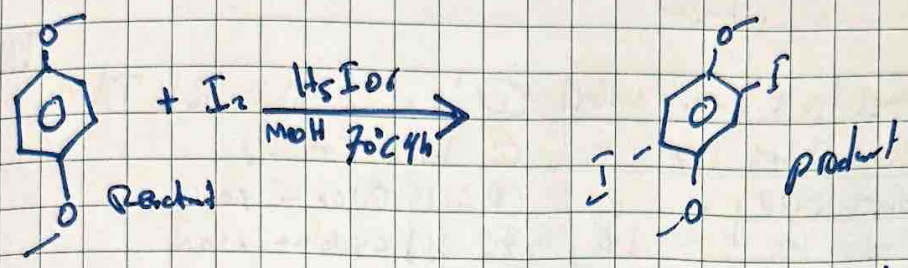
ARH-143



chemical	fw(g)	mass(mg)	mmol	mM	vol(ml)	eq
Reactant	138	3000	21.74	869.6	-	+
I_2	253.8	7.720	30.4	1217.4	-	1.4
H_5IO_6	227.9	2.970	13.03	521.76	-	0.6
product	391.8	8520	21.74	869.6	-	+
MeOH	-	-	-	-	-	24

① in a 100 mL RBF Add Reactant
② Add I_2
③ Add H_5IO_6 → purple suspension
④ ~~MeOH~~ Add MeOH → brown color
⑤ heat ~~for~~ @ 78C 5h.
⑥ quench rxn with $NaHSO_3$

→ white ppt forms + liquid becomes yellow
→ Add 2mL 1M KOH
- filter → pale pink powder
- high vac
- nmr $CDCl_3$ → pure

97% yield.

I_2 MSDS.

- grey solid
- pungent odor
- 113°C MP / 185°C B.P
- Avoid contact with strong oxidizing agents, finely powdered metals (Al, Zn, Rb, copper
- Causes irritation
- wash with soap if exposed.

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4/21/2026

ARH-1-144-1

+TAM ⟶ COF-300

TAM 77

chemical	Fw(g)	mass(mg)	mmol	mM	eq	vol(mL)
Ter Ald	134	12	0.1089	37.08	1.7	-
TAM	380	20	0.052	21.67	1	-
Diox	-	-	-	-	-	1.6
Cyclohex	-	-	-	-	-	0.4
3MAA	-	-	-	-	-	0.4

① in a 4mL vial Add TAM
② Add Ter Ald
③ Add Diox → sonic
④ Add cyclohex → sonic
⑤ transfer to vacuum
⑥ Add 3MAA
⑦ heat @ 65°C 4d

Diffracted ✓ (moderate xtls)

ARH-1-144-2

+TAM ⟶ COF-300

chemical	Fw(g)	mass(mg)	mmol	mM	eq	vol(mL)
Ter Ald	134	12	0.1089	37.08	1.7	-
TAM	380	20	0.052	21.67	1	-
Diox	-	-	-	-	-	1.6
DMA	-	-	-	-	-	0.4
3MAA	-	-	-	-	-	0.4

① in a 4mL vial Add TAM
② Add Ter Ald
③ Add Diox → sonic
④ Add cyclohex → sonic
⑤ transfer to vacuum
⑥ Add 3MAA
⑦ heat @ 65°C 4d.

Diffracted (moderate xtls).

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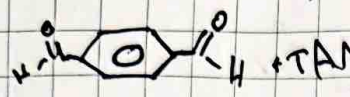
ARH-1-145 -1

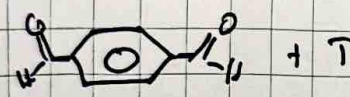


Chemical	Fw(g)	mass(mg)	mmol	mM	eq	vol(ml)
TerAld	134	12	0.0899	37.08	1.7	–
TAM	380	20	0.0526	21.67	1	–
Diox	–	–	–	–		2
3MAA	–	–	–	–		0.4

① in a 4ml vial Add TAM
② Add TerAld
③ Add Diox → sonic
④ ~~Add 3MAA~~ transfer to vacuum
⑤ Add 3MAA → yellow ppt long
⑥ heat @ 65°C → 3d

– Amorphous

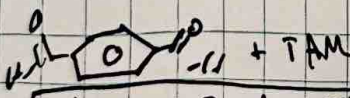
ARH-1-145 -2



Chemical	Fw(g)	mass(mg)	mmol	mM	eq	vol(ml)
TerAld	134	12	0.0899	37.08	1.7	–
TAM	380	20	0.0526	21.67	1	–
Diox	–	–	–	–	–	1.6
Mesitylene	–	–	–	–	–	0.4
3MAA	–	–	–	–	–	–

① An empty vial Add TAM
② Add TerAld
③ Add Diox → Sonic
④ Add Mesitylene → Sonic
⑤ transfer to vacuum
⑥ Add 3MAA → yellow ppt long
⑦ heat @ 65°C 3d →

ARH-1-145 -3



Chemical	Fw(g)	mass(mg)	mmol	mM	vol(ml)	eq
TerAld	134	24	0.179		–	2
TAM	380	20	0.052		–	1
Diox	–	–	–	–	1.6	–
cyclohex	–	–	–	–	0.4	–
3MAA	–	–	–	–	0.4	–

① in a 4ml vial Add TAM
② Add TerAld
③ Add Diox → sonic
④ Add cyclohex → sonic
⑤ ~~Add~~ transfer to vacuum
⑥ Add 3MAA → yellow ppt long
⑦ heat @ 65°C 3d

⑧ Amorphous

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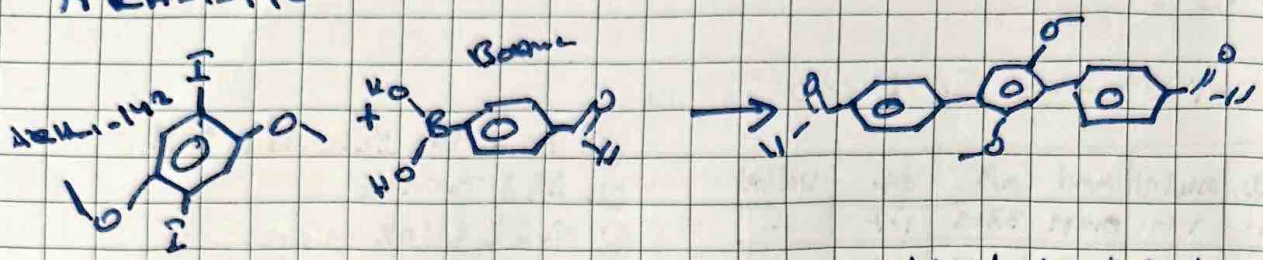
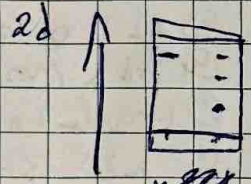
ARH-1-146



chemical	fw g	mmol	mM	eq	volume	mg used
ARH-1-142	391.8	0.766	67.8	1	-	300
Boronic	149	2.14	189	2.79	-	320
Pd(dppf)Cl2	800	0.14	10.848	0.16	-	112
K2CO3	138	2.56	203	3		353
Dioxane	-				9	
Water	-	-	-	-	2.3	-

① Add ARH-1-142 in 25mL RBF
② Add Boronic
③ Add Pd(dppf)Cl2 → Vac→N2 x3
④ Freeze pump thaw Dioxane -Water mix x3
⑤ Add Dioxane water mix to RBF → orange solution
⑥ heat @ 90°C
⑦ after 2h → rxn homogenous black.

TLC



⑧ quench with 1M HCl → bubbles
⑨ extract with 10mL x3 DCM
⑩ filter over celite → brown solution (filtrate)
⑪ rotavap → black gray solid
⑫ dissolve in DCM →
⑬ run through silica plug → yellow soln + black residue on top of silica (Silica packed with hexanes and mobile phase was DCM).
⑭ rotavap → yellow solid → NMR aromatic region pure, but it is not pure on alkane region.

20% yield

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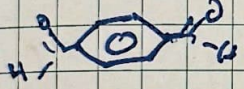
ARH-1-147-1

+ TAM → COF-300

chemical	fw g	mass(mg)	mmol	mM	eq	vol(ml)
Ter Ald	134	10	0.0746	30.98	1.43	–
TAM	380	20	0.0526	21.67	1	–
Diox	–	–	–	–	–	1.6
cyclohex	–	–	–	–	–	0.4
3MAA	–	–	–	–	–	0.4

① In a 4ml vial Add TAM
② Add Ter Ald
③ Add Diox → sonic → orange ppt top sol
④ Add cyclohexanone
⑤ transfer to vacuum vial
⑥ Add 3MAA
⑦ heat ⑤ 85°C 3d.

ARH-1-147-2

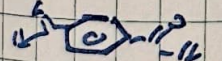


+TAM → COF-300

chemical	fw g	mass(g)	mmol	mM	eq	vol(ml)
TerAld	134	12	0.089	37.08	2	–
TAM	380	17	0.079	18.51	1	–
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Dioxane	–	–	–	–	–	1.6
cyclohex	–	–	–	–	–	0.4
3MAA	–	–	–	–	–	0.7

① in a 4ml vial Add TAM
② Add Ter Ald
③ Add Diox → sonic
④ Add cyclohexanone
⑤ transfer to vacuum vial
⑥ Add 3MAA
⑦ heat ⑤ [illegible]°C 3d

ARH-142-3



+TAM → COF-300

chemical	fw g	mass(mg)	mmol	mM	eq	vol(ml)
TerAld						
TAM						
Dioxane						
3MAA						

Same as ARH-1-144-1

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- quench rxn with 2ml of NH4OH (bubbles)
- pour the rxn mixture in a separatory funnel
- extract x3 toluene. => organic layer is yellow with brown/black
  - aq layer is black
- ~~-~~ @ 3rd extraction organic layer is colorless but with form of ppt.

(10) extract x3 1M, EDTA

(11) extract is brine

(12) Na_2SO_4 → ~~rotavap~~ filter → rotavap orange ppt.

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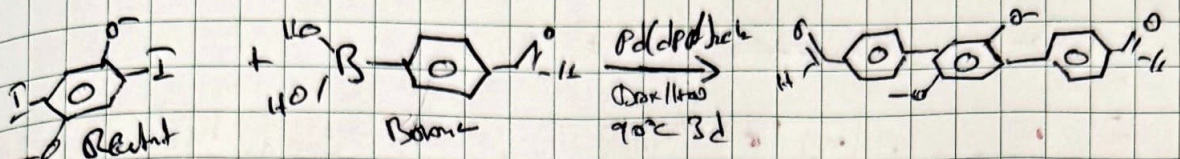
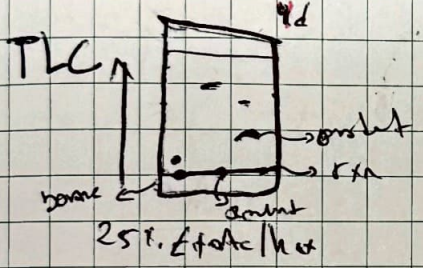
ARH-1449



chemical	Fw (g)	mass (mg)	mmol	mM	eq	vol (mL)
Reactant	391.16	300	0.76	633	1	—
Boronic	149	482	3.23	240	3.179	—
Pd(dppf)Cl2	800	61.29	0.0776	63	0.1	-
K2CO3	138	353	2.56	1910	3	-
Diox	-	-	-	-	-	9
water	-	-	-	-	-	3
Product	346	262.95	0.76		1	—

① in RBF Add Reactant 25 mL
② Add Boronic
③ Add K2CO3
④ Add Pd(dppf)Cl2
⑤ Vac x purge N2 x3
⑥ freeze pump thaw dox/water x3
⑦ Add 12 mL of Mix → orange sol
⑧ heat @ 90°C

@ rxn turns black overnight
No starting material on TLC (reactant)



⑩ ~~quench~~ wait for rxn to cool and quench with 1M HCl → bubbles (add HCl slowly).

⑪ extract with DCM x3 (1 mL each) → organics (organic layer).
⑫ extract with water x3
⑬ extract with brine x3

⑭ dry Na_2SO_4 → ~~filter~~ filter → rotovap → grey brown solid
⑮ pack a column with hexane
⑯ add solid decolorize with chloroform ⑰ run column with 25% EtOAc/Hex
⑱ product is ~~the~~ most polar fraction. NMR → pure.
⑲ yield

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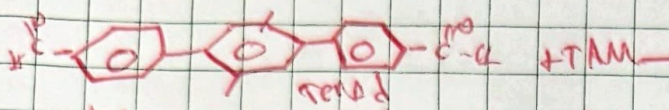
ARH-1-150-1

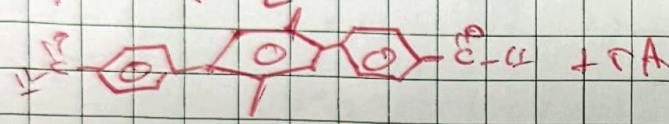


+TAM → COF-330

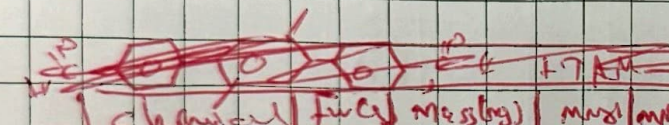
Chemical	Fw(g)	Mass(mg)	mmol	mM	eq	vol(mL)
Terph	3	3/4	0.1	100	2	—
TAM	380	19	0.05	50	1	—
Diox	—	—	—	—	—	2.5
Mesity	—	—	—	—	—	0.5
6MAA	—	—	—	—	—	0.1

① In a ~~beaker~~ tube Add TAM
② " " Add terph
③ Add Dox → sonic
④ Add mesylene → sonic
⑤ Add 6MAA → sonic
⑥ Freeze thaw ×3
⑦ heat 120°C 3d
⑧ good crys.

ARH-150-2



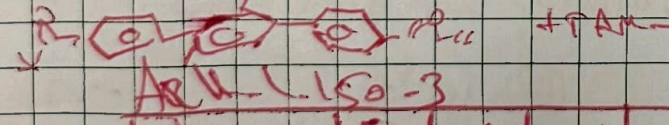
+TAM → cof-330



+TAM → COF-330

Chemical	Fw(g)	Mass(mg)	mmol	mM	eq	vol(mL)
Terph	3	3/4	0.1	100	2	—
TAM	380	19	0.05	50	1	—
Dox	—	—	—	—	—	2.5
Mesity	—	—	—	—	—	0.25
DMA	—	—	—	—	—	0.25
6MAA	—	—	—	—	—	0.1

① In a borosilicate tube Add TAM
② " " Add Terph
③ Add Dox → sonic
④ Add Mesity → sonic
⑤ Add DMA → sonic
⑥ Add 6MAA → sonic
⑦ Freeze thaw ×3
⑧ heat @ 120°C 3d
⑨ Look crys.



+TAM → ~~COF~~ COF-330

ARH-150-3

Chemical	Fw(g)	Mass(mg)	mmol	mM	eq	vol(mL)
Terph	3	3/4	0.1	100	2	
TAM	380	19	0.05	25	1	
Dox	—	—	—	—	—	2
~~Mes~~ 6MAA	—	—	—	—	—	0.2

① Ina borosilicate tube Add TAM
② Add Terph
③ Add Dox sonic
④ Add 6MAA → sonic
⑤ ~~Add~~ Freeze thaw ×3
⑥ heat @ 120°C 3d

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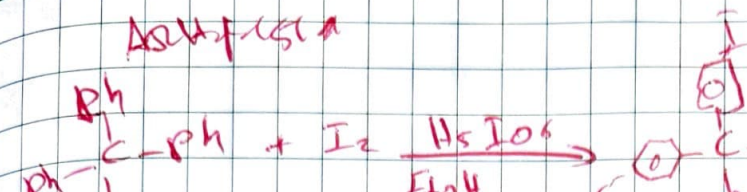

Chemical	FW (g)	mg/drop	mmol	mL	eq	volume
TBPM	320	100	0.312	156	1	-
I_2	253.8	282	0.873	43.68	2.8	-
H_5IO_6	227.9	55	0.374	18.72	1.2	-
EtOH	-	-	-	-	-	2

① in a 10 mL RBF Add TBPM
② Add I_2
③ Add H_5IO_6
④ Add EtOH
⑤ heat @ 85°C
⑥ TLC
⑦

1 day — rxn didn't move (reactant insoluble in EtOH) ⇒ Add 2mL THF

2 day — rxn proceeded

heat

heat

leave for 1 day (maybe I spot disappears) (diiodo ⟶ tetraiodo)?

⑧ rxn quenched with Sodium bisulfite ⟶ green
⑨ Add 2mL KOH 1M.
⑩ filter ⟶ collect residue
⑪ rinse chloroform (solvent overlayed with)
⑫ rinse DMSO ⟶ starting material removed

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ARH-1-152 (Actinometer Synthesis)

$$2\,KOH + H_2C_2O_4 \xrightarrow[80^\circ C]{H_2O} 2\,H_2O + K_2C_2O_4$$

Reagent	Fw (g)	mass (g)	mmol	mM	vol (mL)
H_2O	—	—	—	—	10
KOH	56.1	2.8120	37.49	3700	—
$H_2C_2O_4$	126.07	2.4270	19.24	1924	—
$K_2C_2O_4$	166.22	3.5020	18.01	1800	—

① in a 50 mL RBF Add KOH
② Add $H_2C_2O_4$
③ Add H_2O
④ Heat @ 80°C → 30 mins

$$K_2C_2O_4 + FeCl_3 \xrightarrow[45^\circ C]{H_2O} K_3Fe(C_2O_4)_3$$

Chemical	Fw (g)	mass (g)	mmol	mM	eq	vol (mL)
$FeCl_3 \cdot 6H_2O$	270	1.620	6	600		
$K_2C_2O_4$	166	3.000	18.04	1804		
H_2O	—	—	—	—	—	10

⑤ Add $FeCl_3 \cdot 6H_2O$ and heat to dissolve
⑥ Add H_2SO_4 → some
⑦ Add solvy to rxn mixture in step ④.

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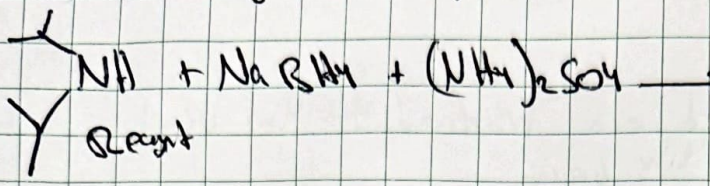
ARH-1-153 synthesis of Amine Borane



Chemical	Fw (g)	mass (g)	mmol	mM	eq	vol (mL)
$NaBH_4$	37.8	1000	26.43	979	1	–
$(NH_4)_2SO_4$	132.15	1750	13.22	489	0.5	–
Reagent	59.11	1.562	26.43	979	1	2.25
THF (SPS)	–	–	–	–	–	27
product	72		26.43		1	

① grind $(NH_4)_2SO_4$
② Add to 100mL RBF
③ grind $NaBH_4$
④ Add to RBF
⑤ Add Reagent
⑥ Add THF (SPS)
⑦ heat @ 80°C

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ARH-1-154 CNMR determination of MTV MIL-125 NH2 ratio

Preparation:

① 2mg of MOF was weighed and transferred to 4ml vial
② Add 3ml DEF → turns brown
③ Add 360μL DMSO → clear yellow solution
④ take NMR

	Input		output
~~ARH-1-12~~	0%		0%
~~ARH-1-48~~	5%		8%
~~ARH-1-49~~	10%		20%
ARH-1-63	15%		36%
ARH-1-64	20%		48%

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